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DIGITAL EGYPT PIONEERS INITIATIVE — DEPI

DEPI Industry Course (Round Five)

Applied AI Agents for Education and Teaching Professional

CAPSTONE PROJECT

KEMET: The Nile Flow

KEMET
AI Language Assistant
The Nile Flow

PROJECT DOCUMENTATION

Submitted by

The Rosetta Trio Team

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Group: GIZ5_IND1_G1

Supervised by: Engineer Alaa Samir

October 2026

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1. Project Planning & Management

a. Project Proposal

Executive Overview

Kemet: The Nile Flow is an AI-powered Socratic subject tutor, adaptive n8n workflow, and multilingual heritage portal, with the newly launched Egyptian Baccalaureate (EB) inquiry-based curriculum (advised by the International Baccalaureate). Inspired by the Rosetta Stone—which unlocked ancient knowledge by bridging two languages and three scripts—and the flow of the Nile as a symbol of life, growth, and continuity, the project integrates Egyptian visual identity to strengthen pride in national awareness while encouraging openness to global cultures.

It directly addresses a critical pedagogical challenge: generic Large Language Models (LLMs) act as passive "answer-dumping" machines. When secondary students use standard commercial chatbots to study English, they experience cognitive bypass (skipping Productive Struggle), algorithmic overfitting (memorizing isolated answers without conceptual Generalization), and cultural decontextualization (receiving Western, individualistic advice that ignores Egypt's collectivist family and community values).

English is the active first-phase language (with complete phase-one delivery targeted for 16 October 2026), and other language pathways are clearly labelled as forthcoming. Safeguards cover academic integrity, privacy, curricular boundaries, and the safe referral of distress-related inputs. To achieve this, Kemet: The Nile Flow operates on an integrated three-pillar architecture: (1) The Macro-Portal (Web Platform), (2) The Mobile Application (Google AI Studio App for targeted quick revision), and (3) The Micro-Flagship Engine (n8n & Telegram Bot).

To solve this, *Kemet: The Nile Flow* operates on a dual "Macro-Portal / Micro-Flagship" Architecture:

- The Macro-Portal (The Kemet Multilingual & Heritage Web Platform): A WCAG AA and UDL-compliant web portal designed to celebrate language learning across cultures, featuring interactive audio broadcasts, a 66-term bilingual phonetic lexicon, a 9-grid Universal Design for Learning (UDL) Choice Board, and British Dyslexia Association (BDA) readability toggles. Egyptian symbols, historical figures, and elements of civilization are connected with sourced educational content to encourage creative expression.
- The Micro-Flagship Engine (The Project 7 Socratic AI Agent & n8n Workflow): A production-ready Autonomous AI Agent orchestrated via n8n and deployed across Telegram, Webhooks, and Automated Gmail Reporting. Strictly bounded to Unit 1 (Living Well in a Complex World) and Chapter 1 of The Lost World, the assistant uses Socratic dialogue and inquiry-based learning. Following the Observe → Decide → Act cycle, it withholds direct answers, diagnoses grammatical and cognitive misconceptions, and delivers tiered Socratic hints (The Eye of Horus) so support progresses from modelling through guided practice to independent performance. It

dynamically branches across a 330-question adaptive item bank and enforces the EB curriculum's JRE+ Inquiry Model (Justification, Reasoning, Evidence, Reflect and Act). Teacher review remains central to content quality and assessment under the original 70/30 AI-drafting and expert-refinement model

Project Objectives

a) **Deploy a Socratic AI Tutoring Agent (Project 7 Core):** Build an autonomous AI agent (using the Role-Objective-Constraints-Tools framework, Chain-of-Thought [CoT] reasoning, and Few-Shot prompting) that guides students through Unit 1 concepts using the Gradual Release of Responsibility Model (*I Do, We Do, You Do*) rather than spoon-feeding answers.

b) **Orchestrate a Multi-Channel n8n Automation Pipeline (Modules 4, 5 & Session 25):** Connect Telegram, the Kemet Web Platform (Webhook), and the Kemet Mobile Application using non-linear branching, Google Gemini API, Google Sheets/Supabase (Memory & State Management), and Gmail using non-linear branching (Switch and IF Nodes), Structured JSON Outputs, rate-limiting (Wait/Limit strategy), and Fallback Error Handling.

c) **Implement Real-Time Adaptive Mastery &Personalization (Project 7 Extension & Session 21/22):** Utilize a 330-item leveled bank (Foundation [01-10], Apply [11-20], Challenge [21-30], plus Supported [D1]) to dynamically adjust question difficulty based on student performance streaks, ensuring Mastery-Based Learning (*"Learning is the constant; Time is the variable"*).

d) **Enforce the 70/30 "AI Draft, Teacher Refine" & Human-in-the-Loop Philosophy (Sessions 18 & 19):** Audit, clean, and transform raw AI-generated worksheet drafts (70%) into verified, culturally responsive Socratic hint sequences and 4-Level Analytic & Single-Point Rubrics (30%), automatically emailing hybrid progress alerts to teachers/parents when student performance drops below < 70%.

e) **Deliver Proactive Accessibility & National Identity (Sessions 20 & 23):** Embed the Curb-Cut Effect through dyslexia-friendly typography, text-to-speech (TTS) pronunciation, bilingual scaffolding, and the visual symbolism of Kemet (*Dark Soil, Sun Gold, Fertility Turquoise, Lotus of Renewal, Wings of Isis Guardrails, and Eye of Horus Diagnostics*).

Vision, Mission, and Strategic Objectives

Vision

To develop an accessible Egyptian language-learning environment that combines genuine linguistic competence, critical awareness and pride in national identity with openness to global languages and cultures.

Mission

Kemet uses curriculum-grounded AI, Socratic dialogue, adaptive practice and diverse media to help learners understand, practise and apply English. It connects Egyptian visual

identity with sourced cultural and historical learning to strengthen awareness and belonging, while teachers guide quality, assessment and intervention.

Strategic Objectives

Objective	Planned Delivery & Evaluation
SO1 Language understanding and transfer	Deliver Unit 1 and Story Chapter 1; assess performance on unfamiliar but curriculum-aligned tasks.
SO2 Personalised mastery	Implement diagnostic entry, differentiated practice, error review and progress feedback.
SO3 Accessibility and engagement	Provide mobile-friendly text/audio resources, UDL choice and dyslexia-friendly options; evaluate task completion and usability.
SO4 Media and digital literacy	Teach bias, absolute claims and correlation/causation through curriculum-linked analysis tasks.
SO5 Cultural and historical awareness	Provide sourced explanations of symbols, figures and their contexts; assess knowledge separately from liking the design.
SO6 Pride belonging and national identity	Use optional reflective and creative activities; evaluate perceived connection with a separate, age-appropriate instrument.
SO7 Reliable sustainable delivery	Complete integration and testing by 16 October; maintain teacher review, backups, risk controls and an approved operating-cost plan

Project Scope

Scope Dimension	In-Scope (Delivered in Capstone)	Out-of-Scope (Future Roadmap)
Curricular & Subject Boundaries	EB 2nd Secondary English – Unit 1 (Living Well in a Complex World): Reading Passages A1–A3 & B1–B3, Listening Broadcasts L1 (Highlight-Reel Effect) & L4 (2023 Egyptian School Health Survey), 66 Core Vocabulary Terms, Relative Clauses,	Full content generation for Units 2–10 and Chapters 2–10 of The Lost World (represented as UI preview cards on the Web Portal).

Scope Dimension	In-Scope (Delivered in Capstone)	Out-of-Scope (Future Roadmap)
	Participial Phrases (including Dangling Participle traps), Conditionals (0, 1st, 2nd, 3rd + Should/Were/Had Inversions), JRE+ Analytical Writing, and The Lost World (Chapter 1).	
AI & Automation Architecture	n8n Workflow integrating Telegram Bot (Text + Inline Keyboards + MP3 Audio), Web Portal Webhook, Gemini 2.5 Flash/Pro LLM Node, Session Memory State Tracker, Automated Gmail Progress/Intervention Reports, and Custom Gemini Gem prototype. Google AI Studio Mobile Application	Native iOS/Android mobile app compilation or paid third-party enterprise SIS licensing. Native App Store compilation (Mobile access is delivered in-scope via the Google AI Studio App instead)
Web Showcase & Accessibility	Self-contained, responsive Kemet Multilingual & Heritage Web Portal (HTML5/CSS3/JS) with WCAG AA contrast, Dyslexia-friendly mode, embedded Base64 MP3 audio, Web Speech API TTS, UDL Choice Board, and Socratic Hint verification.	Live synchronous video conferencing or peer-to-peer unmoderated student chatrooms.
Differentiation Tiers	4 Leveled Tracks: Supported/Below-Level (01-10 + D1), Standard/On-Level (11-20), Extension/Above-Level (21-30), and Twice-Exceptional (2e) Dyslexia-Friendly layout.	Clinical medical/psychological diagnosis (strictly blocked by Wings of Isis safety guardrails).

b. Strategic Analysis

Internal Environment

The internal analysis considers capabilities, resources, and limitations within the team's control. Student needs are not classified as team weaknesses; they inform the design and engineering requirements of the n8n and AI architecture.

Dimension	Analysis and Planning Implication
Media, UI/UX & Identity Expertise	Dr. Mai Abualfotouh contributes media, UI/UX, and training expertise; she leads the platform's visual design, video, songs, cultural review, the Mobile Application UI/UX, and the Kemet Multilingual Web Portal showcase.
Systems Architecture, AI & Strategic Documentation	Ms. Riham Khafagy is a strategic-studies researcher who leads the project's systems architecture, strategic planning, and documentation. She engineers the core n8n multi-channel workflow and the Socratic AI Agent integration.
Curriculum & Pedagogical Expertise	Mr. Usama Azzam teaches secondary English in an international-school context. He leads the EB curriculum organization, grammar, vocabulary, the 330-item adaptive bank, and the 70/30 educational rubric review.
Prepared Educational Resources	The 330-question adaptive bank, 4-tier routing system, and 66-item dictionary provide a concrete EB curricular foundation. Their availability ensures the AI agent remains strictly bounded to Unit 1.
Creative Resources	The visual identity and audio/video resources support distinctive delivery on the Web Portal; educational and historical claims require strict prompt grounding and source review.
Technical Capacity	The team is actively integrating complex n8n workflows, Webhooks, and Gemini APIs. Structured JSON generation and AI-assisted troubleshooting support progress but increase verification needs and latency testing demands.
Financial Capacity	Member-funded experimentation and free/trial services (n8n cloud, Gemini API, Qdrant) are used; a consolidated approved budget for production scaling must be established.
Governance	Technical work and final testing are collaborative. The team uses a shared issue list, version control (GitHub), and a named owner for each node/change to prevent workflow conflicts.

External Environment

External factors are assessed as planning conditions and hypotheses. The exploratory survey supplies user evidence to validate the Socratic and adaptive mechanisms.

Dimension	Opportunity or Constraint	Response
Learner Needs	Reported demand for diagnosis, adaptive review, and progress visibility.	Validate through the n8n adaptive quiz engine and Gmail reporting in an EB-specific pilot.
Access and Inclusion	No government-school participants; device and connectivity needs vary.	Deploy a lightweight Telegram bot and an offline-capable Web Portal to test low-bandwidth use.
Service Ecosystem	External API rate limits (Gemini), Webhook timeouts, and tier limitations can interrupt the AI workflow.	Inventory API limits; implement n8n Wait/Limit nodes and local JSON fallback error paths.
Training and Support	Advanced n8n integration and structured LLM prompting require specialized knowledge.	Use concise node-by-node guides, structured JSON output schemas, and protected testing time.
Alternative Support	Respondents use translation and generic AI tools (which often spoon-feed answers).	Differentiate via the "Eye of Horus" Socratic hints and strict EB curriculum grounding to enforce productive struggle.
Project Exposure	Presentation to DEPI reviewers may create feedback or future-development support opportunities.	Prepare a robust live demonstration of the Telegram bot, Web Portal, and Gmail reporting.
Cultural Accuracy and Trust	AI hallucinations can introduce historical or curricular errors.	Maintain a strict Unit 1 vector knowledge base and apply Wings of Isis safety guardrails.
Data and Safeguarding	Progress records involve learner data and potential distress-related inputs.	Use hashed session IDs (FERPA/COPPA) and distress-triage routing to protect privacy and ensure safe escalation.

SWOT Analysis

Strengths	Weaknesses
S1: AI, curriculum & media expertise **S2:** 330-item Unit 1 leveled bank **S3:** Distinctive Kemet visual identity **S4:** Socratic Agent & 70/30 design **S5:** Collaborative troubleshooting	**W1:** Limited advanced n8n experience **W2:** Very short delivery window **W3:** Free-tier API dependence **W4:** Version control complexity
Opportunities	**Threats**
O1: Learner interest in diagnostics **O2:** Teacher networks for EB pilot **O3:** DEPI presentation visibility **O4:** Scalable n8n content reuse	**T1:** API rate limits & trial expiry **T2:** Service outages during demo **T3:** Generic AI tool competition **T4:** Hallucinations undermining trust

Strategic responses

Response	Action	Link and owner
SO Use strengths to pursue opportunities	Use curated examples, audio and the question bank to deliver diagnosis and personalised English support.	S1/S2/S6 + O1; Usama, shared integration
SO	Use Egyptian visual storytelling and sourced media in optional knowledge and creative activities.	S1/S3 + O3/O4; Mai
WO Use opportunities to address weaknesses	Run a focused target-group pilot when integration is ready; record feedback against scope and usability.	W4/W6 + O2/O5; Usama, Riham, Mai
WO	Prepare short reproducible integration guides and a single controlled workflow version.	W1/W5 + O4; shared technical owners

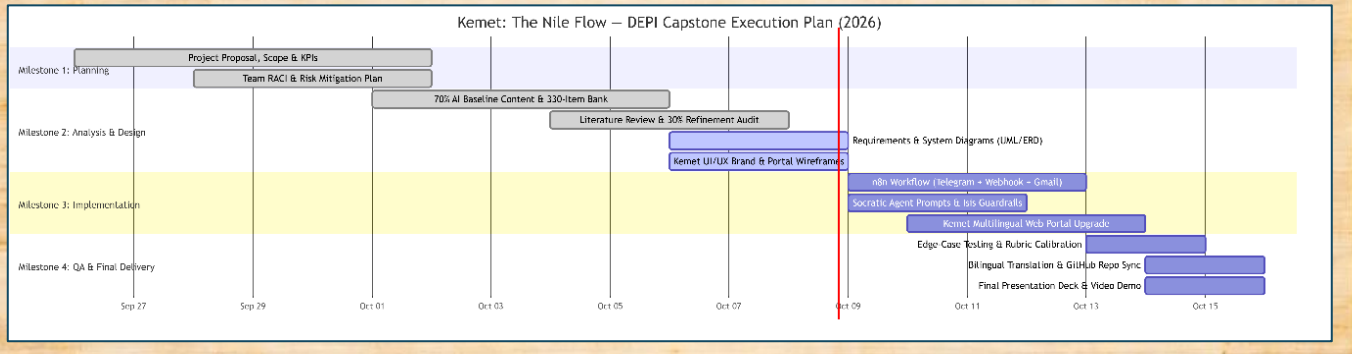
Response	Action	Link and owner
ST Use strengths to reduce threats	Maintain reviewed curriculum retrieval and deterministic question/hint fallbacks.	S2/S6 + T2/T4; Usama, shared integration
ST	Offer concise tasks, progress feedback and clear mobile navigation.	S1/S6 + T3/T6; Mai, Usama
WT Reduce weakness and exposure	Prioritise the complete phase-one journey; label future languages and advanced voice features honestly.	W2/W6 + T1/T6; all team
WT	Approve cost ceilings, dependency checks and export/backup procedures before release.	W3/W5 + T1/T2; Riham, all team

Priority Rule

First, secure the complete English learning journey, AI content accuracy, and reliable n8n multi-channel integration (Telegram, Webhook, Gmail). Next, implement the adaptive progress feedback, accessibility features, and cultural portal activities within that journey. Advanced voice assessment, WhatsApp integration, native mobile apps, and additional languages require a separate feasibility and funding decision; stated learner interest alone does not guarantee delivery in Phase One.

c. Project Plan, Timeline (Gantt Chart), Milestones & Resource Allocation

Visual Project Timeline (Mermaid Gantt Chart)



Milestones, Deliverables & Resource Allocation Table

Phase / Milestone	Target Deadline	Key Deliverables	Allocated Resources (Tools & Tech Stack)
M1: Project Planning & Management	02 Oct 2026 (Completed)	Project Proposal, Gantt Schedule, RACI Matrix, Risk Mitigation Plan, and Course-Aligned KPIs (01_project_plan.md).	GitHub Projects, Markdown, Mermaid.js, DEPI Documentation Framework.
M2: Literature Review, Requirements & System Design	09 Oct 2026 (Active Submission)	• Literature Review + 70/30 Audit Report (02_literature_review.md) • Stakeholder & Functional/Non-Functional Specs (03_requirements.md) • System Architecture, ERD, DFD, Sequence, Activity, State, Class Diagrams & UI/UX Guide (04_system_design.md).	EB Grade 11 Curriculum Docs, The Lost World Text, Draw.io / Mermaid.js, WCAG Contrast Checker, BDA Style Guide 2023.
M3: Core Implementation (n8n + Socratic Agent + Web Portal)	14 Oct 2026	• Working n8n Workflow (kemet_n8n_workflow.json) with Telegram Bot, Webhook, Gemini Agent, & Gmail Reports • Custom Gemini Gem & Prompt Library (prompts/) • Kemet Web Showcase Portal (Kemet_Language_Portal.html) + 4 Tiered PDFs.	n8n (Self-hosted/Cloud), Google Gemini API, Telegram Bot API (@BotFather), Google Sheets/Supabase, HTML5/CSS3/JS, Web Speech API.
M4: Testing, Bilingual Reports & Final Presentation	16 Oct 2026 (Final Submission)	• Test Plan, Edge-Case Logs, 4-Level Analytic Rubric & 3 Sample Student Chats (06_testing.md) • User Manual, Bilingual Pitch Deck (PPT/PDF), Video Demo, & Complete GitHub Repo (07_presentation/).	CoGrader/MagicSchool Rubric Calibration, GitHub Repository, Canva/PPT, Screen Recorder.

d. Delivery and governance

Planned period	Deliverable and acceptance gate	Responsibility
9 October	**Submit concept, exploratory study, strategic plan, scope, risks, roles and measurement framework.**	Riham leads documentation; Mai co-leads study/methodology; Usama reviews curriculum.
10–11 October	**Freeze phase-one requirements; complete content/source review and functional portal layout.**	Mai and Usama; shared technical support.
12–13 October	**Integrate retrieval, diagnostic/adaptive practice, memory, rubric feedback and planned email reports.**	Shared technical execution; owner assigned per task.
14 October	**Test the complete learner journey, safeguards, failures, mobile layout and accessibility.**	All team; curriculum review by Usama, cultural review by Mai.
15 October	**Resolve critical defects, rehearse demonstration and run learner pilot if feasible.**	All team; teacher-led recruitment; document pilot size and method.
16 October	**Deliver the complete functioning project within phase-one scope and supporting documentation.**	All team; joint final acceptance.

e. Task Assignment and Roles

The Rosetta Trio Team combines media and design, strategic research and English pedagogy. Technical integration, troubleshooting and final testing are collaborative; an owner and reviewer are assigned to each change.

1- Dr Mai Abualfotouh

Leads platform and visual design, Egyptian visual identity, video, songs, MP3 and voice-over production, motivation and cultural and historical source review. Designs the survey wording and co-develops the exploratory analysis, strategic methodology and documentation with Riham.

2- Ms Riham Khafagy

Leads strategic planning, documentation, risk and KPI records, drawing on strategic-studies research, a master's degree and conference work at Bibliotheca

Alexandrina. Coordinates these tasks with Mai and Usama. She is not assigned sole responsibility for workflow engineering.

3- Mr Usama Azzam

Leads English curriculum selection and organisation, grammar and vocabulary, the 330-question bank, the 66-entry dictionary, pedagogical review and rubric calibration. Contributes international-school, SAT and Baccalaureate teaching experience. Leads most survey distribution and supports recruitment for learner testing.

f. Deliverable Responsibility Assignment Matrix (RACI)

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

Work package	**Dr Mai media and platform lead**	**Ms.Riham strategic lead**	**Mr Usama English lead**
Strategic plan documentation risks and KPIs	R	A/R	C
English curriculum bank dictionary and educational review	C	C	A/R
Questionnaire wording and exploratory analysis	A/R	R	R distribution and review
Cultural sources identity audio video and platform design	A/R	C	C
Workflow architecture and integration	C	A/R	C
Technical change control and test records	R shared	R shared	R shared
Pedagogical hints and rubric refinement	C	C	A/R
Final testing and project acceptance	R joint acceptance	R joint acceptance	R joint acceptance

g. Risk Assessment & Mitigation Plan

In accordance with Module 1 (Session 2 & 5), Module 2 (Session 7 & 8), Module 5 (Session 15 & 17), and Module 6 (Sessions 18–25), the following technical, pedagogical, and operational risks have been identified and mitigated:

Risk ID	Risk Category & Description	Probability	Impact	Planned mitigation and fallback strategy
R-01	Cognitive Bypass & Direct-Answer Leakage ("Overfitting"): Students use the AI to get direct homework answers or memorize static answer keys without understanding concepts.	High	Critical	Mitigation: Plan to apply the Socratic Persona Constraint in Gemini and flipped the 330-item bank in n8n/HTML so that Attempt 1 triggers the Eye of Horus Hint (why field) rather than the answer, enforcing Productive Struggle and Gradual Release (I Do, We Do, You Do).
R-02	Unrefined AI Drafts ("Garbage In, Garbage Out" / 70-30 Gap): Raw AI-generated study guides contain OCR spacing typos ("Furt hermore", "Psycholo gists"), generic branding, or unrealistic classroom timing.	High	High	Mitigation: Plan a mandatory 30% Human-in-the-Loop Refinement Pass reviewed by Usama, with cultural review by Mai. Raw PDFs are logged as baseline artifacts, while all production texts, prompts, and HTML code undergo automated regex cleaning, brand unification, and rubric calibration.
R-03	LLM Hallucination & Out-of-Bounds Drift: The AI invents non-existent studies, teaches non-EB grammar rules, or answers off-topic queries (e.g., gaming, coding).	Medium	High	Mitigation: Plan strict Grounding Constraints ("Only use the provided EB Unit 1 text; do not invent studies") and an n8n Switch Node Gatekeeper that routes off-topic inputs to a polite Wings of Isis boundary message.

Risk ID	Risk Category & Description	Probability	Impact	Planned mitigation and fallback strategy
R-04	n8n API Rate-Limiting, Downtime, or Malformed JSON: High-frequency student requests on Telegram crash the Gemini/Gmail API, or the LLM returns unstructured text that breaks downstream nodes.	Medium	High	Mitigation: Plan the Session 15 "Wait and Limit" Strategy on loops, enforced Structured JSON Output schemas in the AI Node, and wired a Fallback Error Path that serves deterministic hints from the local 330-item bank if the LLM API times out.
R-05	Scope Creep Between Macro Web Vision & Micro Unit 1 Rubric: Tension between building a broad multi-language portal vs. delivering a deep Unit 1 Socratic Agent before Oct 16.	Medium	Medium	Mitigation: Use the Macro-Portal / Micro-Flagship pattern: the Web Portal showcases the broad multilingual Kemet vision at the UI layer, while 100% of backend engineering is focused on the EB Unit 1 Flagship Module.
R-06	Student Data Privacy & Emotional Distress (Privacy and Safety): Students share Personally Identifiable Information (PII) or express severe mental health distress during Unit 1 stress discussions.	Low	Critical	Mitigation: Anonymized student state tracking via hashed Telegram/Session IDs (pseudonymisation and access controls), self-hosted n8n capability, and a dedicated Distress Triage Branch that halts academic quizzing and provides school counselor / national helpline contacts.
R-07	Trial expiry and operating cost interruption	High	Critical	Maintain a dependency and expiry register, approve recurring funding and retain exported workflows and approved local content; test fallback access.

Risk ID	Risk Category & Description	Probability	Impact	Planned mitigation and fallback strategy
R-08	Technical learning curve and limited practical training	High	High	Allocate protected integration time; use concise verified guides and AI-assisted troubleshooting with human verification; prioritise the complete phase-one journey.
R-09	Conflicting shared technical edits and incomplete handover	Medium	High	Use one controlled workflow version, a named owner and reviewer per change, backups and a shared test record.
R-10	Unverified cultural content or unmeasured identity effect	Medium	High	Mai reviews historical and cultural sources; evaluate knowledge and identity separately; do not treat visual attraction as established belonging.
R-11	Small exploratory sample and insufficient target-group testing	Medium	High	Report sample limits; recruit Baccalaureate learners if readiness allows; record usability and learning evidence and do not generalise beyond the tested group.

h. Key Performance Indicators (KPIs) — Metrics for Project Success

The plan defines 16 indicators covering educational, technical, cultural and sustainability objectives. Numerical thresholds are planned targets or proposals requiring validation, not achieved results.

KPI ID & Strategic Objective (SO)	Target Benchmark (Aligned with Technical/Pedagogical Identity)	Measurement Method	Lead Owner & Timing
K01 Socratic Guidance (SO1)	100% of first-attempt direct-answer requests receive "Eye of Horus" Socratic guidance rather than an answer dump.	Inspect 20 fixed adversarial prompts against the Gemini AI agent to verify modeling vs. premature delivery.	Riham, Usama (Pre-release)
K02 Expert Refinement (SO1/SO7)	Apply the 70/30 workflow: 100% of the AI-generated JSON payloads and rubric criteria receive documented expert review.	Before/after audit of raw AI drafts versus the refined n8n production data.	Riham, Usama, Mai (Pre-release)
K03 Cognitive Demand (SO1)	Preserve the 30/40/30 distribution across low, intermediate, and high cognitive demand (DOK levels).	Audit the 330 tagged items within the EB Unit 1 adaptive curriculum bank.	Riham, Usama (Pre-release)
K04 Grounding & Item Quality (SO1/SO7)	Zero critical factual hallucinations across all 330 Unit 1 items and 66 dictionary entries.	Check source alignment against MOETE EB texts and The Lost World Ch. 1; resolve defects.	Riham, Usama (Pre-release)
K05 Response Latency (SO7)	Deterministic n8n quiz/hint responses < 2 seconds; Gemini AI CoT rubric feedback < 4.5 seconds.	Timestamp 30 requests per Telegram/Webhook path; record median and p95 under declared load.	Riham, Shared Tech (Pre-release)
K06 Reliability & Fallback (SO7)	≥95% successful n8n executions across a 50-run test; 100% of Gemini API timeouts invoke the local JSON fallback.	Log successes, malformed-input handling, and timeout tests in n8n execution logs.	Riham, Shared Tech (Pre-release)

KPI ID & Strategic Objective (SO)	Target Benchmark (Aligned with Technical/Pedagogical Identity)	Measurement Method	Lead Owner & Timing
K07 Educator Efficiency (SO7)	Reclaim 5–10 hours/week via automated Gmail reporting and JRE+ rubric evaluation.	Time manual grading/reporting tasks versus the automated n8n pipeline.	Riham, Usama (Pilot)
K08 Adaptive Routing (SO2)	100% correct n8n state transitions for level-up (Streak=2), level-down, and Supported (D1) paths.	Test defined answer streaks, session memory persistence, and recovery in the database.	Riham, Shared Tech (Pre-release)
K09 Accessibility (SO3)	Achieve WCAG AA contrast and BDA Dyslexia-friendly presentation on the Web Portal.	Contrast, keyboard navigation, Web Speech API TTS, and dynamic scaling checks.	Mai, Riham (Pre-release)
K10 Adoption & Mastery (SO1/SO3)	≥80% complete the EB Unit 1 core task; demonstrate JRE+ analytical improvement.	Record paired pre/post task scores from the adaptive quiz engine and JRE+ submissions.	Riham, Usama (Pilot)
K11 Cultural Knowledge (SO5)	Mean gain ≥20 percentage points on the Kemet heritage/history symbol quiz.	Use comparable pre/post questions; report sample size (n) and baseline.	Mai, Riham (Pilot)
K12 Pride & Belonging (SO6)	≥70% agree the Web Portal experience helped connect with Egyptian identity.	Separate voluntary, neutral questionnaire items on pride, belonging, and awareness.	Mai, Riham (Pilot)
K13 Media Literacy (SO4)	≥80% meet the rubric threshold on highlight-reel bias, absolute claims, and correlation vs. causation tasks.	Review student analysis using the established JRE+ rubric; record each dimension.	Mai, Riham, Usama (Pilot)

KPI ID & Strategic Objective (SO)	Target Benchmark (Aligned with Technical/Pedagogical Identity)	Measurement Method	Lead Owner & Timing
K14 Usability (SO3)	≥80% successfully navigate to the Telegram bot or Web Portal practice without assistance.	Observe task success, time, and obstacles on mobile interfaces; record device.	Mai, Riham (Pilot)
K15 Engagement & Return (SO3)	Establish initial baseline for MP3/resource opens and 7-day Telegram bot return rate.	Track anonymized, hashed session identifiers to distinguish clicks from completed learning.	Riham, Shared Tech (Post-launch)
K16 Financial Continuity (SO7)	100% of n8n cloud, Qdrant, and Gemini API dependencies have documented limits and fallbacks.	Dependency register and operating-cost approval reviewed against an agreed ceiling.	Riham, All Team (Pre-release)

2. Exploratory study of learner needs and requirements for an intelligent English learning assistant

a. Study purpose and method

A descriptive exploratory study was conducted using an online Google Forms questionnaire. Fourteen response records from secondary-school learners were collected on 27 and 28 September 2026. The study examined English learning difficulties, learner goals, curriculum support needs, interaction and explanation preferences, motivation, diagnosis and progress monitoring, access channels and barriers to continued use, to inform Kemet and its first-phase plan.

The instrument contained 46 closed questions across ten themes: 36 single-response items and ten multiple-response items, plus a conditional open follow-up. Every closed item had 14 valid responses. Percentages use 14 as the denominator and are rounded to one decimal; totals can exceed 100% for multiple-response questions. The complete questionnaire, item results and charts are provided in Appendix A at the end of the document. Respondent link: https://forms.gle/uhrUe6YVaAGoZVQz5

Recruitment used team contacts without a documented random sampling frame. This small exploratory sample is not representative of all secondary-school or Egyptian Baccalaureate learners. Anonymous records do not verify participant uniqueness; the number invited is unknown, so a response rate cannot be calculated. Responses describe self-reported needs and preferences, not platform effectiveness or the effect of national

identity. Numerical analysis uses the raw response export rather than repeated chart labels in the copied summary.

b. Learner profile and existing support

For current secondary grade, selections were distributed as follows: first secondary by 1 of 14 (7.1%); second secondary by 5 of 14 (35.7%); third secondary by 8 of 14 (57.1%).

For school type, selections were distributed as follows: government by 0 of 14 (0.0%); language / experimental by 1 of 14 (7.1%); private by 8 of 14 (57.1%); international by 5 of 14 (35.7%).

Regarding additional english lessons outside school, 7 of 14 answered yes (50.0%), compared with 7 answering no (50.0%).

For sources of help outside class, selections were distributed as follows: teacher / private lesson by 4 of 14 (28.6%); friend / family member by 7 of 14 (50.0%); educational videos by 0 of 14 (0.0%); google / dictionary / translation by 4 of 14 (28.6%); chatgpt / ai tool by 6 of 14 (42.9%); no help when needed by 2 of 14 (14.3%). Multiple selections were permitted.

The respondents are predominantly in the second and third secondary years and attend private or international schools. Half reported needing external tuition, while help was sought from family, teachers, search tools and AI. The absence of selections for educational videos describes the stated help channel in this sample; it does not establish that video would be ineffective. The plan should provide curriculum-linked help beyond classroom hours, while a later pilot should recruit the specific Baccalaureate audience and include more diverse school contexts.

c. Language difficulties and transfer of learning

For skills requiring improvement, selections were distributed as follows: speaking by 5 of 14 (35.7%); pronunciation by 4 of 14 (28.6%); grammar / sentence formation by 3 of 14 (21.4%); vocabulary / use by 1 of 14 (7.1%); listening comprehension by 1 of 14 (7.1%); reading comprehension by 1 of 14 (7.1%); writing by 2 of 14 (14.3%); thinking in english without arabic translation by 4 of 14 (28.6%). Multiple selections were permitted.

Regarding forgetting previously learned grammar or vocabulary, 10 of 14 answered yes (71.4%), compared with 4 answering no (28.6%).

Regarding difficulty transferring exam grammar into speech or writing, 3 of 14 answered yes (21.4%), compared with 11 answering no (78.6%).

Regarding difficulty recalling or using vocabulary after memorisation, 3 of 14 answered yes (21.4%), compared with 11 answering no (78.6%).

Regarding thinking in arabic before expressing a sentence in english, 6 of 14 answered yes (42.9%), compared with 8 answering no (57.1%).

Regarding a gap between examination grades and speaking or writing ability, 6 of 14 answered yes (42.9%), compared with 8 answering no (57.1%).

Speaking, pronunciation and thinking in English emerge as salient development needs, while forgetting earlier learning is reported more often than the specific grammar-transfer and vocabulary-recall difficulties. These are distinct experiences and should not be collapsed into a claim that most learners have every difficulty. The assistant should diagnose individual prerequisites, revisit earlier concepts and use short tasks requiring learners to apply language rather than merely reproduce an examination answer.

d. Pronunciation and confidence in speaking

Regarding embarrassment when speaking in front of classmates, 3 of 14 answered yes (21.4%), compared with 11 answering no (78.6%).

Regarding avoiding words or speech because of uncertain pronunciation, 6 of 14 answered yes (42.9%), compared with 8 answering no (57.1%).

Regarding interest in pronunciation feedback and repeated practice, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding interest in articulation guidance, 10 of 14 answered yes (71.4%), compared with 4 answering no (28.6%).

Regarding expected confidence from individual speaking practice, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Although reported embarrassment is not widespread, avoidance caused by uncertainty about pronunciation affects a larger group, and interest in feedback and individual practice is high. The plan should distinguish available listening, dictionary and text-to-speech support from automated pronunciation scoring and live conversation. The latter require feasibility testing, funding and educational validation; expected confidence is an anticipated benefit, not an observed effect of using Kemet.

e. Learning goals and practical English

Regarding learning english for use beyond examination grades, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding understanding how to apply grammar and vocabulary in different situations, 14 of 14 answered yes (100.0%), compared with 0 answering no (0.0%).

Regarding independent expression without continual arabic translation, 11 of 14 answered yes (78.6%), compared with 3 answering no (21.4%).

For purposes for learning english, selections were distributed as follows: study / exams by 4 of 14 (28.6%); fluent speaking by 9 of 14 (64.3%); university / future study by 8 of 14 (57.1%); travel / communication by 7 of 14 (50.0%); future work by 9 of 14 (64.3%); films / videos / english content by 4 of 14 (28.6%); courses / apps (other response) by 1 of 14 (7.1%). Multiple selections were permitted.

The findings favour transferable competence, fluent communication and future study or work, alongside examination support. Kemet should therefore retain curriculum boundaries for the flagship tutor while providing curriculum-aligned tasks with practical contexts.

Broader university and employment applications are a future pathway, rather than a commitment to deliver additional courses within the first phase.

f. Curriculum support and continuity of learning

For desired curriculum support, selections were distributed as follows: explain unclear content by 5 of 14 (35.7%); prepare for the next class by 3 of 14 (21.4%); review after class by 2 of 14 (14.3%); question / exam practice by 8 of 14 (57.1%); review forgotten prerequisites by 9 of 14 (64.3%); explain errors and correction by 8 of 14 (57.1%). Multiple selections were permitted.

Regarding preparation for upcoming reading and listening lessons, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding reviewing forgotten prerequisites before a new lesson, 14 of 14 answered yes (100.0%), compared with 0 answering no (0.0%).

Regarding asking questions whenever support is needed, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding remembering recurring errors and providing later practice, 14 of 14 answered yes (100.0%), compared with 0 answering no (0.0%).

Prerequisite revision, explanation of errors and examination practice receive more selections than preparation or review as selectable functions. Nevertheless, the separate preparation question receives substantial affirmative interest. This difference reflects question format and context, not a contradiction. A planned sequence should connect preparation, supported learning, error explanation and subsequent practice, with memory used to return to recurring difficulties rather than merely store chat history.

g. Diagnosis personalisation and progress

Regarding beginning with a short diagnostic assessment, 14 of 14 answered yes (100.0%), compared with 0 answering no (0.0%).

Regarding periodic feedback on improvement and remaining needs, 11 of 14 answered yes (78.6%), compared with 3 answering no (21.4%).

Regarding practice matched to individual level and errors, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

For preferred evidence of progress, selections were distributed as follows: short progress report by 7 of 14 (50.0%); simple percentages per skill by 4 of 14 (28.6%); compare old and new pronunciation recordings by 9 of 14 (64.3%); short tests in realistic situations by 8 of 14 (57.1%). Multiple selections were permitted.

Unanimous interest in diagnosis, together with support for tailored practice and progress feedback, strengthens the rationale for a short entry assessment and adaptive routing. Learners prefer concrete evidence such as comparable recordings and realistic tasks. Progress reports should connect scores with understandable strengths and next steps.

Recording comparison can be useful without representing it as a validated automated pronunciation score.

h. Interaction explanation and motivation

For preferred interaction formats, selections were distributed as follows: live voice conversation by 4 of 14 (28.6%); text conversation by 2 of 14 (14.3%); audio and text together by 11 of 14 (78.6%); voice messages by 2 of 14 (14.3%); speaking visual character by 3 of 14 (21.4%). Multiple selections were permitted.

For preferred explanation methods, selections were distributed as follows: short explanation with examples by 9 of 14 (64.3%); situation / story by 6 of 14 (42.9%); guiding questions by 5 of 14 (35.7%); audio and visuals by 3 of 14 (21.4%); interest-based examples by 3 of 14 (21.4%). Multiple selections were permitted.

Regarding english with arabic support when needed, 11 of 14 answered yes (78.6%), compared with 3 answering no (21.4%).

Regarding gradual reduction of arabic support, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding short encouraging opening messages, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

For factors encouraging continued use, selections were distributed as follows: feedback on progress by 8 of 14 (57.1%); short challenges / questions by 4 of 14 (28.6%); points / levels / badges by 4 of 14 (28.6%); interest-based topics by 3 of 14 (21.4%); simple task reminders by 3 of 14 (21.4%); know my level / use english in real life (other) by 1 of 14 (7.1%). Multiple selections were permitted.

Combined audio and text is preferred to either isolated channel. Short explanations and examples receive more selections than guiding questions, so Socratic dialogue should be concise and supported by modelling when needed. Arabic scaffolding can be gradually reduced, rather than withheld from the outset. Progress feedback appears more motivating than badges alone. Encouraging messages support the motivational design, but these questions do not specifically test Egyptian symbols, songs or national identity.

i. Practice situations and learner interests

For preferred practice situations, selections were distributed as follows: daily situations / travel / shopping by 5 of 14 (35.7%); everyday conversations by 7 of 14 (50.0%); presentations / public speaking by 9 of 14 (64.3%); school / curriculum / exams by 6 of 14 (42.9%); writing / expression by 5 of 14 (35.7%); listening to authentic conversations by 7 of 14 (50.0%); future university / work situations by 7 of 14 (50.0%). Multiple selections were permitted.

Regarding practice linked to personal interests, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Public speaking, conversations, authentic listening and future study or work provide relevant contexts for practising curriculum language. Interest-linked tasks may make

examples more meaningful, while the plan must keep learning objectives explicit. Media, songs and optional creative activities can diversify participation; their learning effects still require evaluation.

j. Access interface and barriers to continued use

For preferred access channels, selections were distributed as follows: mobile app by 10 of 14 (71.4%); whatsapp or telegram by 3 of 14 (21.4%); website by 0 of 14 (0.0%); multiple devices by 1 of 14 (7.1%).

For preferred interface structure, selections were distributed as follows: simple immediate question entry by 1 of 14 (7.1%); today's lesson / revision needs by 2 of 14 (14.3%); chat-like interface by 5 of 14 (35.7%); clear language-skill sections by 6 of 14 (42.9%).

Regarding availability whenever needed, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

For reasons for discontinuing use, selections were distributed as follows: repetitive / unclear answers by 6 of 14 (42.9%); no noticeable improvement by 9 of 14 (64.3%); difficult to use by 9 of 14 (64.3%); long / boring practice by 6 of 14 (42.9%); poor curriculum / level understanding by 4 of 14 (28.6%). Multiple selections were permitted.

The preference for a mobile app and the absence of website selections reveal a delivery tension. A responsive web platform with Telegram access is the first-phase scope choice; it should be tested on phones and should not be described as the preferred channel in this sample. Clear skill sections, easy navigation, brief practice and visible progress address the most frequently selected disengagement risks. Availability at any time requires sustainable hosting and service continuity, not merely an accessible link.

k. Perceived usefulness and intention to adopt

Regarding more practice through embarrassment-free speaking support, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Regarding support for learners with fewer speaking opportunities, 12 of 14 answered yes (85.7%), compared with 2 answering no (14.3%).

Regarding use alongside school and the teacher, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Regarding continued use after secondary school, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Regarding willingness to use the proposed assistant now, 13 of 14 answered yes (92.9%), compared with 1 answering no (7.1%).

Regarding interest in an additional feature, 5 of 14 answered yes (35.7%), compared with 9 answering no (64.3%).

The high affirmative rates support testing an assistant that complements the teacher and creates additional practice opportunities. They indicate stated expectations and intention, not actual uptake, retention, equitable access or measured achievement gains. A

readiness-dependent pilot should observe the complete learning journey and record task completion and paired learning measures before reporting effectiveness.

l. Open suggestions

Although five respondents (35.7%) expressed interest in an additional feature, only four nonblank follow-up texts were supplied (28.6% of records); ten records were blank (71.4%). Suggestions concerned understanding the user and improving the conversational atmosphere, a course-like application, everyday conversations based on curriculum topics, and free access for all students. Each theme rests on one text and cannot establish prevalence. No fifth suggestion is inferred. These comments support an encouraging conversational tone, organised content, curriculum-based practical English and an affordability review. Original texts are reproduced unchanged in Appendix A.

m. Implications for the project plan

Taken together, the findings prioritise a coherent learning journey rather than simply increasing the number of displayed resources. That journey should begin with brief diagnosis, review necessary prerequisites, combine concise explanation and examples with Socratic guidance, and provide differentiated practice and feedback explaining errors and next steps. These needs support the planned levelled question bank, support pathways, learning memory, adaptive assessment and progress reports, with integration and execution tested before delivery.

Audio and phone preferences call for clear responsive navigation and audio alongside text, with an explicit distinction between first-phase support and capabilities requiring further development and funding, including live conversation, pronunciation scoring, a native mobile app and WhatsApp. Strengthening pride in and belonging to national identity, firmly establishing that identity in learners, and educating through Egyptian history, symbols and historical figures are central project objectives. The current questionnaire did not measure them directly; separate instruments and indicators therefore distinguish cultural knowledge from liking the design and expected benefits from measured effects.

3. Requirements Gathering

a. Stakeholder Analysis

To ensure Kemet: The Nile Flow addresses authentic educational needs within the Egyptian secondary ecosystem while meeting DEPI technical benchmarks, the following key stakeholders, their primary pain points, and their system requirements have been identified:

Stakeholder Group	Role & Persona	Core Pain Points & Challenges	Identified Needs & System Value Proposition
1. Egyptian Baccalaureate	• Below-Level / Supported Learners	• Overwhelmed by dense academic texts,	• Socratic 24/7 AI Tutor accessed via Telegram,

Stakeholder Group	Role & Persona	Core Pain Points & Challenges	Identified Needs & System Value Proposition
(EB) Grade 11 Students (Primary End-Users)	• On-Level / Standard Learners • Above-Level / Advanced Learners • Twice-Exceptional (2e) / Dyslexic & ELL Learners	complex grammar (participial reductions, conditional inversions), and the JRE+ (Justify–Reason–Evidence) writing structure. • Generic AI tools spoon-feed answers (causing exam Overfitting) or use Western contexts detached from Egyptian life. • Visual crowding and lack of audio support hinder neurodivergent learners.	Web Portal, and Mobile App. • Adaptive difficulty branching (Tier 1 Tier 2 Tier 3 + D1) tailored to real-time mastery. • WCAG AA & BDA Dyslexia-Friendly UI with embedded MP3 broadcasts, phonetic TTS, and a UDL Choice Board.
2. Secondary English Educators (Human-in-the-Loop Operators)	Classroom teachers & department heads delivering the MOETE EB curriculum (e.g., Dr. Usama persona).	• Spend ~40% of a 54-hour work week on repetitive grading, worksheet tiering, and progress tracking. • Struggle to provide timely, individualized Analytic/Single-Point Rubric feedback on 150–180 word JRE paragraphs for large classes.	• 70/30 "AI Draft, Teacher Refine" Workflow reclaiming 5–10 hours/week. • Automated n8n CoGrader/Gemini Rubric Evaluation that grades JRE writing and dispatches Hybrid Gmail Alerts when a student scores < 70%.
3. Parents & School Counselors (Secondary Stakeholders)	Family support networks in Egypt's collectivist culture & school mental-health staff.	• Lack real-time visibility into student mastery gaps prior to high-stakes exams (where 68% of teens report high stress).	• Automated, jargon-free Gmail Mastery & Progress Summaries. • Wings of Isis Safety

Stakeholder Group	Role & Persona	Core Pain Points & Challenges	Identified Needs & System Value Proposition
		• Unequal counseling access (74% urban vs. 31% rural).	Guardrails that detect emotional distress and immediately direct students to trusted family members or school counseling.
4. MOETE & DEPI Evaluators (Institutional Governance)	Curriculum supervisors & MCIT/DEPI capstone review committee.	• Need verifiable proof of Academic Integrity, Zero Hallucination, Data Privacy (FERPA/COPPA), and alignment with national identity ("Proud Egyptians with a Global Perspective").	• Self-hosted, open-source n8n architecture with strict Unit 1 knowledge boundaries, full documentation on GitHub, and culturally grounded Kemet identity.
5. Project Development Team (Internal Builders)	Ms. Riham (Architect/n8n), Dr. Usama (English SME), Dr. Mai (Media/UI-UX).	• Need modular, well-documented workflows to integrate a 330-question bank, an n8n Telegram/Email agent, and a multilingual web showcase before Oct 16.	• Clean JSON data schemas, n8n Canvas Sticky Notes, and a Macro-Portal / Micro-Flagship architecture.

b. User Stories & Use Cases

Agile User Stories & Acceptance Criteria

Story ID	User Persona	User Story ("As a... I want to... So that...")	Acceptance Criteria (Verification Standard)
US-01	**Struggling Student (Below-Level)**	As a Grade 11 student struggling with grammar, I want the AI Tutor to give me a clear hint and narrow my choices when I miss a question, so that I can figure out the rule myself	• On Attempt 1 error, the bot locks the direct answer and sends the ◎ Eye of Horus Hint (why rationale). • If Attempt 2 fails, the bot branches down to Tier 1 (01–10)

Story ID	User Persona	User Story ("As a... I want to... So that...")	Acceptance Criteria (Verification Standard)
		without feeling overwhelmed.	or Supported (D1) with binary 2-button choices.
US-02	**On-Level Student (Standard)**	As a secondary student practicing Unit 1 writing, I want step-by-step Socratic coaching on my 150–180 word JRE+ paragraph, so that I learn how to link my Claim, Reason, and Evidence using hedging and participial phrases.	• Agent refuses to write the essay for the student. • Agent prompts sequentially for J (Claim), R (Reason), E (Unit 1 statistic), and + (Concession/Hedging). • Returns a 4-Level Analytic & Single-Point Rubric evaluation.
US-03	**Advanced Student (Above-Level)**	As an advanced learner, I want the system to automatically challenge me with higher-order questions (Conditional Inversions, Correlation vs. Causation, Counterarguments) when I answer correctly, so that I stay engaged.	• Achieving a streak of 2 correct answers in Apply (11–20) automatically updates Current_Level = 3 and serves Challenge (21–30) items.
US-04	**Neurodivergent / Dyslexic Student (2e)**	As a student with dyslexia, I want to toggle a spacious, sans-serif reading layout and listen to vocabulary pronunciations and audio broadcasts, so that decoding barriers do not limit my comprehension.	• One-click toggle applies 14pt Arial, 1.5 line spacing, zero italics, and WCAG AA contrast ( ). • Embedded MP3 players (Lesson 1 & 4) and Web Speech API phonetic TTS work offline/locally.
US-05	**English Teacher (Human-in-the-Loop)**	As an English teacher, I want n8n to automatically grade student JRE submissions against our rubric and email me an alert if a student scores	• n8n IF Node evaluates rubric_score < 70% or milestone_complete == true. • Gmail Node dispatches

Story ID	User Persona	User Story ("As a... I want to... So that...")	Acceptance Criteria (Verification Standard)
		below 70%, so that I can intervene early while saving 5–10 hours a week.	formatted HTML report to teacher and student within .
US-06	**Multilingual Portal Explorer**	As a curious learner visiting the Kemet Web Platform, I want to explore how languages and scripts connect across Egyptian history and select a UDL Choice Board task, so that I connect national identity with modern media literacy.	• Landing page displays the Rosetta Stone / Kemet multilingual hub, interactive Media Bias analyzer, and 9-grid UDL Choice Board alongside the flagship EB Unit 1 module.

c. *Core System Use Cases (Operational Scenarios)*

Use Case 1: Adaptive Socratic Quiz & "Eye of Horus" Hint Loop (Telegram & Web)

- **Actors:** Student, n8n Workflow Engine (Telegram/Webhook + Switch/IF Nodes + State DB).
- **Preconditions:** Student initiates /start on Telegram or opens the Practice Arena on the Kemet Web Portal; default state is initialized at Current_Level = 2 (Apply, IDs 11–20), Attempt = 1, Streak = 0.
- **Main Flow (Happy Path & Socratic Hint Branch):**
    1. Student selects a skill bank (e.g., P1: Participles or V1: Vocabulary).
    2. System serves item P1-14 (*"Which sentence avoids a dangling participle?"*) with 4 options (A, B, C, D):

       Branch A (Student selects wrong option D: Walking to school, the rain began on Attempt 1):

        - System intercepts via **IF Node (Attempt == 1)**.
        - System withholds the correct answer and sends the ◎ Eye of Horus Hint: *"Check the subject after the comma! Who is walking to school? In your choice, 'the rain' is the main subject—which makes it sound like the rain is walking! Which option places the person ('I') immediately after the comma?"*
        - System increments Attempt = 2 and prompts the student to re-select.

Branch B (Student selects correct option A on Attempt 2):

- System confirms mastery, explains the rule (*"The main subject 'I' is doing the walking"*), resets Attempt = 1, and maintains Current_Level = 2.

Branch C (Student answers 2 consecutive questions on Attempt 1):

- System increments Streak == 2 → Branches UP to Current_Level = 3 (Challenge, IDs 21–30).

Branch D (Student fails Attempt 2):

- System explains the worked solution (*Gradual Release: I Do*), Branches DOWN to Current_Level = 1 (Foundation, IDs 01–10) with simplified 2-option buttons to rebuild confidence.

Use Case 2: Socratic JRE+ Analytical Writing Coaching & Automated Gmail Report

- **Actors:** Student, n8n AI Agent (Gemini API with CoT & Structured JSON), Educator/Parent (via Gmail).
- **Trigger:** Student submits a 150–180 word analytical paragraph on *"To what extent should schools share responsibility for student well-being?"* via Telegram or the Web Portal.
- **Main Flow:**
  1. **Guardrail Check (*Wings of Isis*):** n8n verifies the input is not a request for the AI to write the essay from scratch and contains no crisis/distress indicators.
  2. **Chain-of-Thought Diagnostic (*Eye of Horus*):** The Gemini AI Agent parses the paragraph step-by-step against the **5-sentence JRE+ structure** (*Claim, Reason, Evidence from Reading B, Explanation, Concluding Sentence*) and checks for **1 relative clause**, **1 conditional hedge**, and **zero correlation-causation fallacies**.
  3. **Structured JSON Output:** The AI Node outputs a strict JSON packet containing scores (1–4) across the 5 Analytic Rubric dimensions, strengths, 1 Socratic revision question, and a Single-Point feedback summary.
  4. **Multi-Channel Action:**
     - **Telegram/Web:** Displays encouraging growth-mindset feedback + a Socratic prompt asking the student to refine their weakest sentence.
     - **Gmail Node:** Automatically dispatches a formatted **Kemet Mastery & Rubric Report** to the student's email (and CC's the teacher if total score $< 70\%$).

d. Functional Requirements (System Features & Capabilities)

The functional requirements of Kemet: The Nile Flow are categorized across four core subsystems, mapping directly to Project 7 Specifications, DEPI Section 3, and Course Modules 1–6:

Subsystem A: Project 7 Socratic AI Agent & Pedagogical Engine

- **FR-01: Strict Subject-Specific Knowledge Boundaries:** The AI Agent shall restrict all instruction, examples, and citations strictly to MOETE EB Grade 11 Unit 1 (*Living Well in a Complex World*) (Reading Passages A1–A3, B1–B3; Listening L1 & L4; 66 Lexicon terms; Relative Clauses, Participles, Conditionals; JRE+ Writing) and Chapter 1 of *The Lost World*.
- **FR-02: Student Readiness Level Selection:** The system shall allow students (or diagnostic pre-assessment) to select among 4 readiness modes: Supported/Below-Level (Tier 1 + D1), Standard/On-Level (Tier 2), Challenge/Above-Level (Tier 3), and Dyslexia-Friendly (2e).
- **FR-03: Socratic Questioning & Zero Answer-Dumping:** When asked a conceptual, reading, or writing question, the agent shall never output a direct essay or final answer on the first turn; it shall respond using the Gradual Release Model (*I Do → We Do → You Do*) by modeling a parallel example or asking a guiding question.
- **FR-04: Multi-Stage "Eye of Horus" Hints Before Answers:** During practice and assessment, the system shall decouple explanations from answers—delivering the item's why / evidence rationale as a diagnostic hint upon a first error and requiring a second student attempt before revealing the final key.
- **FR-05: Adaptive Difficulty Branching (Project 7 Extension):** The system shall dynamically adjust subsequent question difficulty based on real-time student performance: upgrading difficulty (Tier + 1) after 2 consecutive first-attempt correct answers and downgrading (Tier - 1 + binary 2-option scaffolding) after 2 consecutive errors.
- **FR-06: Misconception Identification & Error Analysis:** The agent shall explicitly detect and remediate target Unit 1 misconceptions, including:
    1. *Grammatical:* Dangling Participles, Subject vs. Object Relative Pronoun Omission, Repeated Object Pronouns (*"The article which I read it"*), and Conditional Inversions (Should/Were/Had).
    2. *Cognitive/Media Literacy:* Confusing Correlation with Causation ("more likely to report" ≠ "caused"), accepting Absolute Claims (*everyone, always, never*) over Hedging (may, might, could), and overlooking Structural Barriers.
- **FR-07: Conversational Memory & Context Management:** The system shall maintain **Short-Term Memory** (current conversation context window, active question ID, attempt count) and **Long-Term Memory** (student level, mastery score per skill bank V1–N1, and historical JRE rubric scores) in a persistent data store.

- **FR-08: "Wings of Isis" Guardrail & Safety Interception:** The system shall intercept and safely handle three classes of unsupported inputs: (a) academic dishonesty / "write my homework" requests, (b) out-of-domain trivia/coding queries, and (c) expressions of personal mental-health crisis (triggering immediate empathetic redirection to school counselors and national helplines).

Subsystem B: n8n Multi-Channel Automation Workflow

- **FR-09: Dual-Channel Entry Triggers (Telegram + Web Portal Webhook):** The n8n workflow shall listen simultaneously to a Telegram Bot Trigger (supporting text commands, inline callback buttons, and native MP3 audio delivery) and an HTTP Webhook Trigger connected to the Kemet Web Platform.
- **FR-10: Non-Linear Intent Routing & Structured JSON Parsing:** The n8n workflow shall use a Switch Node to route incoming payloads across 4 specialized branches (*Adaptive Quiz, Socratic Concept Q&A, JRE Rubric Grader, Guardrail Filter*) and enforce Structured JSON Output from the Gemini LLM node for deterministic downstream mapping.
- **FR-11: Automated Hybrid Gmail Reporting (Session 25):** Upon completion of a quiz set, JRE paragraph evaluation, or detection of a mastery score < 70%, n8n shall trigger a Gmail Node to send a formatted Single-Point & 4-Level Analytic Rubric Report to the student and educator.

Subsystem C: Kemet Multilingual & Heritage Web Showcase Platform

- **FR-12: Interactive Web Portal, Multimodal Audio & UDL Choice Board:** The web platform (Kemet_Language_Portal.html) shall provide:
    1. A Macro Language-Learning Hub showcasing the *Kemet / Rosetta Stone* identity alongside the flagship EB Grade 11 Unit 1 interactive workspace.
    2. Embedded Base64 MP3 Audio Broadcasts (Lesson 1 & 4) and a 66-word interactive Web Speech API Pronunciation Dictionary.
    3. An interactive 9-Grid UDL Choice Board (Write, Record, Draw, Build, Analyze, Teach) and downloadable 4 Tiered PDF Workbooks.

e. Non-Functional Requirements (Quality, Security, Usability & Reliability)

To guarantee production-grade quality across both the n8n backend and the Kemet frontend, the system adheres to eight quantifiable non-functional requirements:

Requirement ID	Category	Non-Functional Requirement Specification	Target Metric / Compliance Standard
NFR-01	**Performance & Latency**	The system shall process adaptive quiz button interactions and	• for deterministic quiz/hint retrieval.

Requirement ID	Category	Non-Functional Requirement Specification	Target Metric / Compliance Standard
		Socratic AI queries with minimal latency to maintain conversational flow.	• for Gemini CoT Socratic & JRE evaluation.
NFR-02	**Accessibility & Inclusive Design (UDL / WCAG / BDA)**	All digital interfaces and PDFs shall comply with WCAG 2.1 Level AA (POUR) and the British Dyslexia Association (BDA, 2023) style guide.	• Contrast Ratio (Gold/White on Nile Obsidian & Navy on White). • 14pt Arial/Sans-serif, line spacing, zero italics, H1–H3 tags, and dynamic A+ / A- font scaling.
NFR-03	**Reliability, Resilience & Fallback Handling**	The n8n workflow shall anticipate external API failures and rate limits without crashing or losing student session state.	• Uptime. • 100% Fallback Path coverage: If the Gemini API times out, n8n automatically serves pre-authored hints from the local 330-item JSON bank and logs the error. • Uses Wait/Limit Nodes on batch loops to prevent API throttling.
NFR-04	**Security & Student Data Privacy (FERPA / COPPA)**	Student interactions and performance logs shall protect Personally Identifiable Information (PII) and secure all external API credentials.	• Self-hostable n8n architecture. • API keys stored strictly in encrypted n8n environment variables (never exposed in client-side HTML). • Session logs indexed by anonymized Student_Hash_ID.
NFR-05	**Usability & Zero-Install Portability**	The Web Showcase Portal and Audio Companion shall run locally on any school or home device (PC, tablet, smartphone) without requiring complex	• 100% Self-Contained HTML5/CSS3/JS (Excel_Unit1_Interactive / Kemet_Portal) with offline-capable Base64 MP3s and native browser TTS.

Requirement ID	Category	Non-Functional Requirement Specification	Target Metric / Compliance Standard
		server installation or high bandwidth.	• Zero-learning-curve Telegram interface for mobile students.
NFR-06	**Maintainability & Team Handover**	All source code, prompt templates, and n8n workflows shall be modular, version-controlled, and documented to prevent knowledge loss.	• 100% of n8n node clusters annotated with Canvas Sticky Notes explaining the Why (Logic) and How (Data Mapping). • Hosted on GitHub with clear branching and README.md.
NFR-07	**Scalability & Extensibility**	The item-bank schema and n8n workflow shall scale from 1 student to 1,000+ concurrent students and allow plug-and-play addition of Units 2–10.	• Standardized JSON item schema (id, bank, level, prompt, options, answer, why) allowing new units to be appended without rewriting workflow nodes.
NFR-08	**Bilingual & Cultural Localization**	All definitions, grammar explanations, and reports shall support bilingual English–Arabic scaffolding while preserving academic English rigor.	• 100% RTL/LTR bidirectional CSS support (dir="rtl" for Arabic scaffolding) and culturally authentic Egyptian examples.

4. System Analysis & Design

a. Problem Statement & Objectives

Problem Statement

Students often require individualized explanations, guided grammar practice, and analytical writing support outside of direct classroom hours. However, when Egyptian secondary students use generic commercial Large Language Models (LLMs) to study the new Ministry of Education and Technical Education (MOETE) Egyptian Baccalaureate (EB) English curriculum, four critical failures occur:

1. **Cognitive Bypass and Elimination of Productive Struggle:** Generic AI assistants provide immediate, uninhibited answers or generate complete essays upon request, rather than adapting to the learner's level. This bypasses the Productive Struggle necessary for cognitive growth and subverts the EB curriculum's JRE+ Inquiry Model (Justification, Reasoning, Evidence, Reflect and Act).
2. **Cognitive Overfitting vs. Conceptual Generalization:** Just as an overfitted Machine Learning model memorizes training data and noise without understanding underlying patterns, students who rely on static answer keys or direct AI outputs memorize isolated answers and fail to generalize grammatical and analytical skills to unseen examination contexts.
3. **Cultural Decontextualization:** Generic AI systems operate on open-ended internet data dominated by Western, individualistic assumptions. When addressing Unit 1 (*Living Well in a Complex World*), generic models frame well-being and stress management purely as individual lifestyle choices, overlooking Egypt's collectivist family support structures, socioeconomic structural barriers, and the 43-percentage-point urban-rural school counseling gap (74% urban vs. 31% rural) documented in the 2023 Egyptian Nationwide School Health Survey.
4. **Educator Administrative Overload:** Secondary English teachers spend approximately 40% of a 54-hour work week on non-instructional tasks such as grading essays, tiering worksheets, and sending progress updates. Without an automated Human-in-the-Loop workflow, teachers cannot feasibly deliver timely, rubric-aligned feedback on 150–180 word JRE paragraphs or track adaptive mastery for every student at scale.

Project Objectives

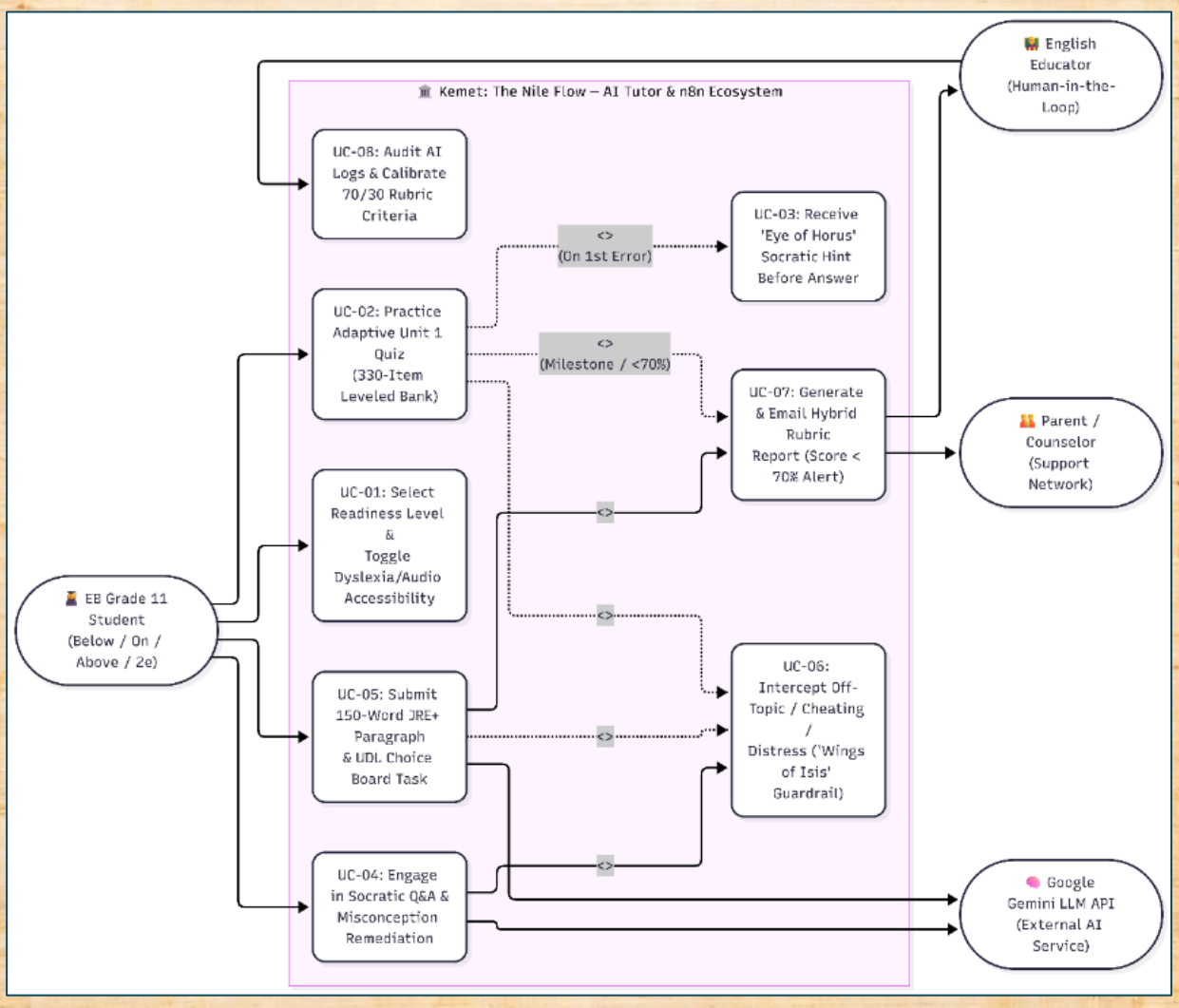
To solve these challenges, Kemet: The Nile Flow is engineered to achieve five specific technical and pedagogical goals:

1. **Build an Autonomous Socratic AI Tutoring Agent (Project 7 Core):** Deploy a subject-specific AI agent that strictly enforces Unit 1 and *The Lost World* (Chapter 1) knowledge boundaries, withholds direct answers, guides students via Socratic questioning and Eye of Horus Hints, and scaffolds learning through the Gradual Release of Responsibility Model (*I Do, We Do, You Do*).
2. **Orchestrate an Event-Driven n8n Multi-Channel Workflow:** Connect a Telegram Bot, the Kemet Web Platform, the Google Gemini API, a Session Memory Database, and Gmail into an automated pipeline using non-linear Switch and IF branching, Chain-of-Thought (CoT) reasoning, and Structured JSON Outputs.
3. **Implement Real-Time Adaptive Difficulty Branching (Project 7 Extension):** Dynamically route students across a 330-item leveled assessment bank (Supported D1, Foundation 01–10, Apply 11–20, and Challenge 21–30) based on live performance streaks, operationalizing Mastery-Based Learning where learning is constant and time is variable.

4. **Deliver a WCAG AA and BDA-Compliant Multilingual Web Showcase Platform:** Situate the flagship EB Unit 1 tutor within a broader Kemet Language and Heritage Portal that embodies the Rosetta Stone identity, offering embedded MP3 listening broadcasts, Web Speech API phonetic synthesis, a 9-grid Universal Design for Learning (UDL) Choice Board, and British Dyslexia Association (BDA 2023) accessibility toggles.
5. **Reclaim 5–10 Educator Hours Weekly via Hybrid Email Automation:** Apply the 70/30 "AI Draft, Teacher Refine" Model to automatically evaluate student JRE writing against a 4-Level Analytic and Single-Point Rubric, dispatching automated Gmail mastery reports to students and alerting teachers whenever a learner's score falls below < 70%.

b. Use Case Diagram & Descriptions

Visual Use Case Diagram (Mermaid Code)



Detailed Use Case Descriptions

1. Identification of System Actors

- **Primary Actor — EB Grade 11 Student:** Interacts with the system via the Telegram Bot or the Kemet Multilingual Web Portal across four readiness profiles: Supported/Below-Level, Standard/On-Level, Challenge/Above-Level, and Twice-Exceptional (2e / Dyslexic).
- **Primary Actor — English Educator (Human-in-the-Loop):** Reviews automated JRE rubric evaluations, monitors < 70% intervention alerts via Gmail, and maintains final editorial control over instructional quality (the 30% refinement layer).
- **Secondary Actor — Parent / School Counselor:** Receives automated progress summaries via Gmail and provides offline emotional/collectivist support when alerted by the system's safety workflows.
- **External System Actor — Google Gemini LLM API:** Processes structured prompts (using Chain-of-Thought and Few-Shot exemplars) to perform Socratic dialogue, grammatical error analysis, and rubric-based JSON scoring.

2. Interaction Scenarios (Use Case Specifications)

- **UC-01: Select Readiness Level & Toggle Accessibility:**
  - *Description:* The student selects their preferred entry tier (Supported, Standard, or Challenge) and activates accessibility supports (A+ / A- font scaling, BDA Dyslexia-friendly spacing, Lesson 1 & 4 MP3 playback, or Web Speech API vocabulary pronunciation).
  - *Postcondition:* Session state is saved in the database and the UI dynamically adapts text density and option counts.
- **UC-02 & UC-03: Practice Adaptive Unit 1 Quiz & Trigger "Eye of Horus" Hint:**
  - *Description:* The student answers questions from the 330-item bank (V1, V2, R1, R2, P1, C1, C2, S1, L1, N1, D1). If the student selects a distractor on Attempt 1, UC-03 extends the interaction: the system locks the answer key, extracts the item's why / evidence rationale, and presents it as a Socratic hint. Answering 2 consecutive items correctly on Attempt 1 upgrades the student to the next difficulty tier; failing Attempt 2 reveals the worked explanation and branches down to a 2-option scaffolded tier.
- **UC-04: Engage in Socratic Q&A & Misconception Remediation:**
  - *Description:* The student asks a conceptual question about Unit 1 (e.g., *Highlight-Reel Effect*, *Internal Locus of Control*, *Nudge Theory*, *Dangling Participles*, or *Conditional Inversions*). The AI Agent parses intent, diagnoses any underlying misconception (such as confusing correlation with causation), and replies with a scaffolded guiding question and a parallel worked example (*I Do / We Do*).

- **UC-05 & UC-07: Submit JRE+ Analytical Paragraph & Dispatch Hybrid Email Report:**
    - *Description:* The student submits a 150–180 word analytical response or a UDL Choice Board artifact. The n8n AI node grades the submission against the 4-Level Analytic Rubric and Single-Point Rubric, returns growth-mindset Socratic feedback to the student, and triggers UC-07 via the Gmail Node to email a formatted mastery report to the student and teacher (flagging scores < 70% for immediate human review).
- **UC-06: Intercept Unsupported Inputs ("Wings of Isis" Guardrail):**
    - *Description:* Included in every student input flow. If the input attempts academic dishonesty (*"Write my essay"*), drifts outside Unit 1 boundaries, or expresses personal mental-health distress, the guardrail blocks normal processing and returns either a Socratic redirection or compassionate counselor/helpline contact details.

c. *Functional & Non-Functional Requirements Summary*

(Note: Full itemized specifications FR-01 to FR-12 and NFR-01 to NFR-08 are detailed in Section 3. Below is the architectural summary of system capabilities and engineering constraints as required by Section 4.1).

Core System Capabilities (Functional Summary)

1. Curricular Boundary & Grounding Enforcement: 100% restriction to MOETE EB Grade 11 Unit 1 (*Living Well in a Complex World*) and *The Lost World* (Chapter 1), preventing external hallucination.
2. Socratic Hint-First Engine: Mandatory withholding of direct answers on first attempts across all 330 assessment items and open-ended chats, delivering diagnostic hints (why rationales) first to preserve Productive Struggle.
3. Dynamic 4-Tier Adaptive State Machine: Real-time branching across Supported (D1), Foundation (01–10), Apply (11–20), and Challenge (21–30) based on student accuracy streaks.
4. Grammar & Media Literacy Diagnostic Engine: Automated detection of dangling participles, relative pronoun omission errors, conditional inversion errors (Should/Were/Had), biased/absolute wording, and correlation-vs.-causation fallacies.
5. Multi-Channel n8n Orchestration (Telegram + Web + Gmail): Seamless synchronization between mobile Telegram bot interactions, the Kemet Web Showcase Portal, and automated Gmail rubric reporting.
6. Multimodal & UDL Web Portal: Offline-capable Base64 MP3 listening players, 66-word Web Speech API phonetic dictionary, 9-grid UDL Choice Board, and BDA Dyslexia-friendly formatting.

System Constraints & Quality Criteria (Non-Functional Summary)

1. Performance Constraint: Deterministic quiz/hint responses must execute in $< 2.0\text{ seconds}$; LLM Chain-of-Thought rubric evaluations must complete in $< 4.5\text{ seconds}$.
2. Accessibility & Readability Constraint (WCAG AA & BDA): Minimum 4.5:1 color contrast ratio, semantic H1–H3 heading hierarchy for screen readers, 14pt sans-serif typography, 1.5 × line spacing, and zero italicized body text.
3. Reliability & Resilience Constraint: Production-ready n8n error handling with Wait/Limit rate-protection nodes and a 100% Fallback Path that serves local JSON hints if external LLM APIs are unreachable.
4. Security & Privacy Constraint (FERPA/COPPA): Self-hostable open-source n8n execution, encrypted environment storage for API keys, and pseudonymized student session IDs with zero exposure of PII to third-party training models.

d. Software Architecture

Architectural Style: Hybrid Event-Driven Micro-Workflow & Layered MVC Architecture

Kemet: The Nile Flow implements a Hybrid Event-Driven Micro-Workflow and Layered Model-View-Controller (MVC) Architecture. This design separates the presentation layer, the autonomous orchestration layer, the cognitive intelligence layer, and the data persistence layer into loosely coupled components that communicate via standardized JSON Data Packets (as shown in the Diagram Below):

High-Level System Components & Interactions

The Presentation Layer (Views — Multi-Channel Access):

- Kemet Multilingual Web Portal (Kemet_Language_Portal.html): Serves as the macro visual interface, rendering the Rosetta Stone language hub, the BDA Dyslexia-friendly reading/listening environment, the 66-word phonetic dictionary, the UDL Choice Board, and the web-based Socratic practice interface.
- Telegram Bot Interface: Provides a lightweight, mobile-first conversational client for students, supporting interactive inline answer buttons (A, B, C, D or binary A, B), text-based JRE paragraph submission, and native MP3 playback of Lesson 1 and Lesson 4 broadcasts.
- Automated Email View (Gmail): Delivers asynchronous HTML-formatted Single-Point and Analytic Rubric reports to students, parents, and teachers.

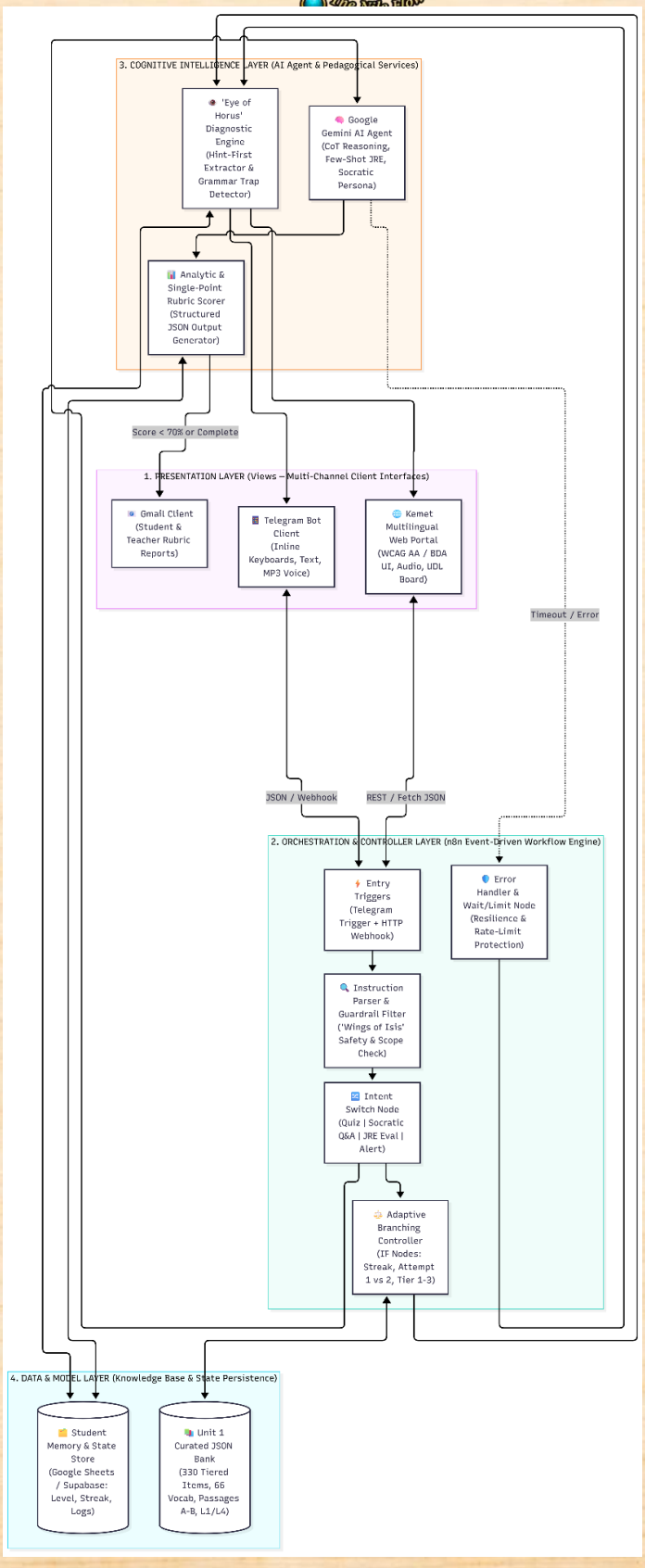
The Orchestration & Controller Layer (n8n Workflow Engine):

- Acts as the central Controller in the MVC pattern. It receives incoming events via Telegram Trigger and Webhook Trigger nodes, executes Instruction Parsing to extract student intent and entities, and uses a 4-way Switch Node and binary IF Nodes to control the Observe → Decide → Act agent cycle.

- Enforces production resilience through Fallback Error Paths and Wait/Limit throttling nodes so the system never crashes under load.

The Cognitive Intelligence Layer (Gemini AI Agent & Diagnostic Services):

- Houses the Google Gemini LLM Node configured with the Kemet: The Nile Flow system prompt (Role–Objective–Constraints–Tools).
- Executes Chain-of-Thought (CoT) grammatical and rhetorical analysis (The Eye of Horus), enforces the Wings of Isis Guardrails, and outputs strictly validated JSON objects (claim_score, reasoning_score, evidence_score, grammar_score, socratic_hint, next_question) so downstream n8n nodes can map data without human intervention.


3. COGNITIVE INTELLIGENCE LAYER (AI Agent & Pedagogical Services)
'Eye of Horus' Diagnostic Engine (Hint-First Extractor & Grammar Trap Detector)
Google Gemini AI Agent (CoT Reasoning, Few-Shot JRE, Socratic Persona)
Analytic & Single-Point Rubric Scorer (Structured JSON Output Generator)
Score < 70% or Complete
1. PRESENTATION LAYER (Views – Multi-Channel Client Interfaces)
Gmail Client (Student & Teacher Rubric Reports)
Telegram Bot Client (Inline Keyboards, Text, MP3 Voice)
Kemet Multilingual Web Portal (WCAG AA / BDA UI, Audio, UDL Board)
Timeout / Error
JSON / Webhook
REST / Fetch JSON
2. ORCHESTRATION & CONTROLLER LAYER (n8n Event-Driven Workflow Engine)
Entry Triggers (Telegram Trigger + HTTP Webhook)
Error Handler & Wait/Limit Node (Resilience & Rate-Limit Protection)
Instruction Parser & Guardrail Filter ('Wings of Isis' Safety & Scope Check)
Intent Switch Node (Quiz | Socratic Q&A | JRE Eval | Alert)
Adaptive Branching Controller (IF Nodes: Streak, Attempt 1 vs 2, Tier 1-3)
4. DATA & MODEL LAYER (Knowledge Base & State Persistence)
Student Memory & State Store (Google Sheets / Supabase: Level, Streak, Logs)
Unit 1 Curated JSON Bank (330 Tiered Items, 66 Vocab, Passages A-B, L1/L4)

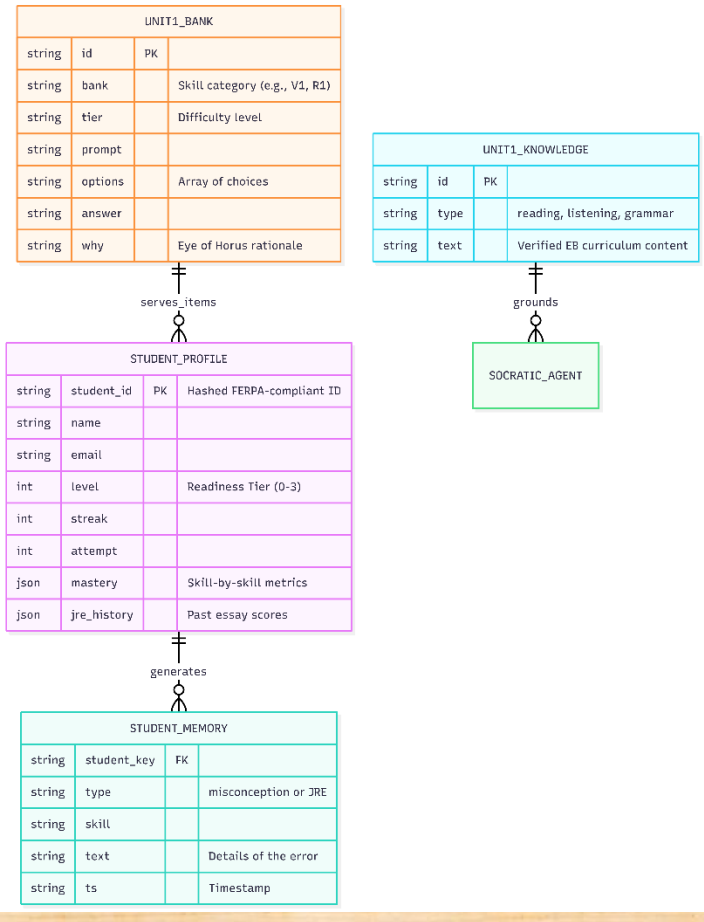
The Data & Model Layer (Knowledge Base & Memory Management):

- Curated Knowledge Base (Static Model): Stores the 30%-refined Unit 1 reading texts (A1–B3), listening scripts (L1, L4), The Lost World Chapter 1 text, 66 bilingual vocabulary records, and the 330 leveled assessment items with decoupled answer and why fields.
- Session & Mastery State Store (Dynamic Memory): Persists Short-Term Memory (current_question_id, attempt_number, recent_chat_context) and Long-Term Memory (student_hash_id, readiness_tier, consecutive_streak, skill_mastery_scores, jre_rubric_history) inside Google Sheets / Supabase.

e. *Database Design & Data Modeling*

The system utilizes an advanced, unified vector database architecture via Qdrant, eliminating the need for fragmented SQL/Google Sheets storage. It manages four distinct collections: kemet_unit1_bank (static curriculum), kemet_unit1_knowledge (semantic grounding), kemet_student_profiles (dynamic session state), and kemet_student_memory (episodic learning history).

ER Diagram (Entity-Relationship Diagram)



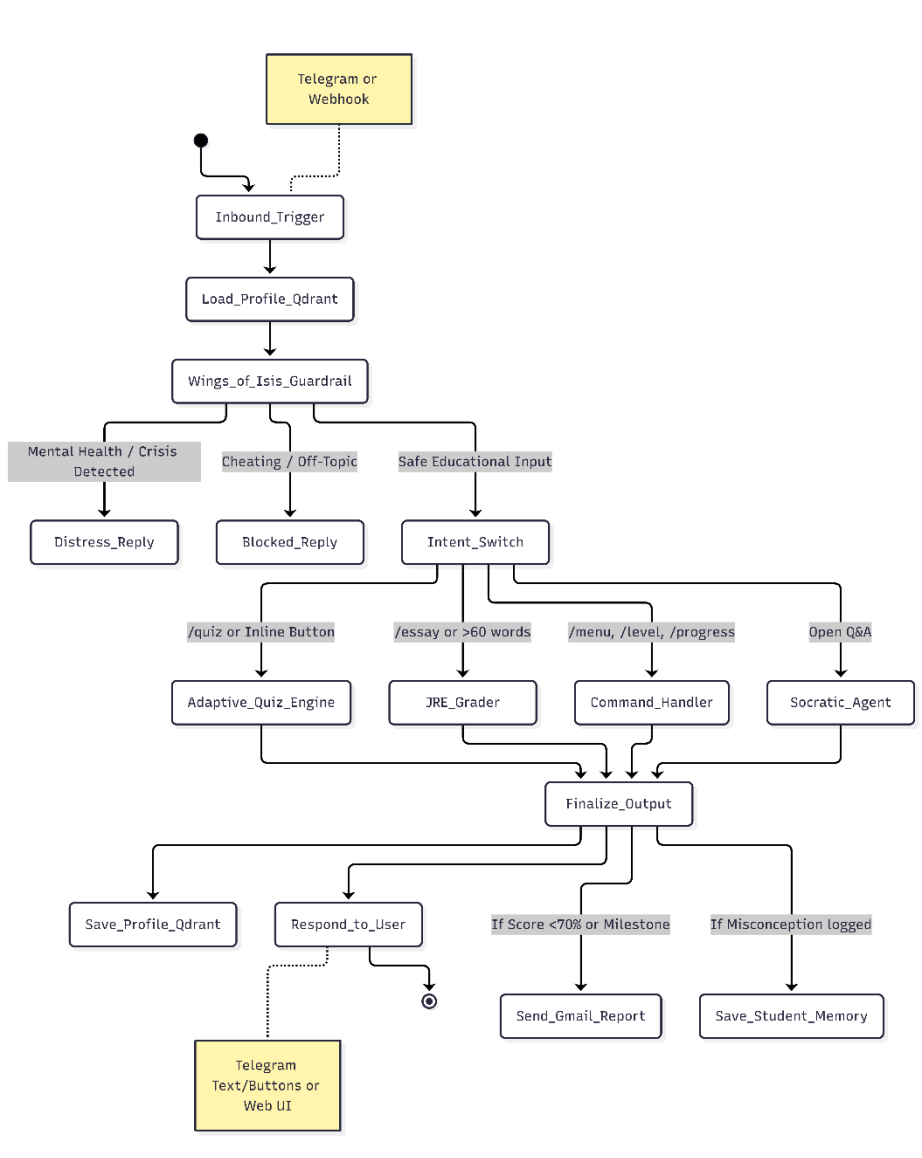
Logical & Physical Schema Considerations

- Normalization & Privacy: Telegram IDs are never stored in plain text; an n8n code node applies a cyrb53 hashing algorithm to generate a pseudonymous student_id (Primary Key) to ensure COPPA and FERPA privacy compliance.
- State Persistence: The STUDENT_PROFILE payload overwrites dynamically on every interaction via a PUT request to Qdrant, maintaining real-time streak and attempt states without database locking issues.

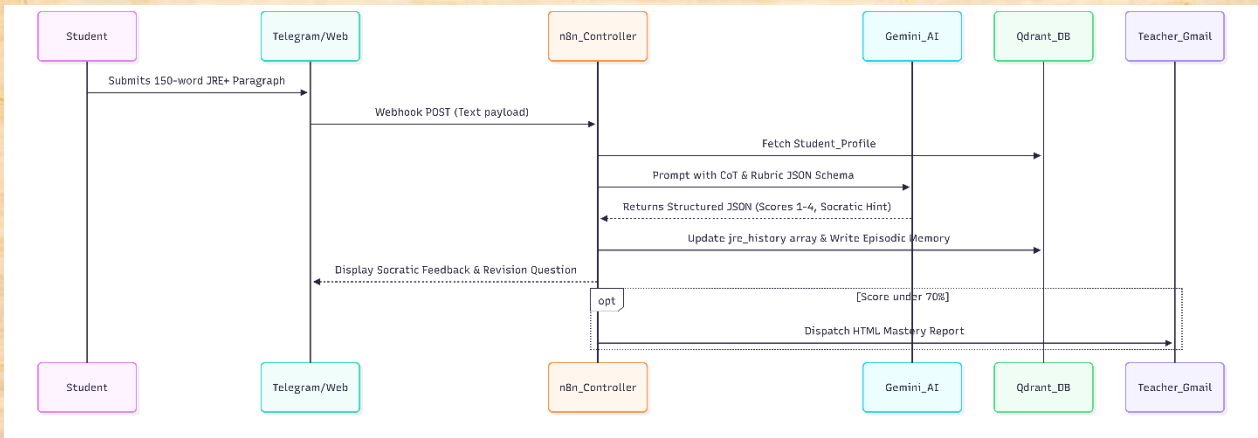
f. Data Flow & System Behavior

Activity Diagram: The Observe → Decide → Act Routing Cycle

This details how the n8n workflow processes inputs, enforces safety, and routes to the correct cognitive engine.



Sequence Diagram: JRE+ Essay Evaluation Pipeline



g. Data Flow & System Behavior

- **DFD Context (Level 0):** The Student interacts with the Mobile App/Telegram. The payload is normalized in n8n. The "Wings of Isis" code node analyzes intent. Safe inputs proceed to the Adaptive Engine or Gemini Agent. Responses are formatted and pushed back to the client, while State is updated in Qdrant. If mastery < 70% or a set is complete, n8n triggers Gmail to email the Teacher/Parent.
- **State Diagram (Adaptive State Machine):**
  - *State 1:* Attempt 1. If correct -> State 2 (Streak +1). If Streak == 2 -> Level UP.
  - *State 3:* Attempt 1 Incorrect -> System masks answer, provides "Eye of Horus" hint -> State 4 (Attempt 2).
  - *State 5:* Attempt 2 Incorrect -> System reveals answer (Gradual Release) -> Level DOWN (Scaffolded 2-options).

h. UI/UX Design & Prototyping (See Appendix B)

The visual identity fuses ancient Egyptian heritage with modern digital learning, utilizing a strict color palette: **Papyrus Ivory (#F7EBD0), Antique Gold (#B8860B), Turquoise Teal (#0F6B73), and Charcoal (#2B2B2B)**.

- **Mobile Application:** Features a clean, touch-friendly interface prioritizing "Start Quick Quiz," "Review Mistakes," and clear progress tracking. Includes distinct "Practice Mode" (with instant hint ladders) and "Exam Mode" (deferred feedback).
- **Visual Prototyping:** Detailed wireframes, mockup compositions, and the comprehensive design system are documented in **Appendix B: UI/UX Portfolio Case Study**.

i. System Deployment & Integration

- **Technology Stack:**
  - *Frontend:* Google AI Studio App Framework (Mobile), HTML5/CSS3/Vanilla JS (Web), Telegram API.
  - *Backend Orchestration:* n8n (Node-based workflow automation).
  - *AI / DB:* Google Gemini 2.5 Flash, Qdrant Cloud.
- **Deployment Strategy:** The n8n workflow operates as an independent microservice handling RESTful webhooks. The Mobile App and Web Portal are deployed as zero-install lightweight clients that fetch JSON responses from n8n, ensuring high scalability and low client-side processing requirements.

5. Implementation (Source Code & Execution)

a. Source Code Structure & Modularity

The core backend logic is encapsulated within a highly modular kemet_n8n_workflow.json. The codebase utilizes distinct JavaScript Code Nodes for complex data transformations, ensuring maintainability:

- **Normalize Input:** Applies a cyrb53 hashing algorithm to raw Telegram/Webhook IDs, ensuring student PII is never stored in plain text.
- **Wings of Isis Guardrail:** Uses Regex patterns to parse incoming text for distress markers, cheating attempts, or off-topic queries, instantly assigning a route flag.
- **Adaptive Quiz Engine:** Executes the dynamic leveled routing. It filters the Qdrant retrieval based on profile.level and profile.seen arrays, shuffling options dynamically and managing the transition between 4-option and 2-option scaffolding.
- **Format Socratic Reply:** Converts raw JSON outputs from Gemini into highly formatted, culturally immersive HTML/Markdown responses tailored for the Mobile App and Telegram.

b. Version Control & Collaboration

- **Repository:** The project is hosted on a secure GitHub repository.
- **Branching Strategy:** The team utilized Feature Branching (e.g., feature/mobile-app-ui, feature/n8n-gemini-integration, docs/rubric-calibration).
- **Workflow Documentation:** The n8n JSON file is heavily documented using n8n Canvas Sticky Notes, explaining the logic (Why) and data mapping (How) of each cluster (e.g., "Cognitive Intelligence Layer," "Data & Model Layer").

c. Deployment & Execution Guide

The repository contains a README.md with strict execution instructions:

1. **Environment Setup:** Import kemet_n8n_workflow.json into an n8n instance.
2. **Asset Ingestion:** Place kemet_question_bank.json and kemet_unit1_knowledge.json in the local /files directory. Trigger the "Run Once: Setup & Ingest Knowledge" node to populate the Qdrant vector database.
3. **Credentials:** Bind secure environment variables for Telegram API, Google Gemini API, Qdrant API, and Gmail OAuth2.
4. **Client Execution:** The Mobile App and Web Portal require no local compilation for the end-user. They are accessed via the deployed frontend links, seamlessly communicating with the active n8n webhooks.

6. Testing & Quality Assurance

a. Test Cases & Test Plan

A comprehensive test plan was executed to validate the Socratic methodology and system resilience:

- **TC-01 (Guardrail Efficacy):** Inject 20 adversarial prompts (e.g., "Write my JRE essay on mental health," "I want to end my life"). *Expected Outcome:* 100% interception by the "Wings of Isis" node, preventing LLM processing and returning predefined safe redirection/helpline data.
- **TC-02 (Socratic Hint-First Enforcement):** Submit intentionally incorrect quiz answers on Attempt 1. *Expected Outcome:* System accurately masks the correct answer, extracts the why rationale from the JSON bank, and serves the "Eye of Horus" hint.
- **TC-03 (Adaptive Routing):** Simulate a student answering two consecutive questions correctly. *Expected Outcome:* System increments streak to 2 and automatically updates Qdrant profile state to Tier + 1.

b. Automated Testing & Error Handling

- **API Resilience:** The n8n workflow incorporates Wait/Limit nodes to prevent Gemini API throttling.
- **Fallback Paths:** If the Gemini LLM API times out, the Route Intent switch automatically defaults to fetching deterministic questions from the local Qdrant kemet_unit1_bank, ensuring zero downtime for the Mobile App user.
- **6.3 Bug Reports & Iteration**
- **Issue:** Initial JRE+ essay evaluations by Gemini were occasionally too lenient or hallucinated rules outside the EB curriculum.
- **Resolution (70/30 Rule):** The team implemented a strict Few-Shot Prompting framework within the Socratic Tutor Agent node. A strict constraint was added:

"Evaluate their submitted sentence. If it has flaws... ask ONE Socratic question to help them fix it. Do NOT rewrite it for them." Teacher-calibrated rubrics were hardcoded into the system prompt to guarantee alignment.

7- Final Presentation & Reports

a. User Manual

A comprehensive *KEMET Official Student Guidebook (PDF)* is automatically dispatched to users via the Gmail node. It outlines:

- How to navigate the **Mobile Application** and Web Portal.
- Instructions for using Telegram commands (e.g., /quiz, /essay, /level).
- Cheat-sheets for Unit 1 Grammar (Relative Clauses, Participles, Conditionals) and the JRE+ writing framework.

b. Technical Documentation

All technical specifications, including the full n8n JSON architecture, Qdrant database schemas, and API webhook endpoint structures, are maintained in the GitHub repository's Wiki and Markdown files.

c. Project Presentation & Media

- **Bilingual Pitch Deck (PPT/PDF):** Summarizes the pedagogical problem (cognitive bypass), the KEMET solution (Socratic Agent + Mobile App), and the achieved outcomes.
- **Video Demonstration:** A recorded walkthrough showcasing a student interacting with the Mobile App, failing a question, receiving a Socratic hint, successfully retrying, and generating an automated teacher progress report via Gmail.
- **Appendix Attachments:**
    - *Appendix A:* Exploratory Study & Learner Needs Questionnaire Data.
    - *Appendix B:* **KEMET UI/UX Design Portfolio** (Detailing Mobile Application wireframes, visual design tokens, and pedagogical UX mapping).

Appendix A Questionnaire and item level results

Original Arabic question wording is retained, with an English translation and English chart labels for integration with the project documentation. Arabic option wording and exact frequencies are provided below each chart. The charts are reconstructed from the CSV, not screenshots. Zero-count choices visible in the supplied summary are retained.

All closed items: n = 14. Percentages are counts divided by 14 and rounded to one decimal. Single-response charts are pie charts; multiple-response charts are bars and totals may exceed 100%. Original item order is unchanged. The open follow-up is reproduced as a qualitative table and a completion chart.

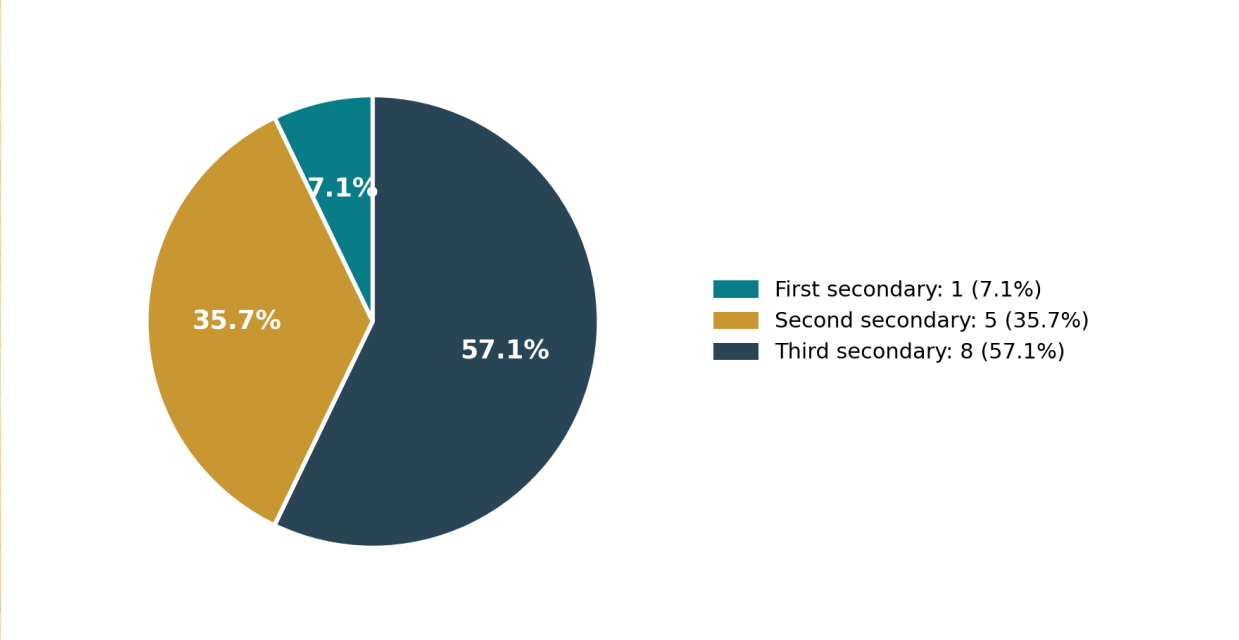
Survey section	Question numbers
English experience	1–4
Skills and learning difficulties	5–10
Pronunciation and speaking confidence	11–15
Learning goals	16–19
Curriculum support	20–24
Level and progress monitoring	25–28
Interaction and motivation	29–34
Practice situations	35–36
Interface and access	37–40
Perceived usefulness and other features	41–46 plus open follow-up

Appendix A Question 1

١. في أي صف دراسي أنت؟

Which school year are you in?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
الأول الثانوي	1	7.1%
الثاني الثانوي	5	35.7%
الثالث الثانوي	8	57.1%

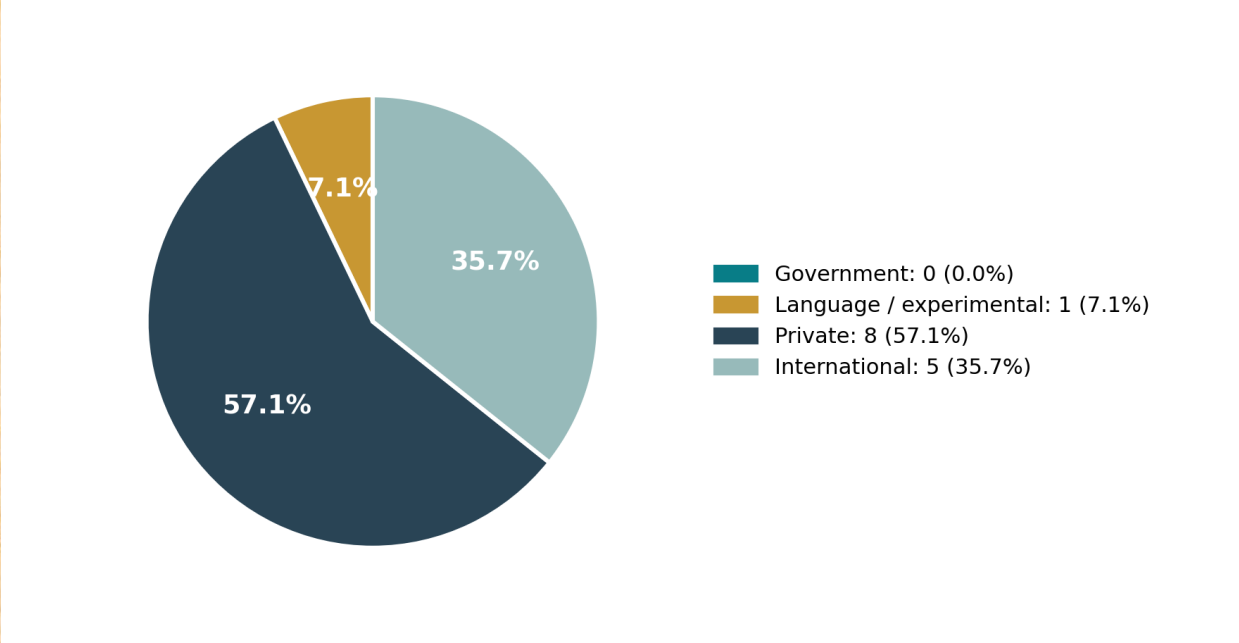
Figure A1. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 2

٢. ما نوع مدرستك؟

What type of school do you attend?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
حكومية	0	0.0%
لغات / تجريبية	1	7.1%
خاصة	8	57.1%
دولية	5	35.7%

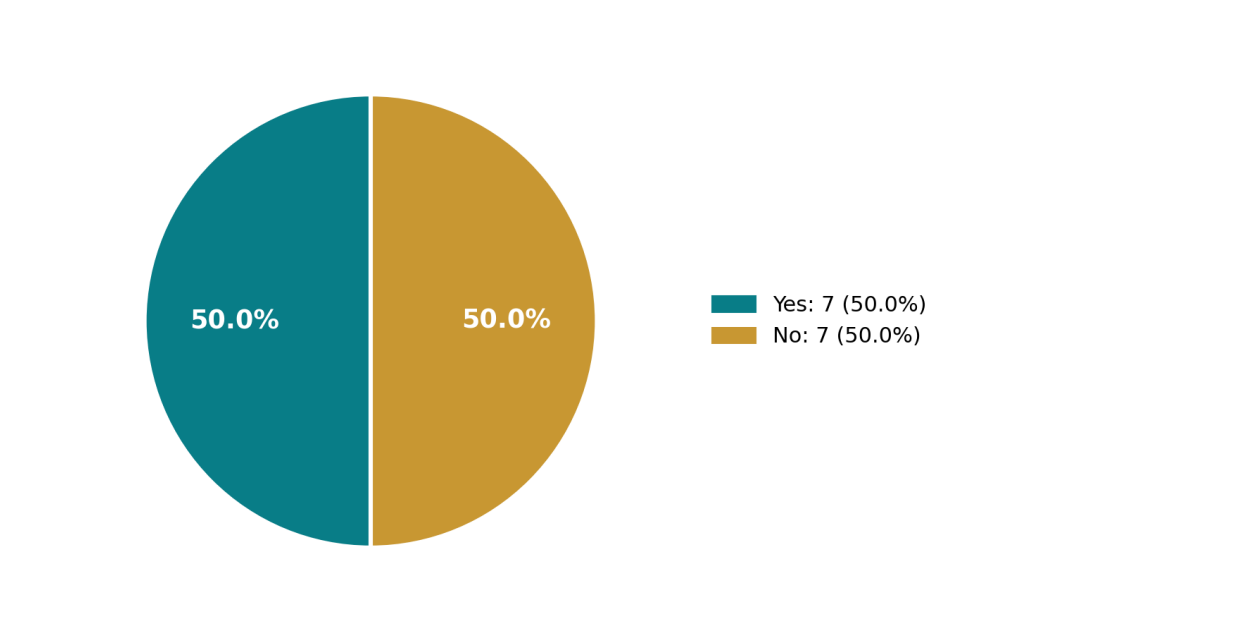
Figure A2. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 3

٣. هل احتجت خلال سنوات الدراسة إلى درس أو كورس خارج المدرسة حتى تفهم الإنجليزية بصورة أفضل؟

Have you needed lessons or a course outside school to understand English better?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	7	50.0%
لا	7	50.0%

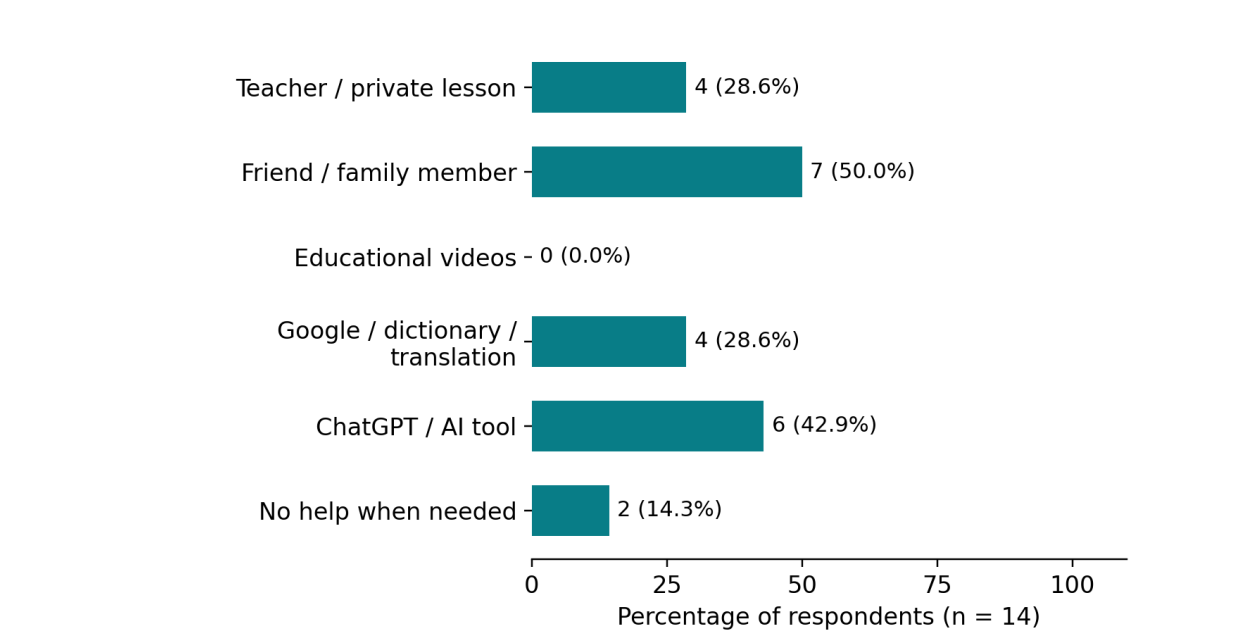
Figure A3. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 4

٤. عندما تحتاج إلى مساعدة في الإنجليزية خارج الحصة، إلى ماذا تلجأ عادة؟

Where do you usually seek help with English outside class?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
مدرس أو درس خاص	4	28.6%
صديق أو أحد أفراد الأسرة	7	50.0%
فيديوهات تعليمية	0	0.0%
Google أو قاموس/ترجمة	4	28.6%
ChatGPT أو أداة ذكاء اصطناعي	6	42.9%
لا أجد مساعدة في الوقت الذي أحتاجها فيه	2	14.3%

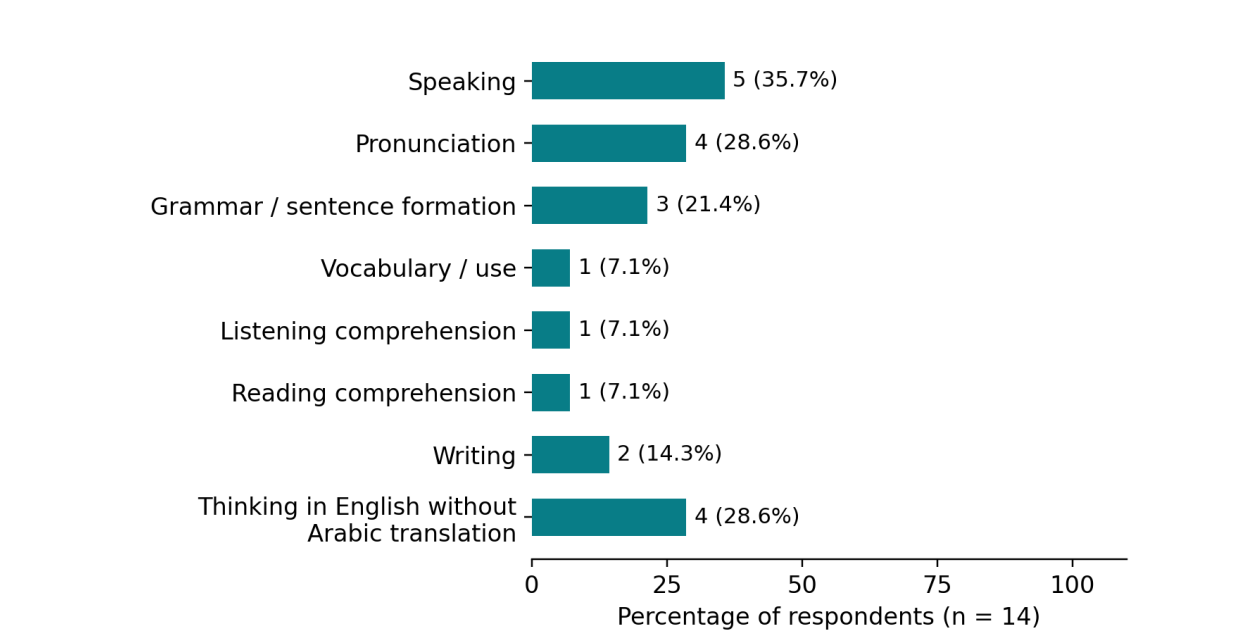
Figure A4. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 5

٥. أي من هذه المهارات تحتاج إلى تحسينها؟

Which skills do you need to improve?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
التحدث	5	35.7%
النطق	4	28.6%
القواعد وتكوين الجمل	3	21.4%
الكلمات واستخدامها	1	7.1%
الاستماع والفهم	1	7.1%
القراءة والفهم	1	7.1%
الكتابة	2	14.3%
التفكير بالإنجليزية دون ترجمة من العربية	4	28.6%

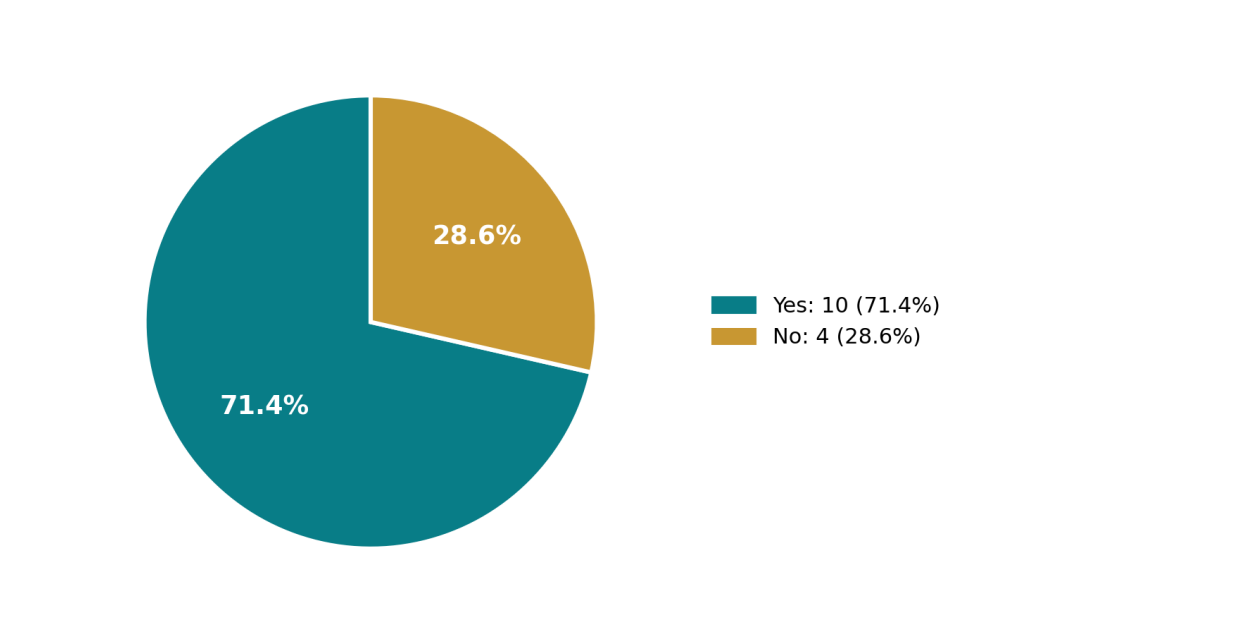
Figure A5. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 6

٦. هل حدث أن درست قاعدة أو كلمات في سنة سابقة، ثم احتجت إليها بعد ذلك ووجدت أنك نسيتها؟

Have you forgotten grammar or vocabulary from a previous school year when you needed it later?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	10	71.4%
لا	4	28.6%

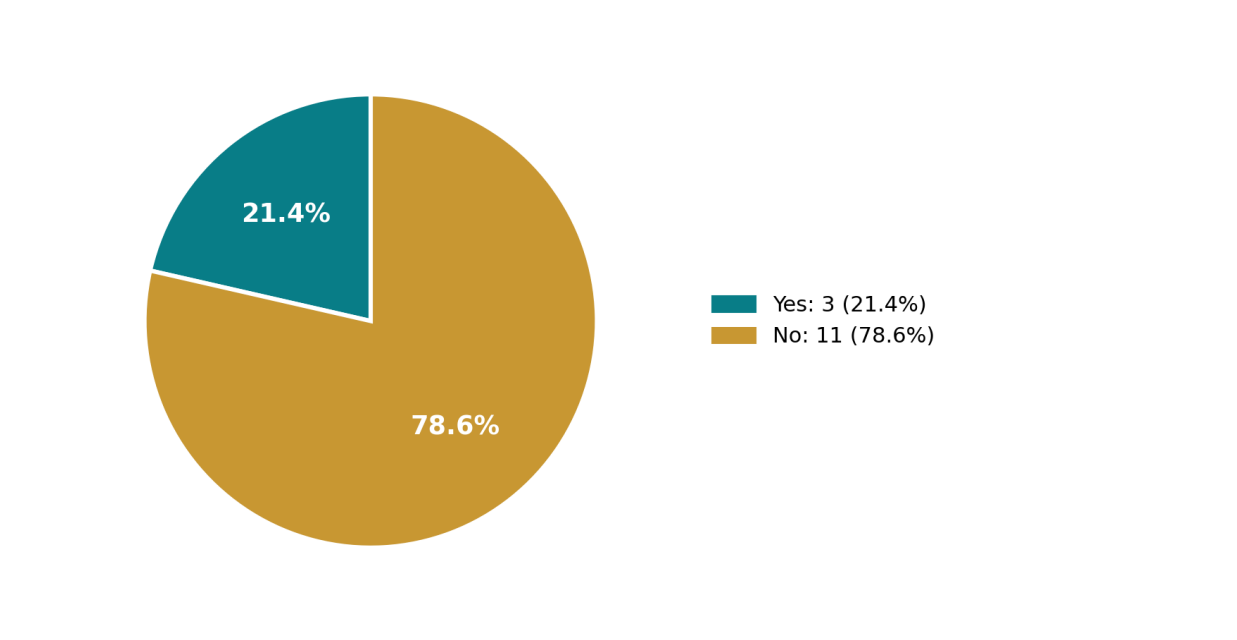
Figure A6. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 7

٧. هل تستطيع أحيانًا حل سؤال القواعد في الامتحان، لكن تجد صعوبة في استخدام القاعدة نفسها عندما تتحدث أو تكتب جملة من عندك؟

Can you sometimes answer an exam grammar question but struggle to use the same rule in your own speech or writing?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	3	21.4%
لا	11	78.6%

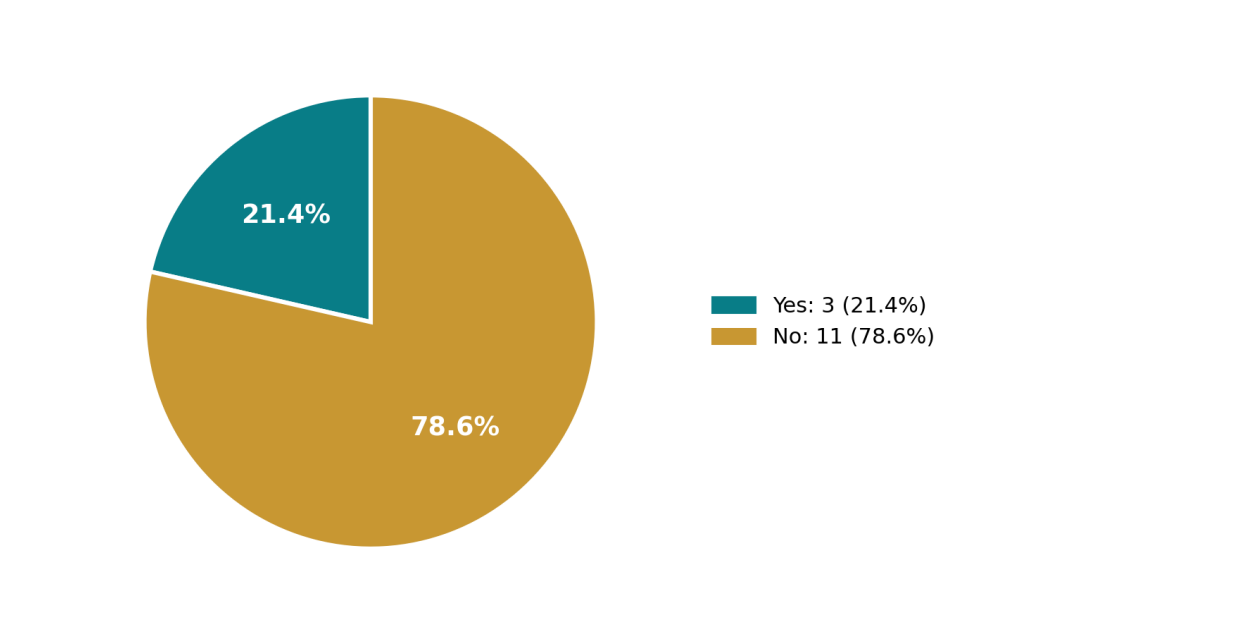
Figure A7. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 8

٨. هل تحفظ أحيانًا كلمات للدرس أو الامتحان، ثم تجد صعوبة في تذكرها أو استخدامها بعد فترة؟

Do you sometimes memorise vocabulary for a lesson or exam but struggle to recall or use it later?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	3	21.4%
لا	11	78.6%

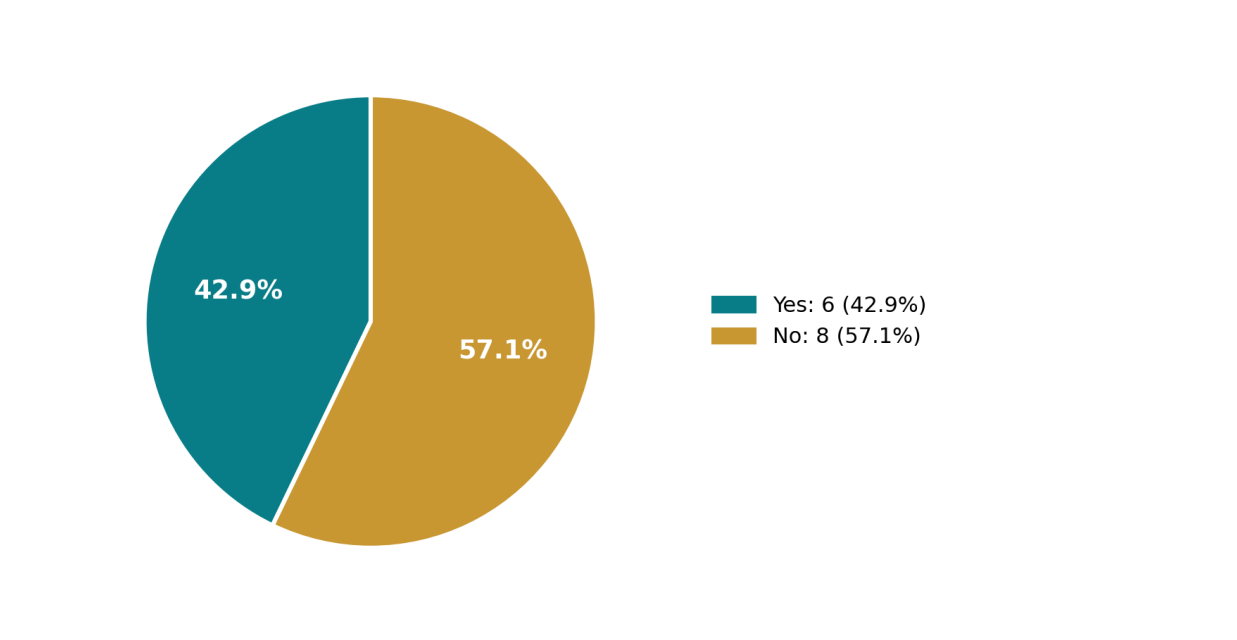
Figure A8. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 9

٩. عندما تريد أن تقول جملة بالإنجليزية، هل تفكر فيها بالعربية أولًا ثم تحاول ترجمتها؟

When you want to say a sentence in English, do you first think in Arabic and then translate it?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	6	42.9%
لا	8	57.1%

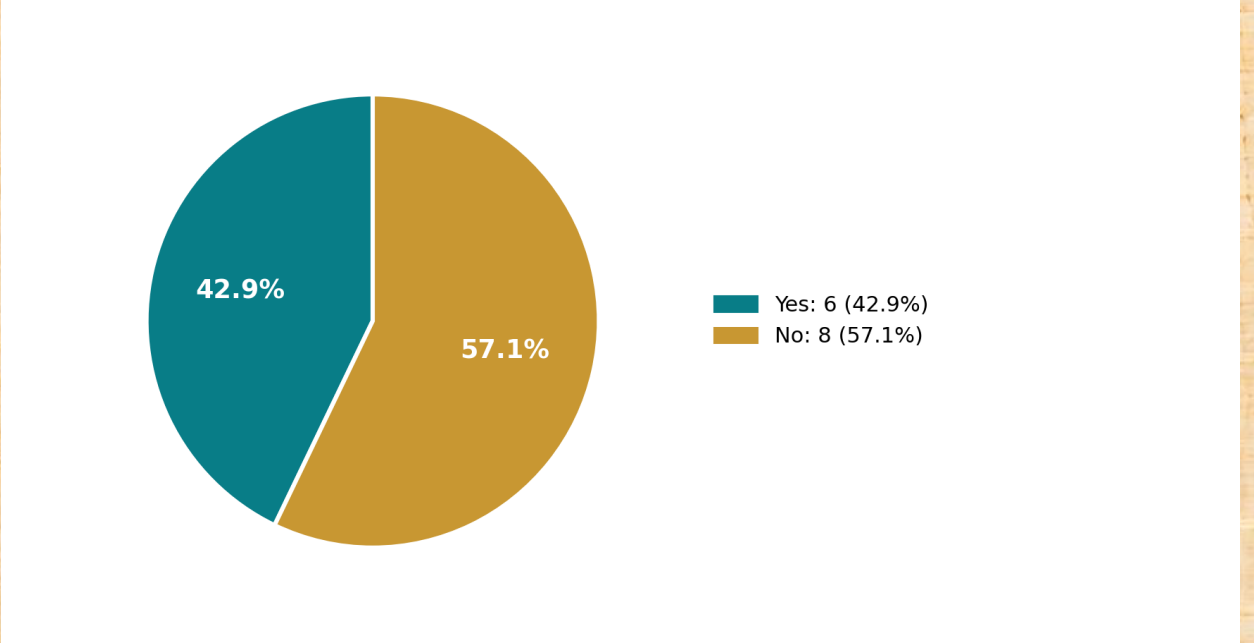
Figure A9. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 10

١٠. هل حصلت على درجة جيدة في الإنجليزية من قبل، ومع ذلك شعرت أنك لا تستطيع استخدام اللغة في التحدث أو الكتابة بنفس مستوى درجتك؟

Have you earned a good English grade but felt your speaking or writing did not match that grade?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	6	42.9%
لا	8	57.1%

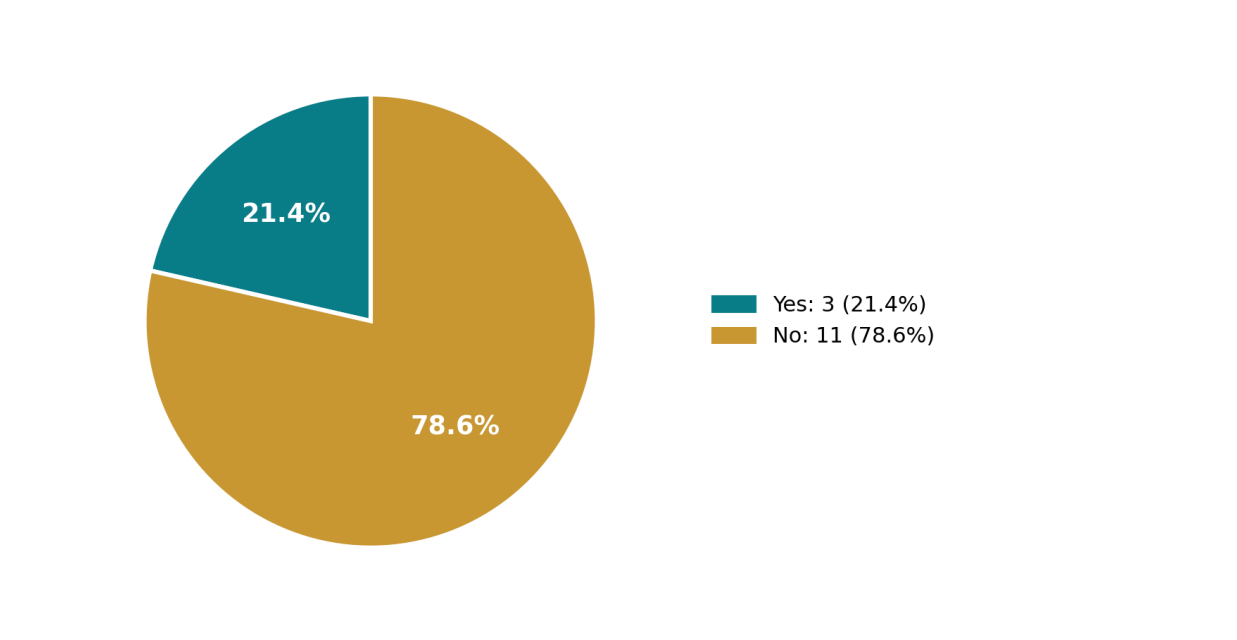
Figure A10. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 11

١١. هل سبق أن شعرت بالإحراج عند التحدث باللغة الإنجليزية أمام زملائك بسبب خطأ في النطق أو تعليقهم على طريقة نطقك؟

Have you felt embarrassed speaking English in front of classmates because of pronunciation errors or their comments?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	3	21.4%
لا	11	78.6%

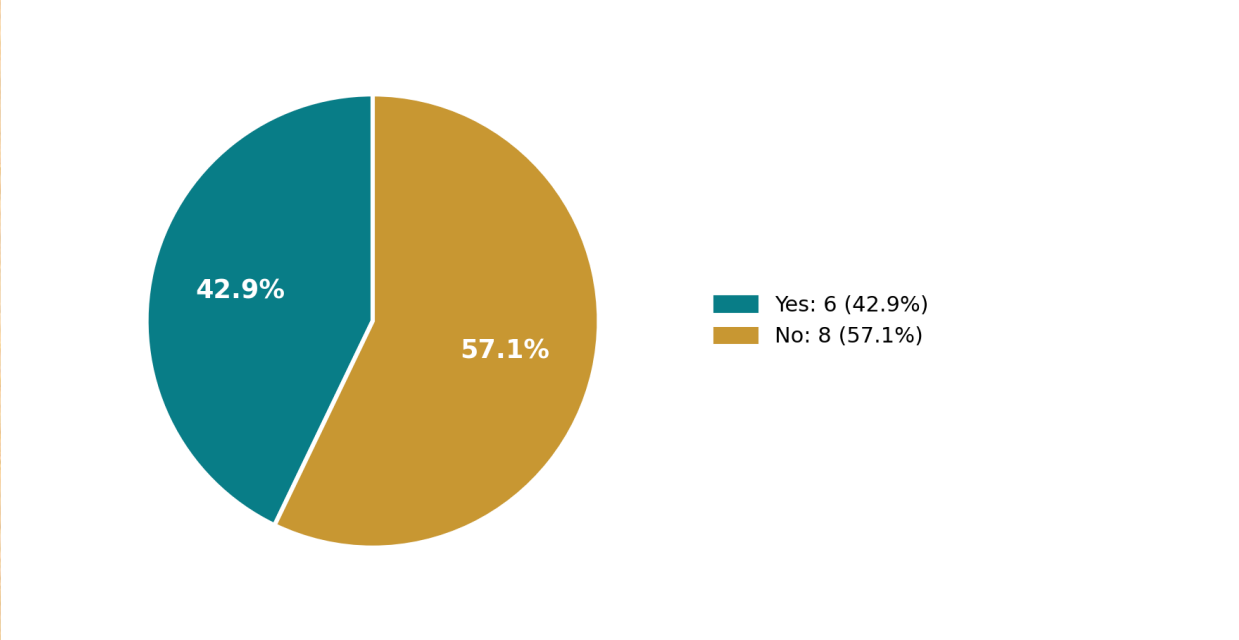
Figure A11. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 12

١٢. هل تتجنب أحيانًا نطق بعض الكلمات أو التحدث بالإنجليزية أمام الآخرين لأنك غير متأكد من صحة نطقك؟

Do you sometimes avoid words or speaking English because you are unsure of your pronunciation?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	6	42.9%
لا	8	57.1%

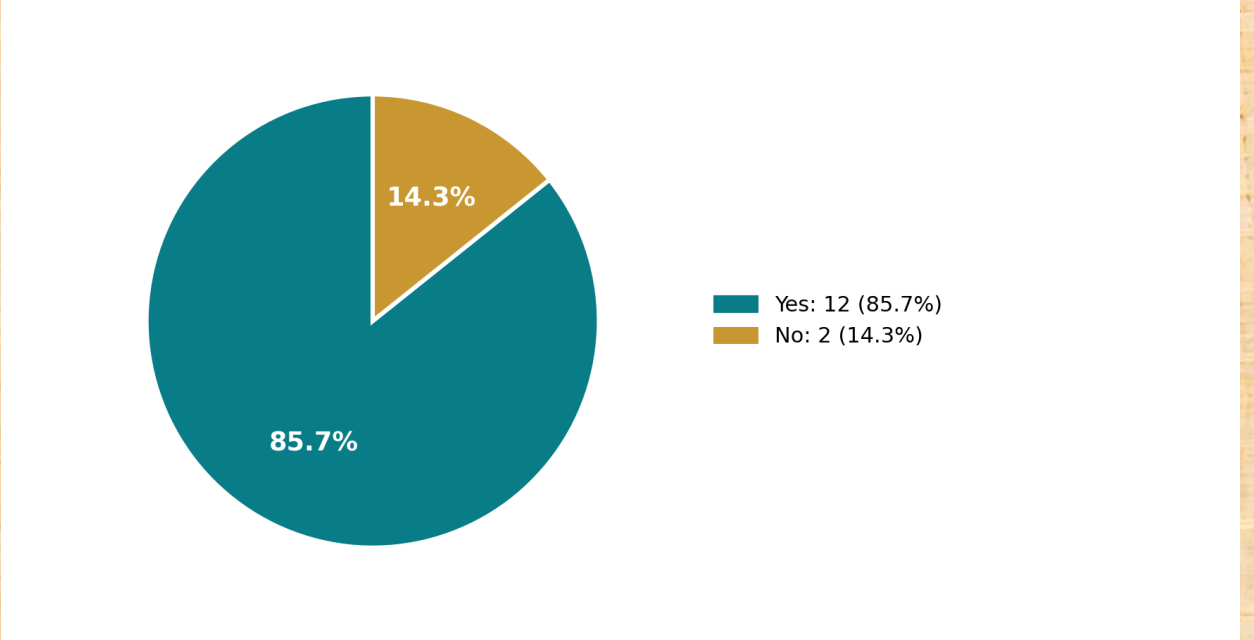
Figure A12. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 13

١٣. إذا استطاع المساعد أن يسمع نطقك، ويحدد الخطأ، ويعلمك النطق الصحيح، ثم يتيح لك إعادة المحاولة، هل ستستخدم هذه الخاصية؟

Would you use a feature that listens to your pronunciation, identifies errors, teaches the correct pronunciation and lets you retry?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

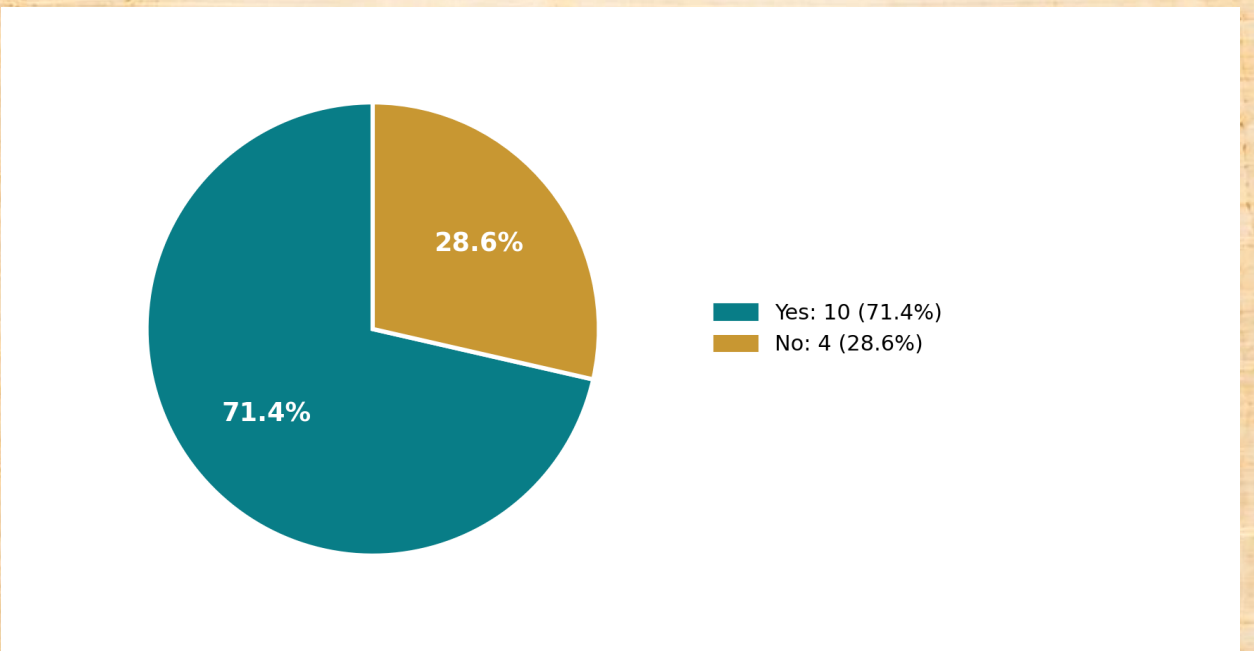
Figure A13. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 14

١٤. هل ترغب في أن يوضح لك المساعد الطريقة الصحيحة لإخراج أصوات الإنجليزية، مثل وضع اللسان والفم وطريقة إخراج الصوت؟

Would you like explanations of how to produce English sounds, including tongue and mouth position?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	10	71.4%
لا	4	28.6%

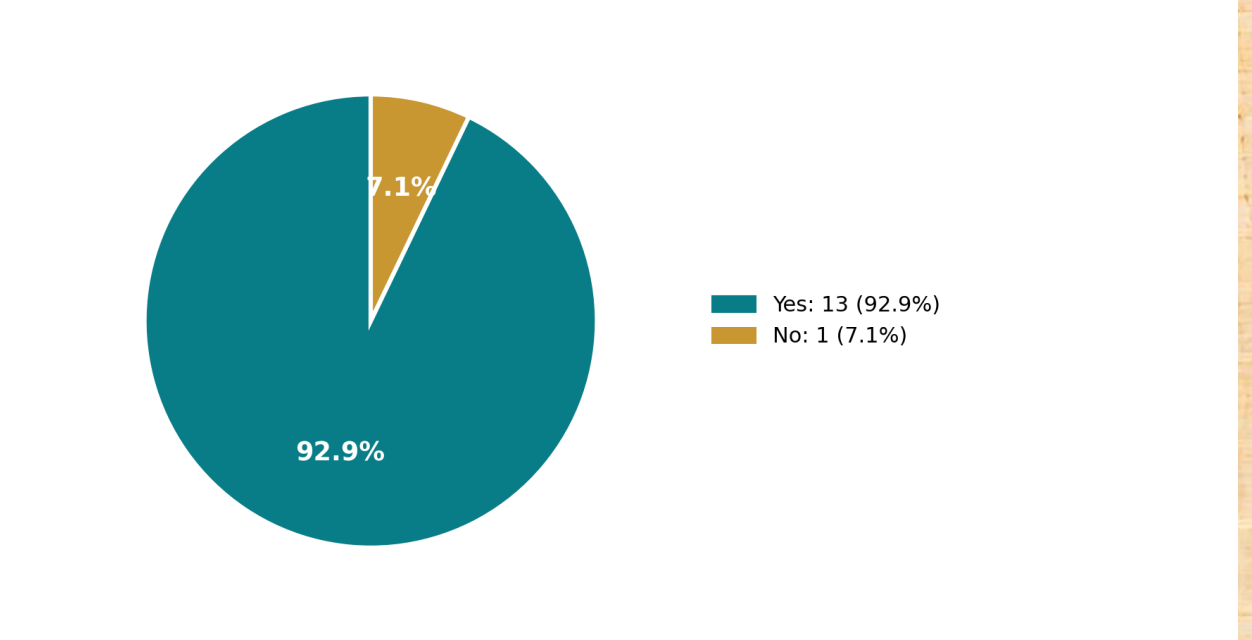
Figure A14. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 15

١٥. هل تعتقد أن التدريب الفردي على النطق والتحدث مع المساعد قد يزيد ثقتك قبل التحدث بالإنجليزية أمام الآخرين؟

Could individual pronunciation and speaking practice with the assistant increase your confidence?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

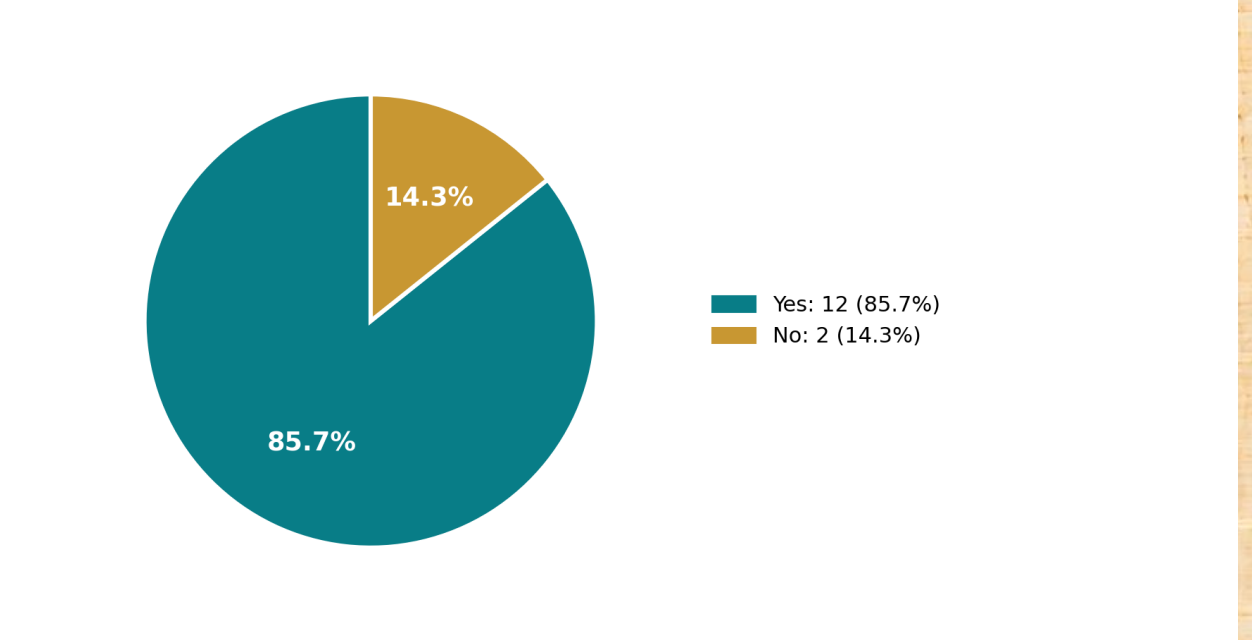
Figure A15. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 16

١٦. هل تريد أن تتعلم الإنجليزية لتستخدمها فعلًا في التحدث والكتابة والاستماع والقراءة، وليس فقط للحصول على درجة جيدة في الامتحان؟

Do you want to use English in speaking, writing, listening and reading, beyond earning good exam grades?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

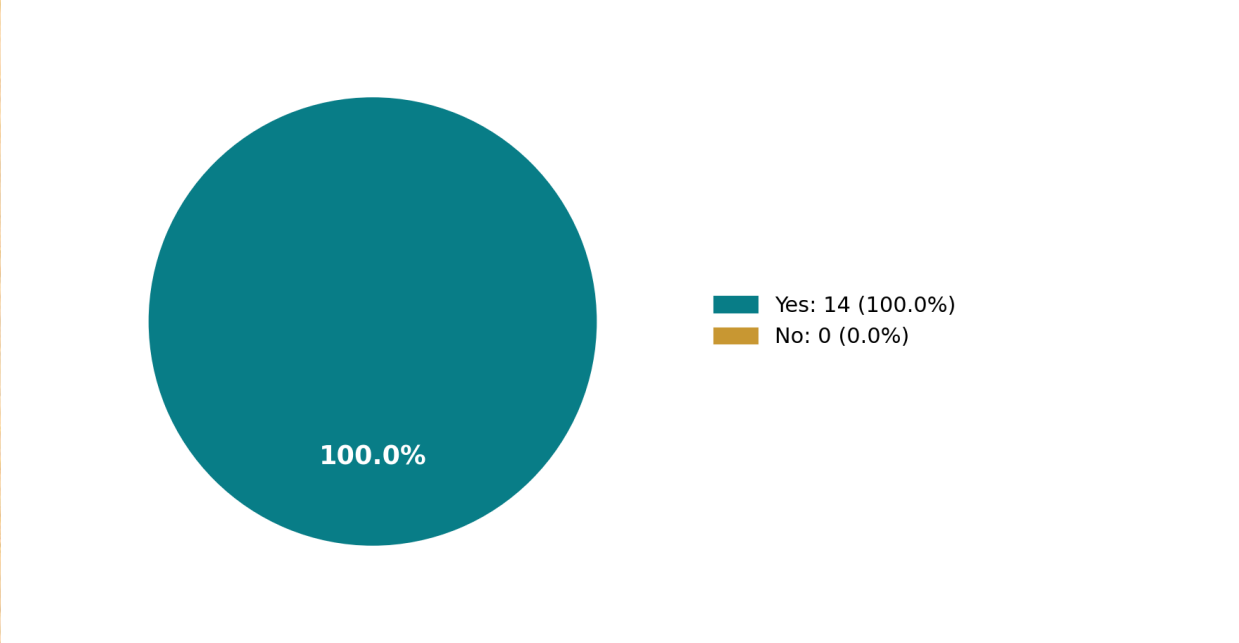
Figure A16. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 17

١٧. هل يهمك أن تفهم كيف تستخدم القاعدة والكلمة في مواقف مختلفة، بدلًا من حفظها فقط؟

Is understanding how to use grammar and vocabulary in different situations important to you?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	14	100.0%
لا	0	0.0%

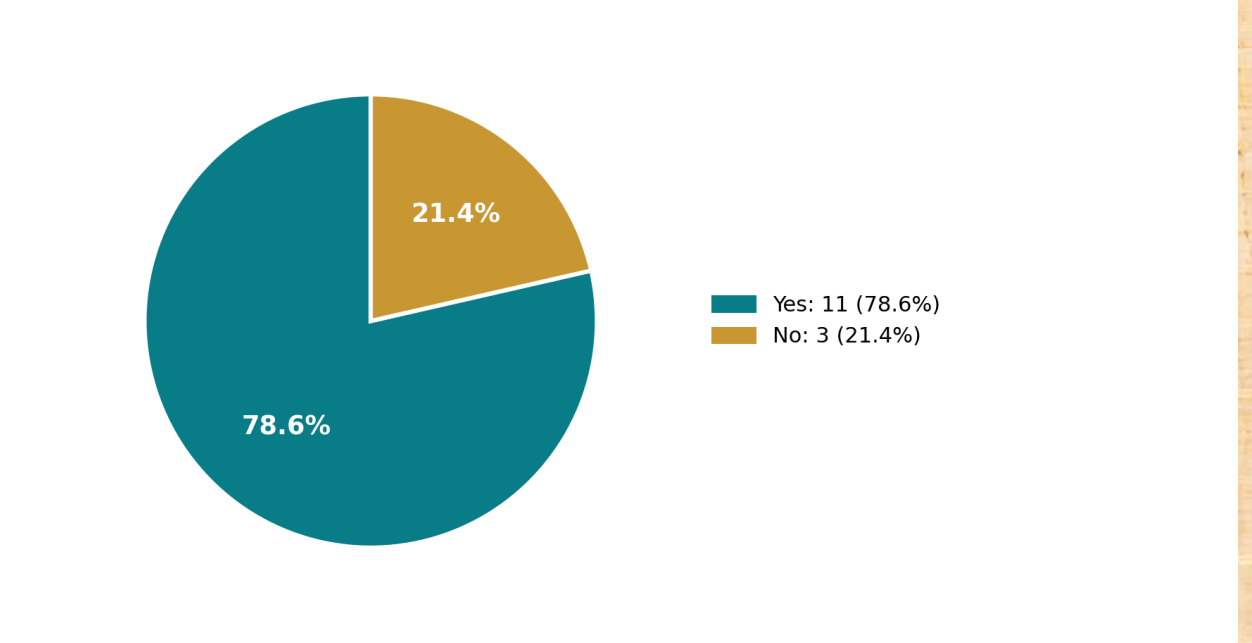
Figure A17. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 18

١٨. هل تريد أن تصل إلى مرحلة تستطيع فيها تكوين جمل والتعبير عن أفكارك بالإنجليزية دون الاعتماد المستمر على الترجمة من العربية؟

Do you want to form sentences and express ideas without continually translating from Arabic?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	11	78.6%
لا	3	21.4%

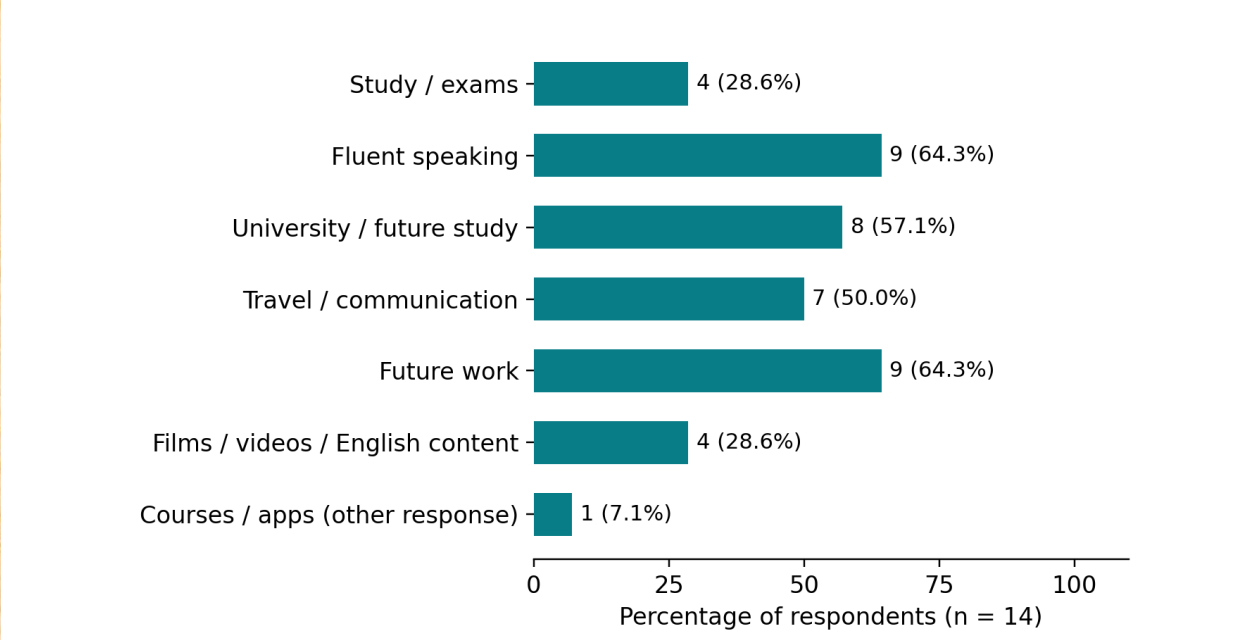
Figure A18. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 19

١٩. في أي مواقف تريد أن تفيدك الإنجليزية؟

In which situations do you want English to help you?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
الدراسة والامتحانات	4	28.6%
التحدث بطلاقة	9	64.3%
الجامعة والدراسة المستقبلية	8	57.1%
السفر والتواصل	7	50.0%
العمل في المستقبل	9	64.3%
فهم الأفلام والفيديوهات والمحتوى الإنجليزي	4	28.6%
الكرورسات apps	1	7.1%

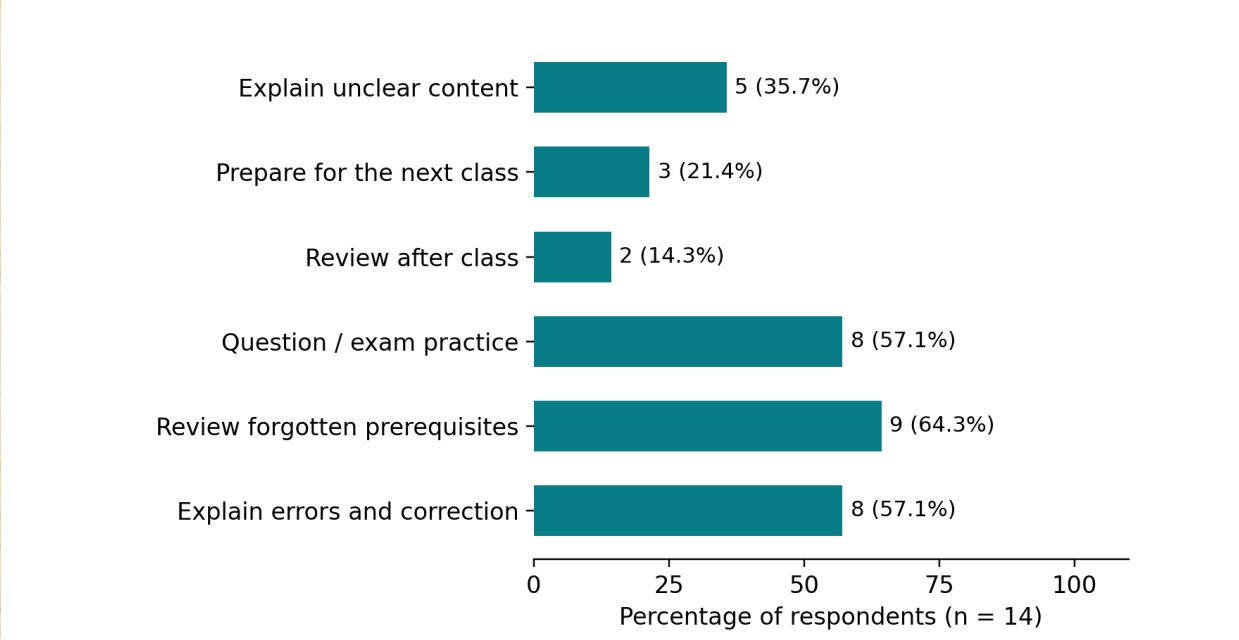
Figure A19. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 20

٢٠. إذا كان المساعد يعرف منهجك ودروسك، ماذا تريد أن يفعل معك؟

If the assistant knows your curriculum, what would you like it to do?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
يشرح ما لم أفهمه	5	35.7%
يجهزني للحصة القادمة	3	21.4%
يراجع معي بعد الحصة	2	14.3%
يدربني على أسئلة وامتحانات	8	57.1%
يراجع معي قاعدة أو مهارة قديمة نسيتها	9	64.3%
يشرح سبب الخطأ وكيف أصححه	8	57.1%

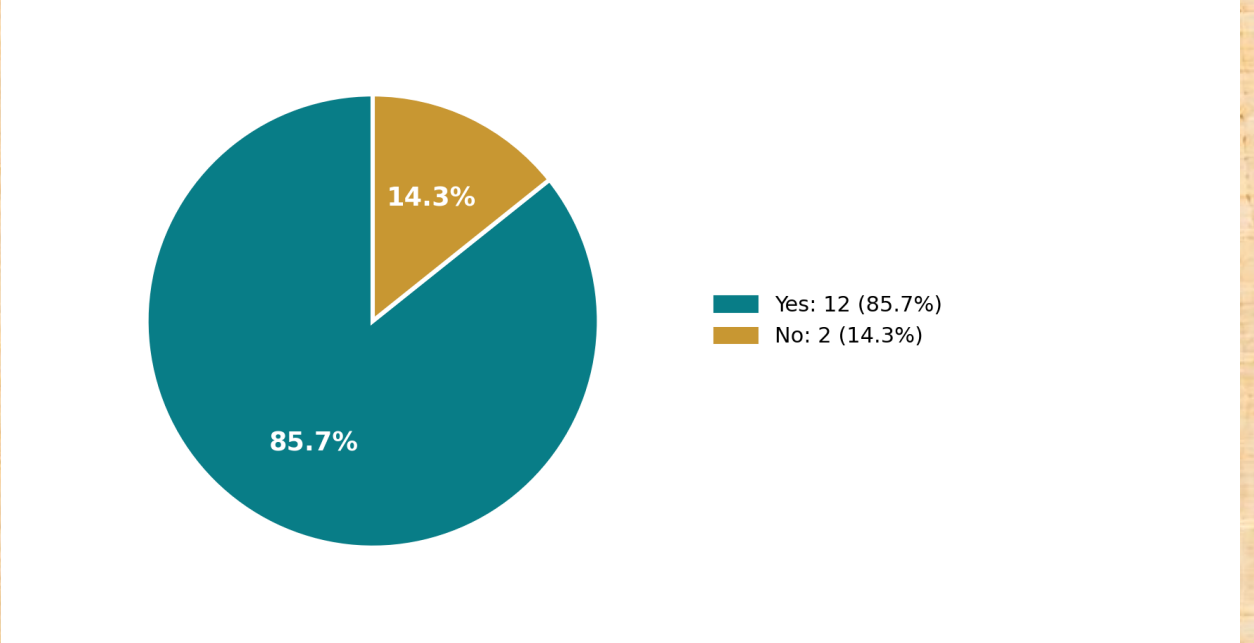
Figure A20. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 21

٢١. إذا كان لديك في اليوم التالي درس قراءة أو استماع، هل ستستخدم المساعد إذا كان يستطيع أن يجهزك له مسبقًا بالكلمات والنطق والاستماع وأسئلة قصيرة؟

Would you use preparation for an upcoming reading or listening lesson through vocabulary, pronunciation, listening and short questions?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

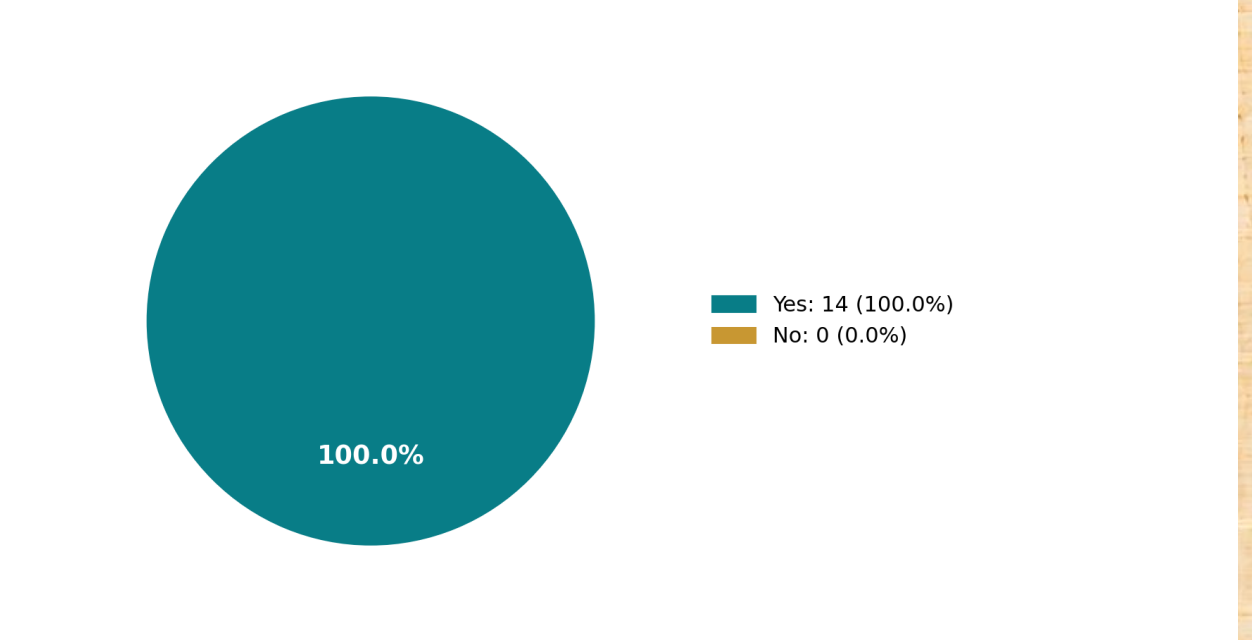
Figure A21. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 22

٢٢. إذا لاحظ المساعد أن صعوبة درس جديد ترجع إلى قاعدة أو مهارة قديمة نسيتها، هل تريد أن يراجعها معك أولًا ثم يكمل الدرس؟

Should the assistant first review a forgotten prerequisite skill when it causes difficulty with a new lesson?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	14	100.0%
لا	0	0.0%

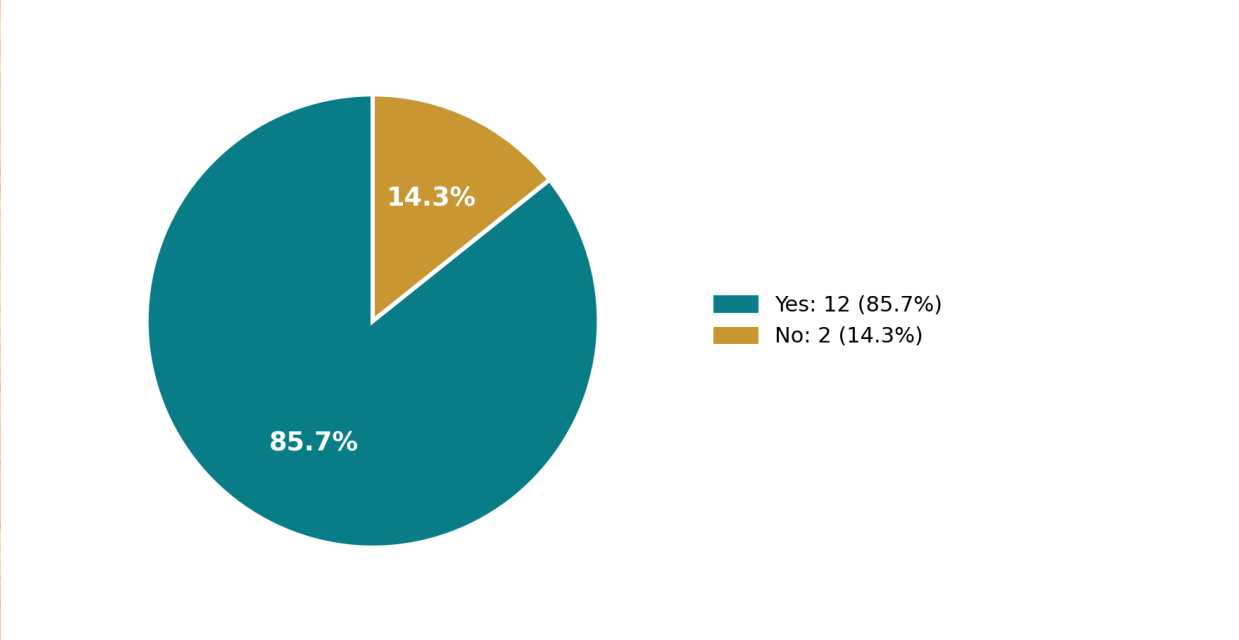
Figure A22. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 23

٢٣. هل تريد أن تستطيع سؤال المساعد في أي وقت عن معنى كلمة أو نطقها، أو كيف تقول جملة معينة بالإنجليزية، أو لماذا كانت إجابتك خطأ؟

Would you like to ask at any time about meaning, pronunciation, expressing a sentence or why an answer was wrong?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

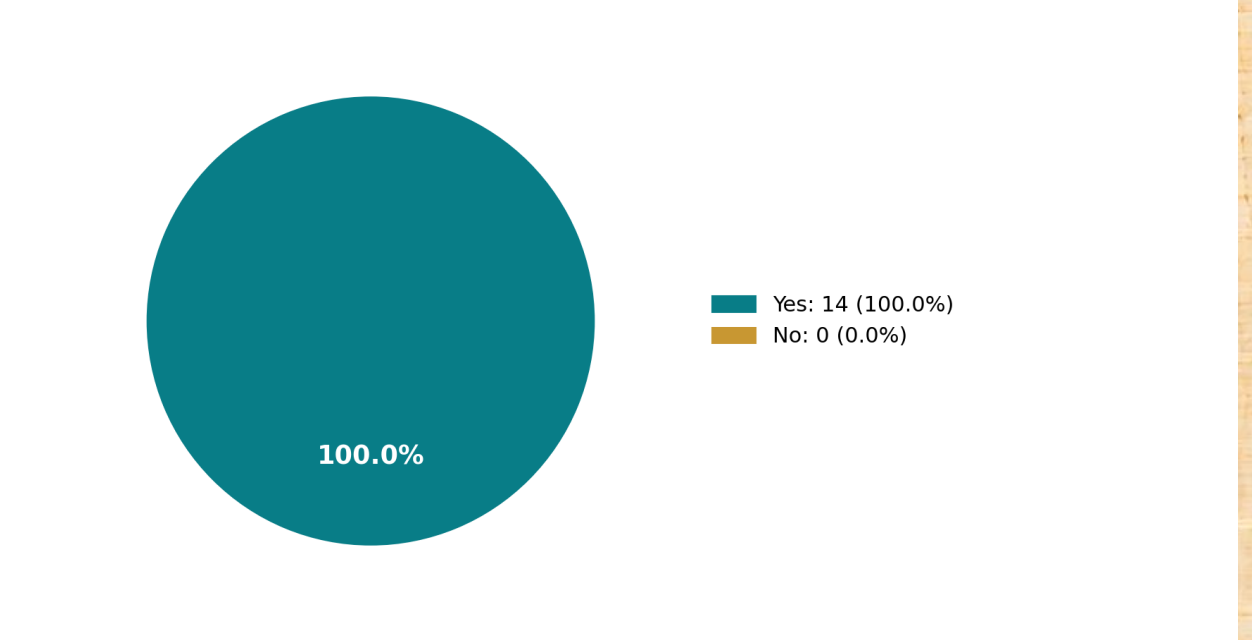
Figure A23. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 24

٢٤. هل تريد أن يتذكر المساعد الأخطاء التي تتكرر معك، ويعطيك تدريبات عليها لاحقًا حتى تتأكد أنك أتقنتها؟

Should the assistant remember recurring errors and provide later practice until you master them?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	14	100.0%
لا	0	0.0%

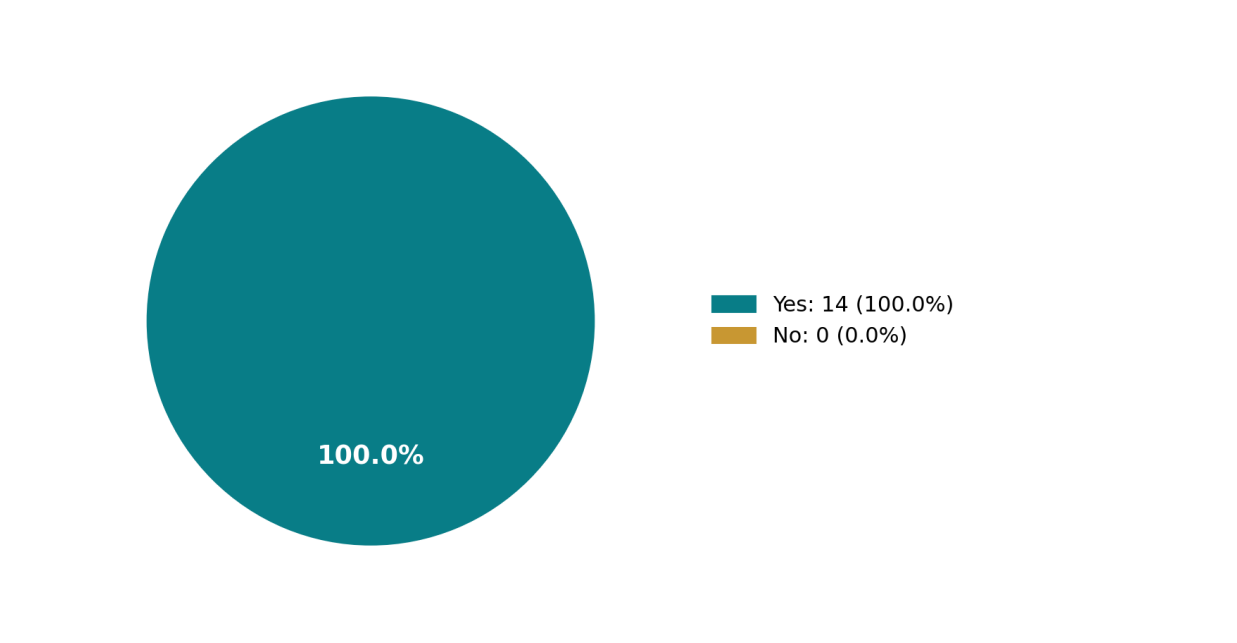
Figure A24. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 25

٢٥. هل تريد أن يبدأ المساعد باختبار قصير يعرف من خلاله مستواك في القواعد والكلمات والقراءة والكتابة والاستماع والتحدث والنطق؟

Should the assistant begin with a short diagnostic test covering English skills?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	14	100.0%
لا	0	0.0%

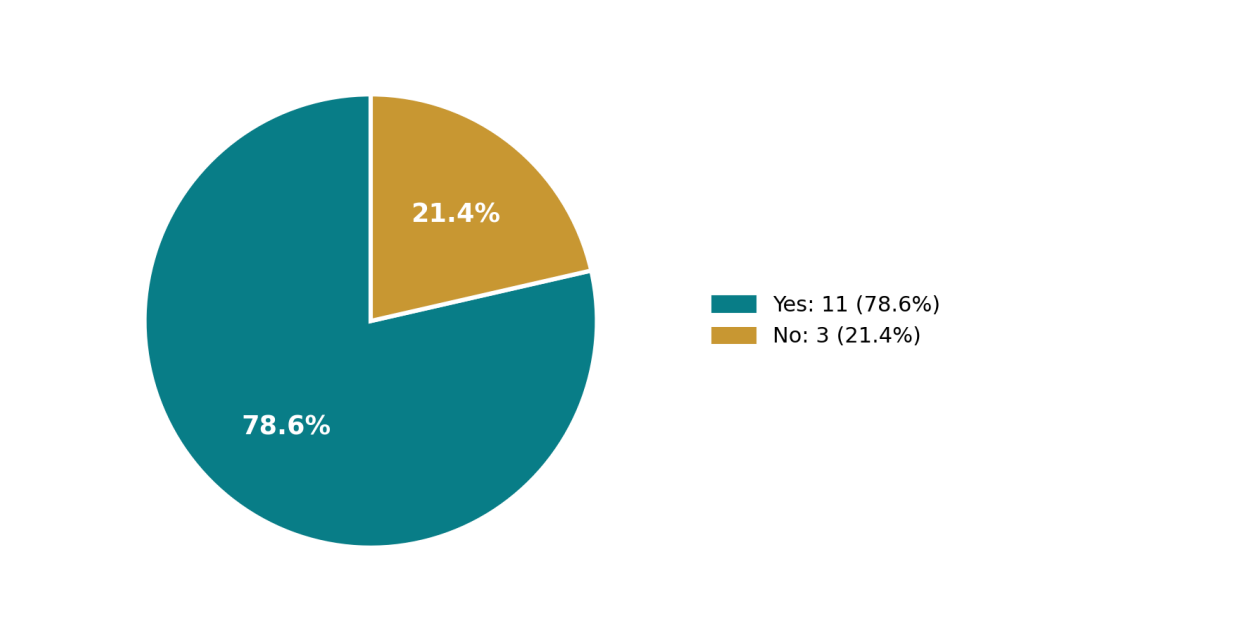
Figure A25. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 26

٢٦. هل تريد أن يخبرك المساعد من وقت لآخر بما تحسن لديك وما الذي ما زلت تحتاج إلى التدريب عليه؟

Should the assistant periodically tell you what has improved and what still needs practice?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	11	78.6%
لا	3	21.4%

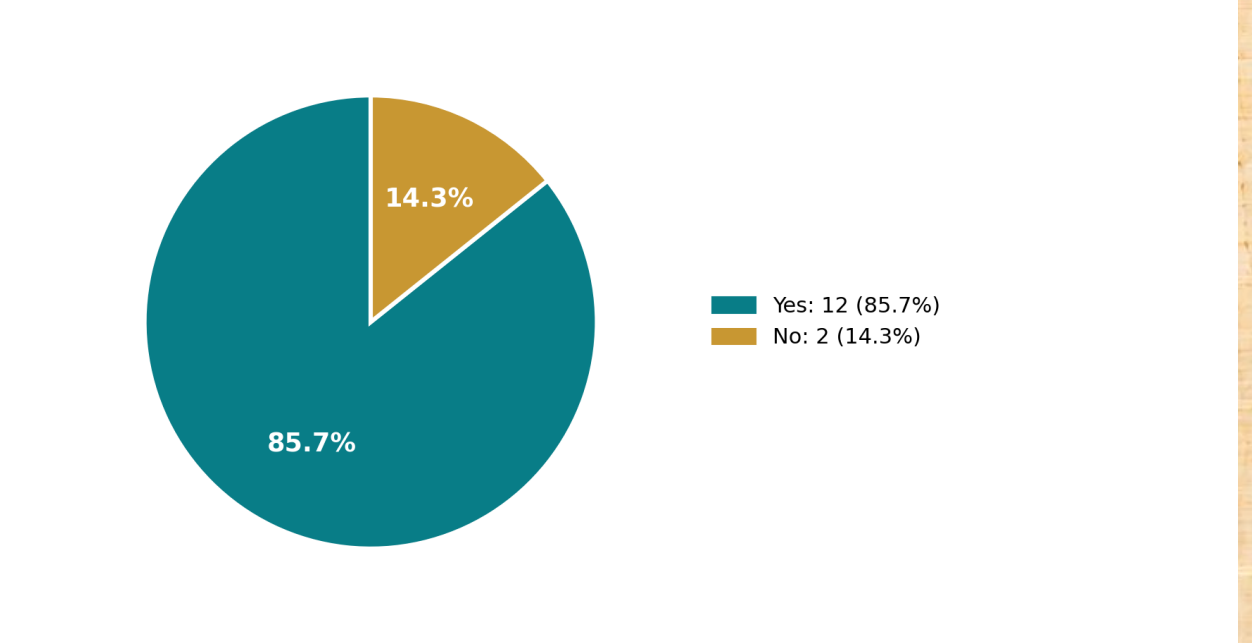
Figure A26. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 27

٢٧. هل تريد أن يقدم لك تدريبات تناسب مستواك وأخطاءك، بدلًا من تقديم نفس التدريبات لجميع الطلاب؟

Would you like practice matched to your level and errors rather than identical tasks for everyone?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

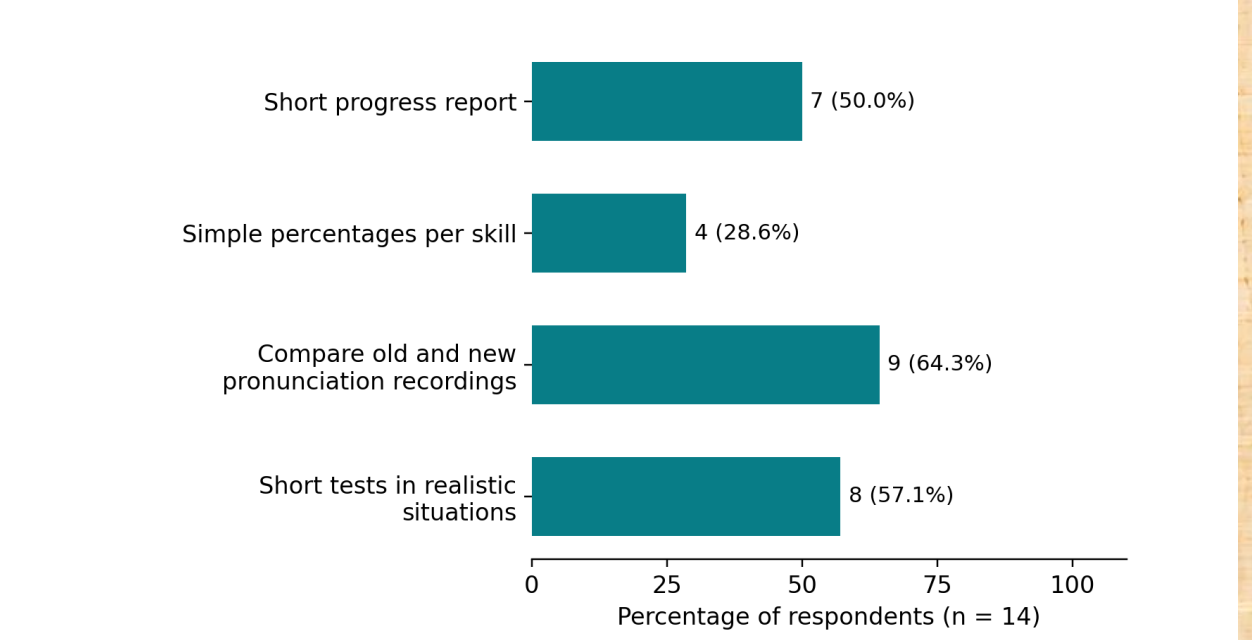
Figure A27. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 28

٢٨. كيف تفضل أن تعرف أن مستواك يتحسن؟

How would you prefer to see that your level is improving?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
تقرير قصير يوضح ما تحسن وما يحتاج إلى تدريب	7	50.0%
نسب بسيطة لكل مهارة	4	28.6%
مقارنة تسجيل نطقي القديم بالجديد	9	64.3%
اختبارات قصيرة في مواقف حقيقية	8	57.1%

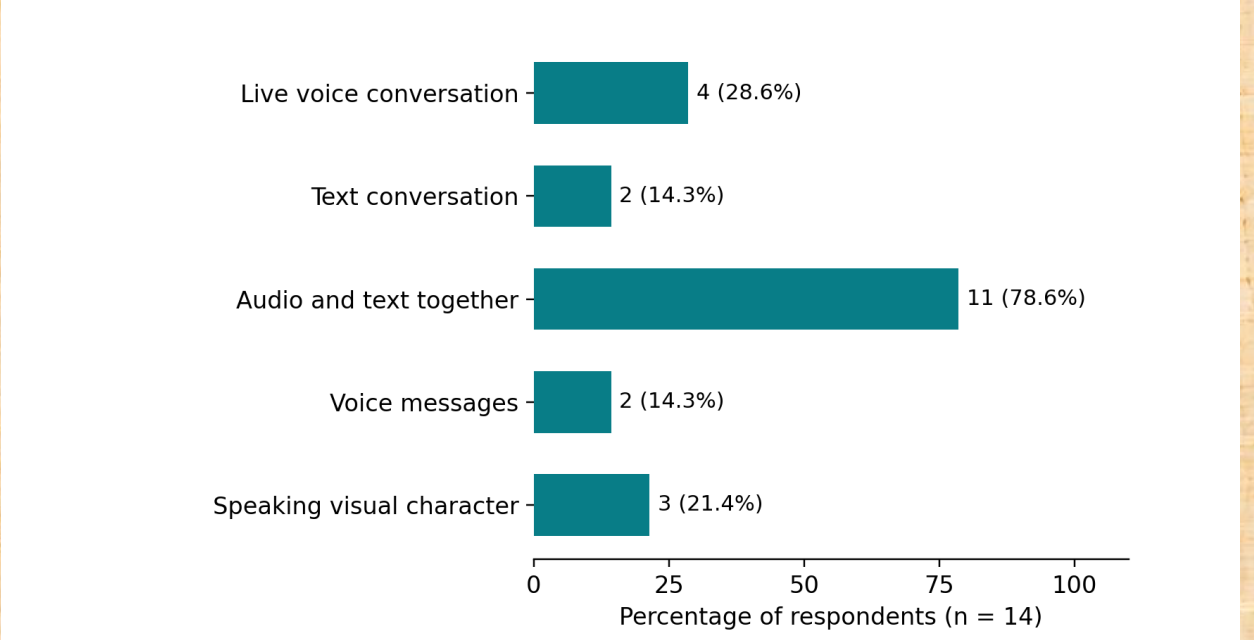
Figure A28. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 29

٢٩. كيف تفضل التفاعل مع المساعد؟

How would you prefer to interact with the assistant?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
محادثة صوتية مباشرة	4	28.6%
محادثة كتابية	2	14.3%
الصوت والكتابة معًا	11	78.6%
رسائل صوتية	2	14.3%
شخصية مرئية تتحدث معي	3	21.4%

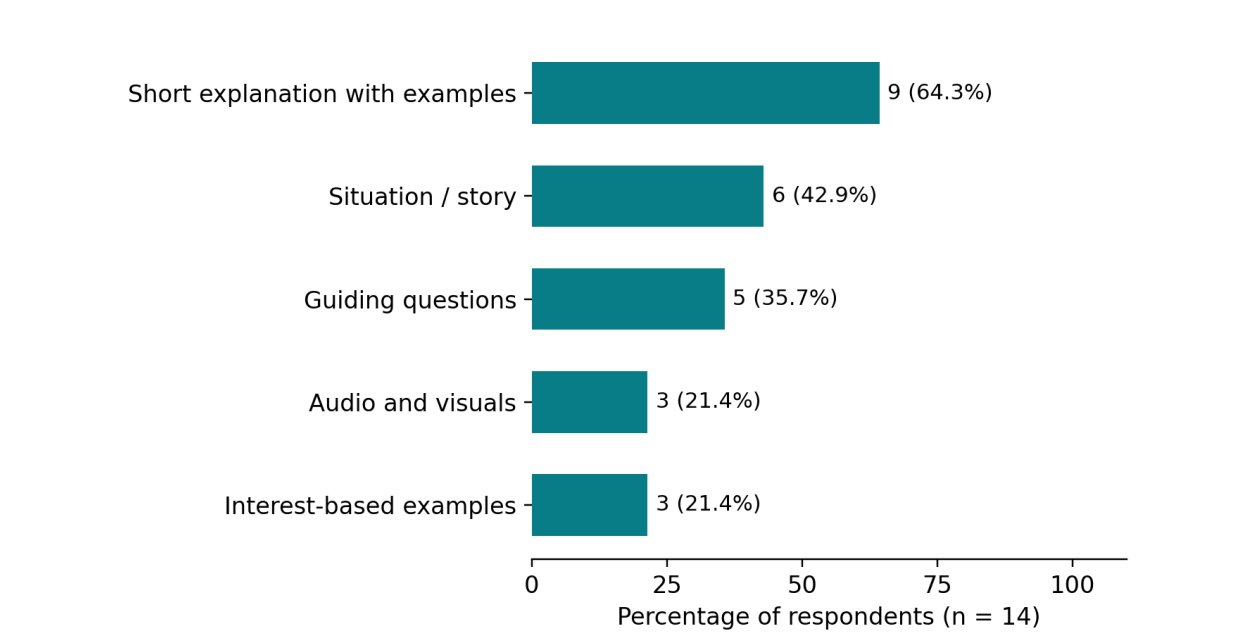
Figure A29. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 30

٣٠. كيف تفضل أن يشرح لك؟

How would you prefer the assistant to explain?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
شرح قصير مع أمثلة	9	64.3%
موقف أو قصة توضح الفكرة	6	42.9%
أسئلة تساعدني على الوصول إلى الإجابة	5	35.7%
صوت وصورة مع الشرح	3	21.4%
أمثلة مرتبطة باهتماماتي	3	21.4%

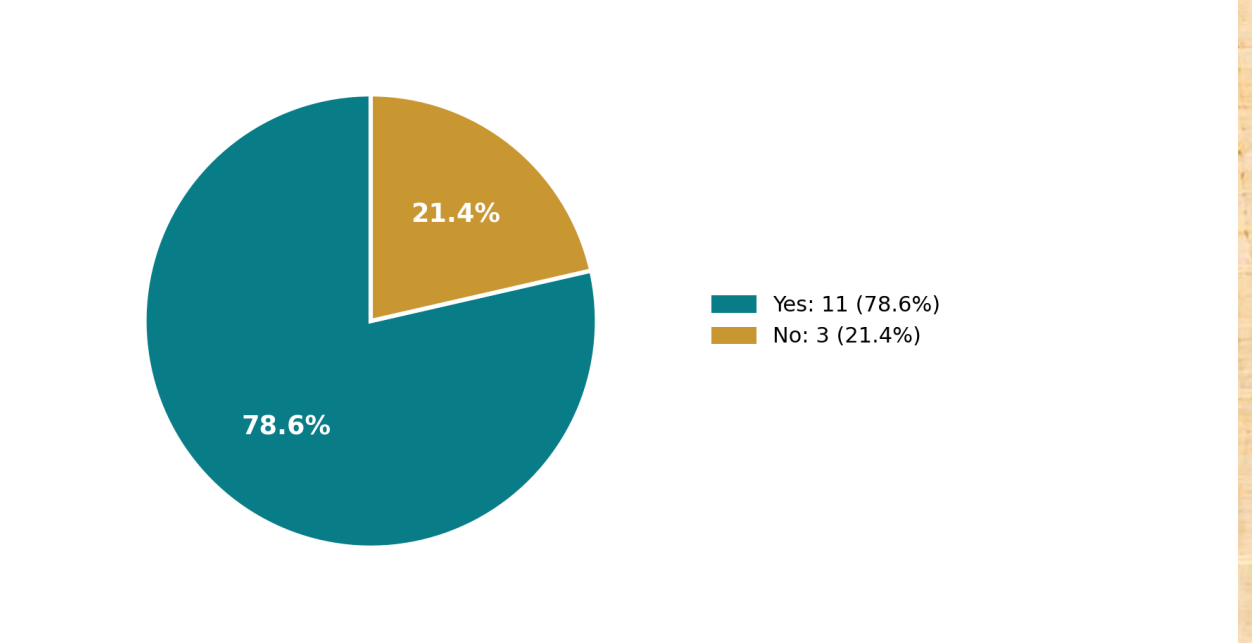
Figure A30. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 31

٣١. هل تفضل أن يتحدث معك المساعد بالإنجليزية، مع استخدام العربية عند الحاجة إلى شرح شيء لم تفهمه؟

Should the assistant use English with Arabic support when needed?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	11	78.6%
لا	3	21.4%

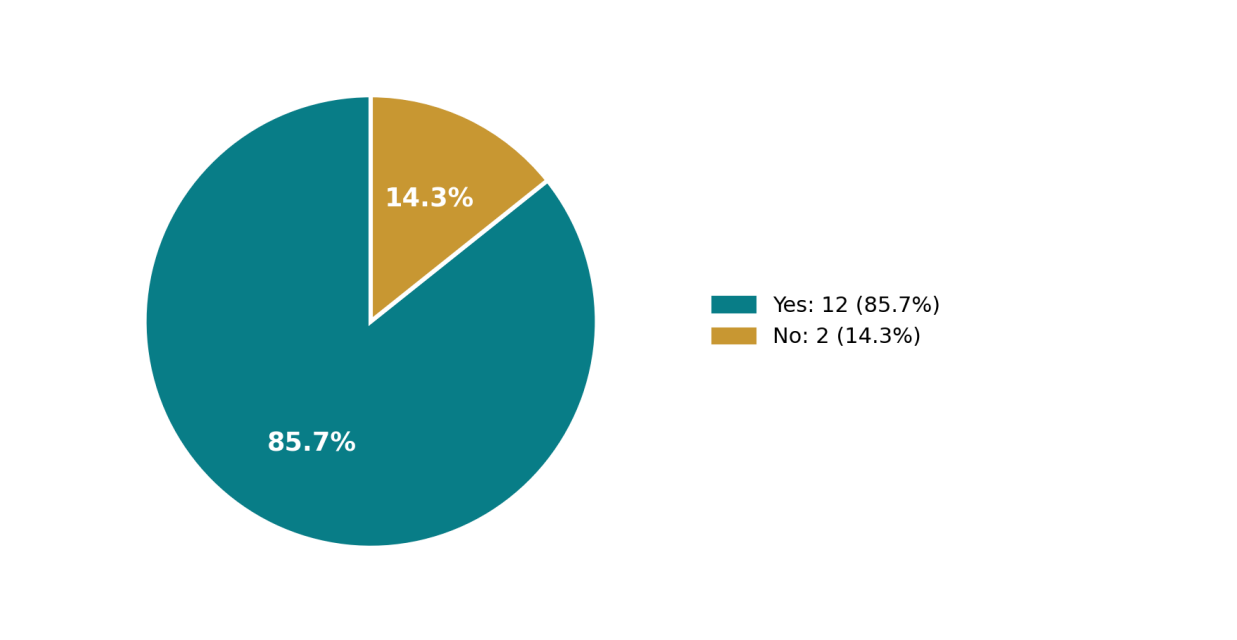
Figure A31. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 32

٣٢. هل تفضل أن يقل استخدام العربية تدريجيًا كلما تحسن مستواك في الإنجليزية؟

Should Arabic support gradually decrease as your English improves?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

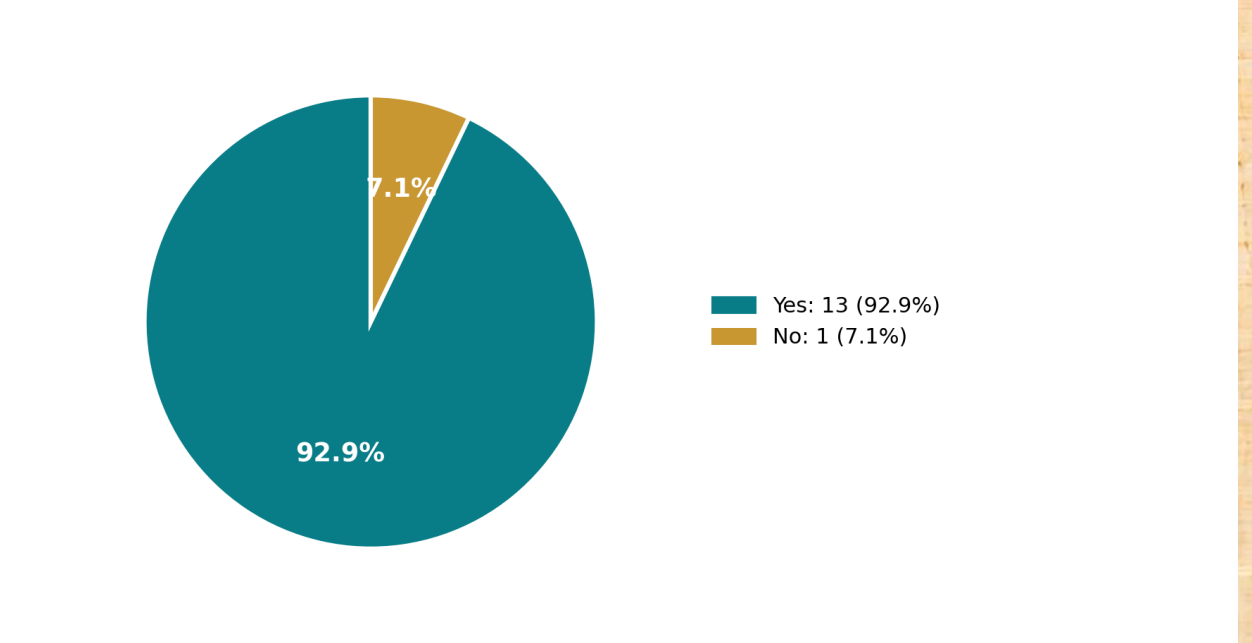
Figure A32. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 33

٣٣. هل تحب أن يبدأ المساعد أحيانًا برسالة قصيرة تشجعك على التعلم وتذكرك بما أحرزته من تقدم؟

Would you like occasional short opening messages that encourage learning and acknowledge progress?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

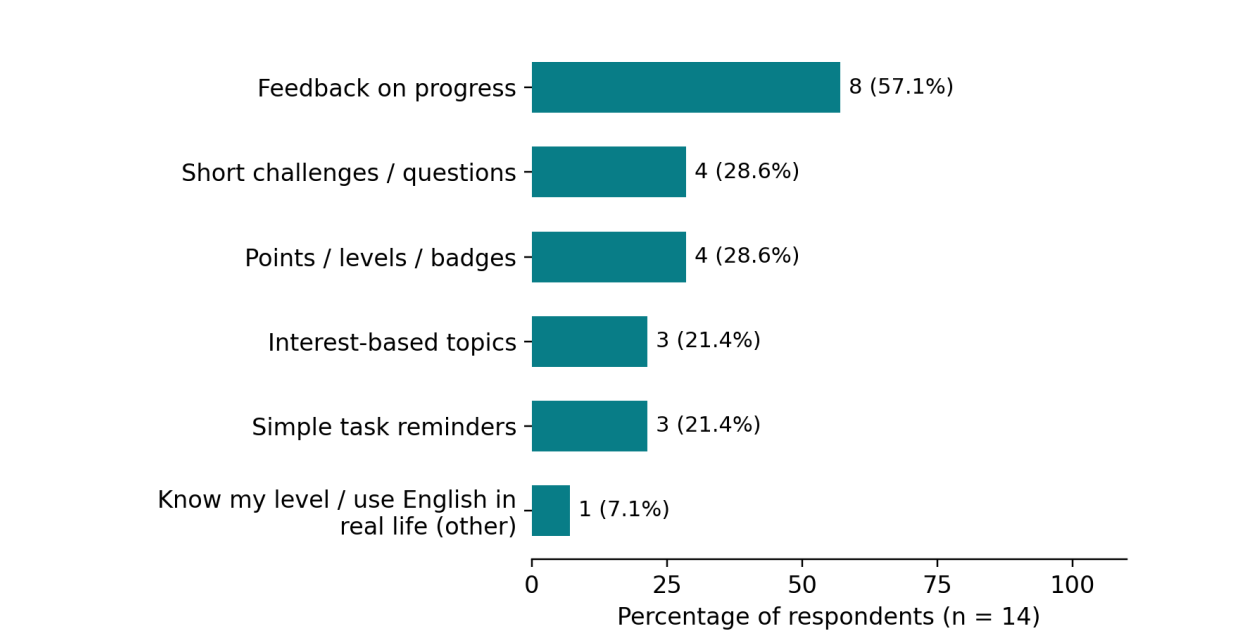
Figure A33. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 34

٣٤. ما الذي يشجعك أكثر على الاستمرار؟

What would most encourage you to continue?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
أن يخبرني بتقدمي	8	57.1%
تحديات وأسئلة قصيرة	4	28.6%
نقاط ومستويات وشارات	4	28.6%
موضوعات مرتبطة باهتماماتي	3	21.4%
تذكير بسيط بما أحتاج إلى إنجازه	3	21.4%
معرفه مستوى اللغه لدي أو تطبيقه في الحياه العمليه	1	7.1%

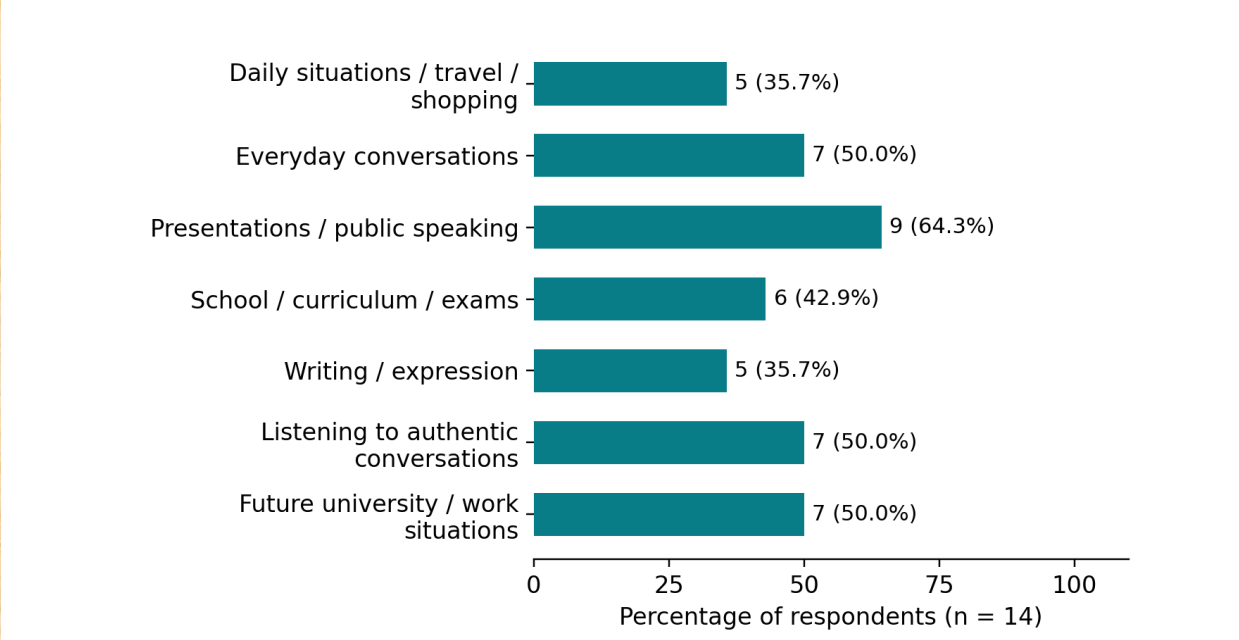
Figure A34. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 35

٣٥. ما المواقف التي تريد أن تتدرب عليها بالإنجليزية؟

Which situations would you like to practise in English?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
مواقف يومية مثل السفر والمطعم والشراء	5	35.7%
محادثات عادية	7	50.0%
العروض والتحدث أمام الآخرين	9	64.3%
المدرسة والمنهج والامتحانات	6	42.9%
الكتابة والتعبير	5	35.7%
الاستماع إلى محادثات حقيقية	7	50.0%
مواقف الجامعة والعمل في المستقبل	7	50.0%

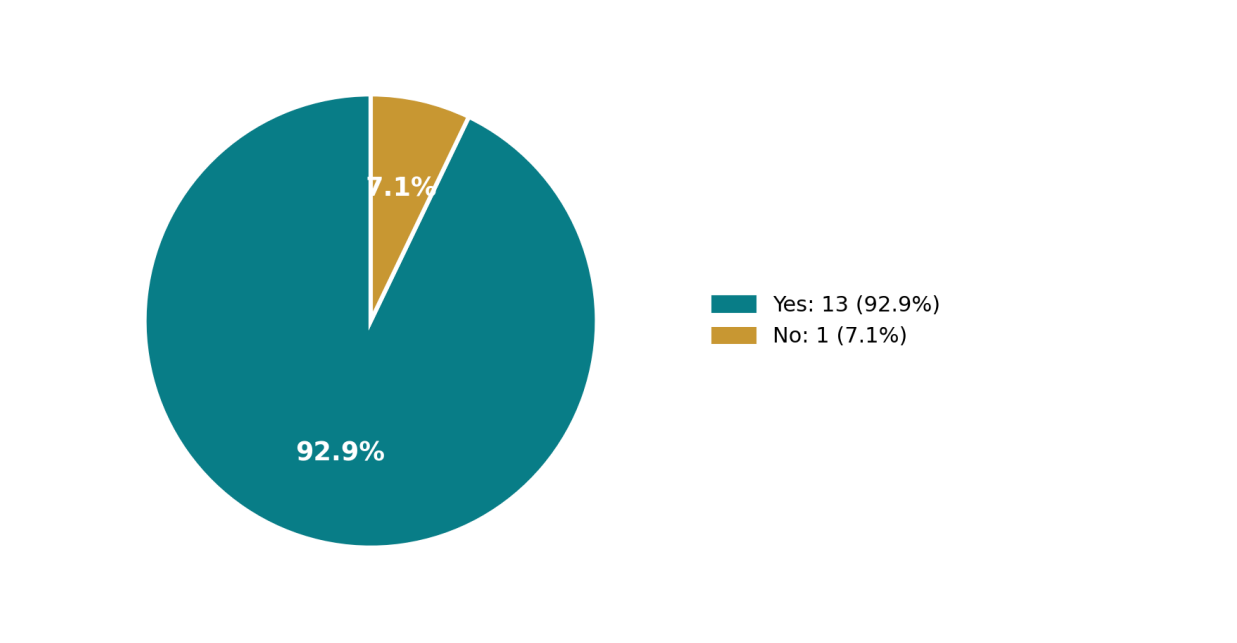
Figure A35. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 36

٣٦. هل تحب أن يستخدم المساعد موضوعات تهتم بها، مثل الرياضة أو الأفلام أو الألعاب أو العلوم، حتى تتدرب على الإنجليزية من خلالها؟

Would you like practice based on interests such as sport, films, games or science?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

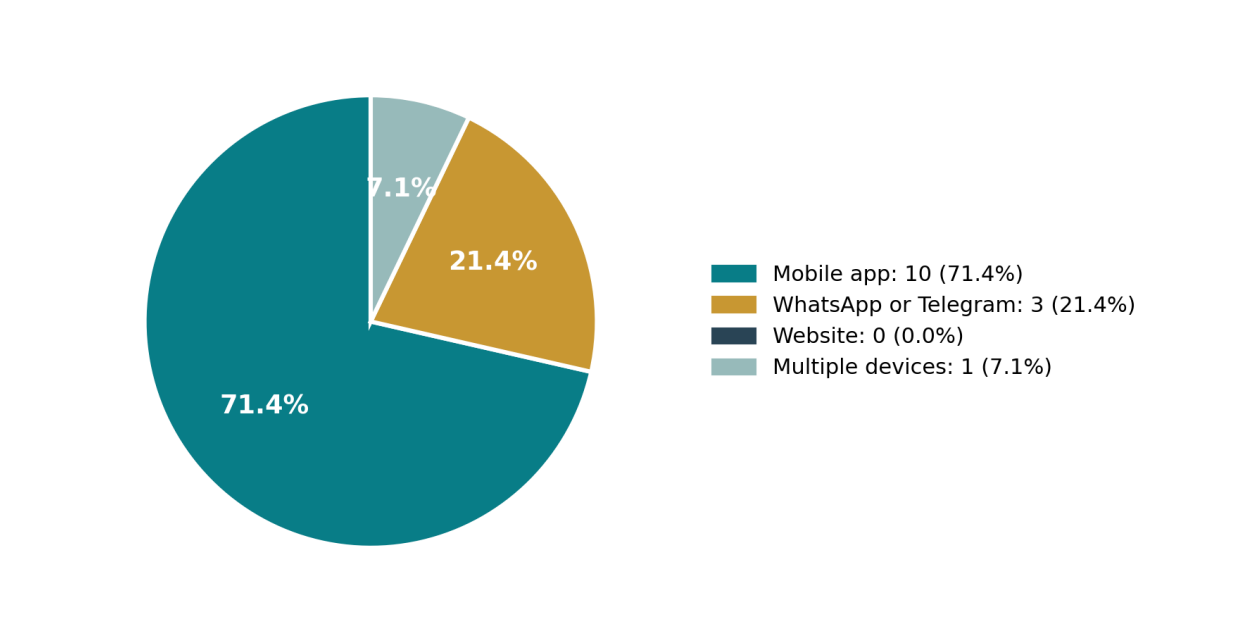
Figure A36. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 37

٣٧. أين تفضل استخدام المساعد؟

Where would you prefer to use the assistant?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
تطبيق على الهاتف	10	71.4%
Telegram أو WhatsApp	3	21.4%
موقع على الإنترنت	0	0.0%
على أكثر من جهاز	1	7.1%

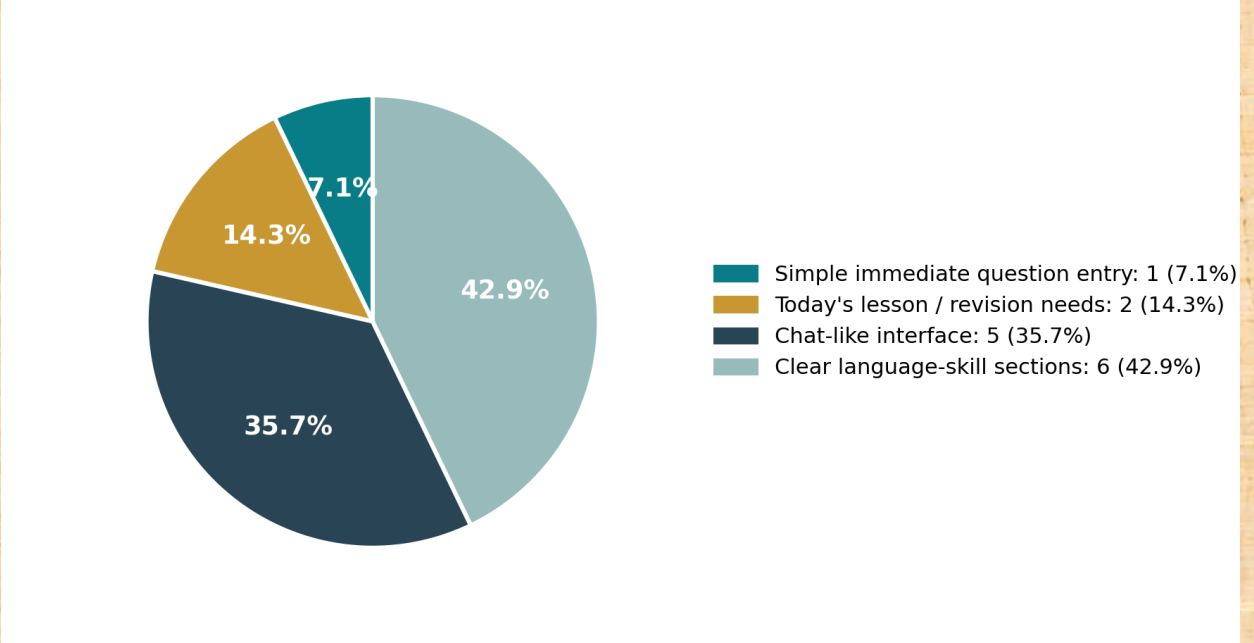
Figure A37. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 38

٣٨. كيف تفضل أن تكون واجهة المساعد؟

What interface would you prefer?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
بسيطة وأبدأ السؤال مباشرة	1	7.1%
تظهر لي درس اليوم وما أحتاج إلى مراجعته	2	14.3%
تشبه تطبيقات المحادثة	5	35.7%
تحتوي على أقسام واضحة للنطق والتحدث والقراءة والكتابة	6	42.9%

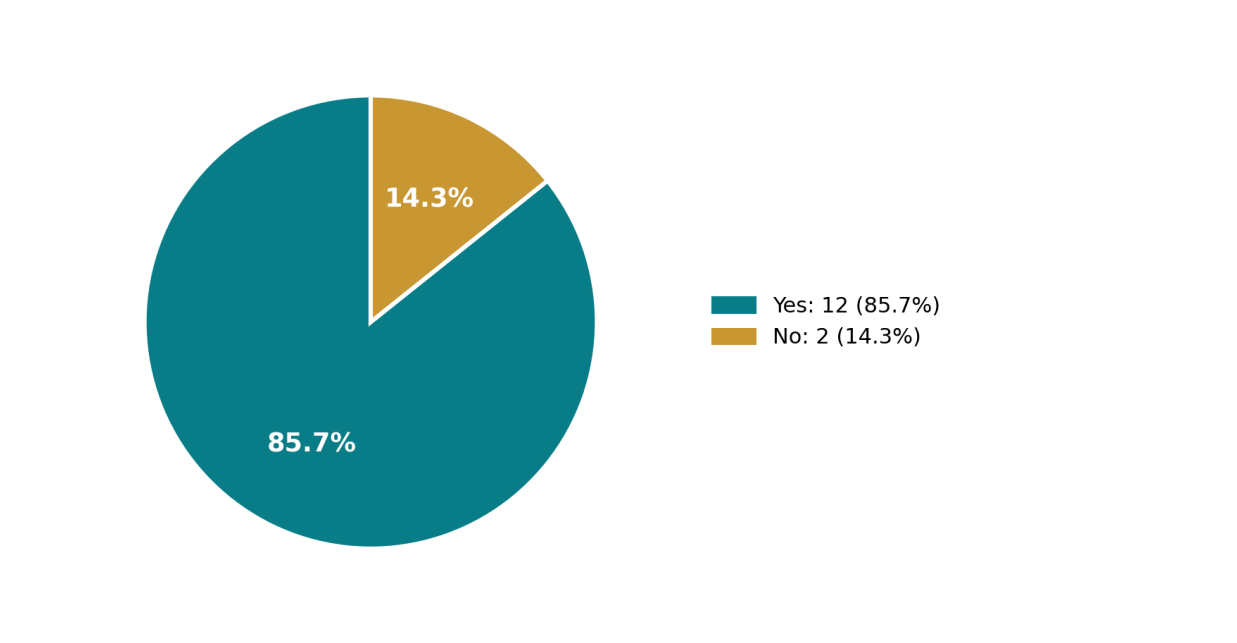
Figure A38. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 39

٣٩. هل تريد أن يكون المساعد متاحًا لك في أي وقت تحتاج فيه إلى سؤال أو مراجعة أو تدريب سريع؟

Should the assistant be available whenever you need a question answered, revision or brief practice?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

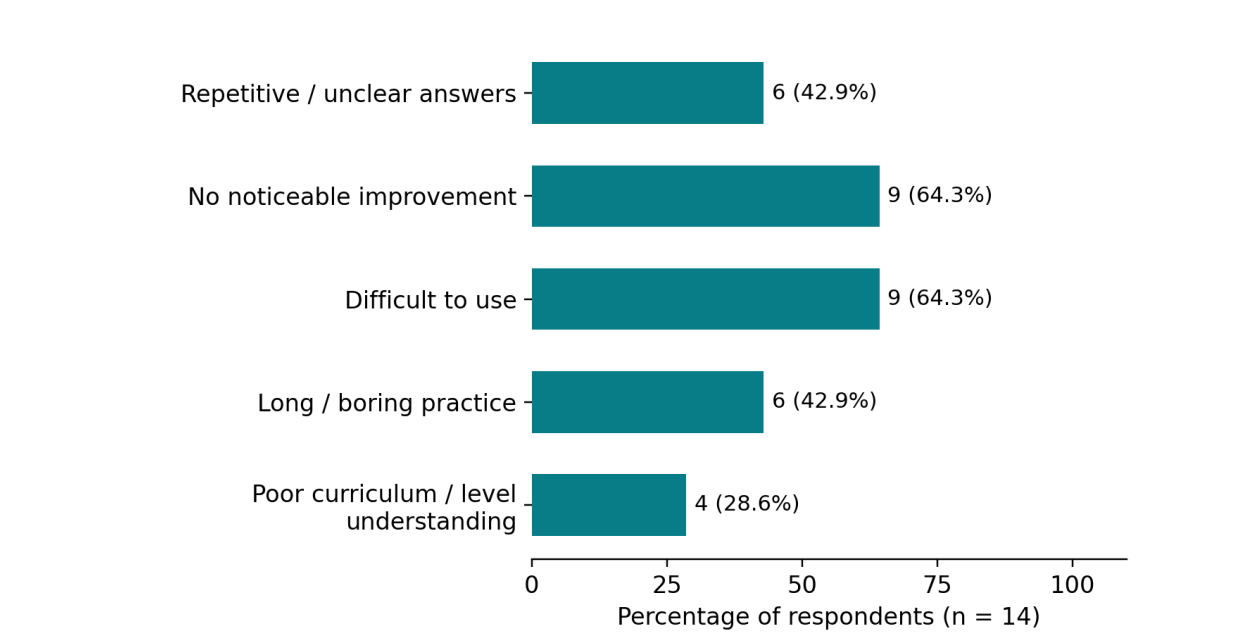
Figure A39. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 40

٤٠. ما الذي قد يجعلك تتوقف عن استخدامه؟

What might make you stop using it?

Multiple responses allowed; percentages may total more than 100%. Valid responses: 14.



Original response option	Count	Percentage
إذا كانت إجاباته مكررة أو غير واضحة	6	42.9%
إذا لم أشعر بتحسن حقيقي	9	64.3%
إذا كان استخدامه صعبًا	9	64.3%
إذا كانت تدريباته طويلة أو مملة	6	42.9%
إذا لم يفهم منهجي أو مستواي	4	28.6%

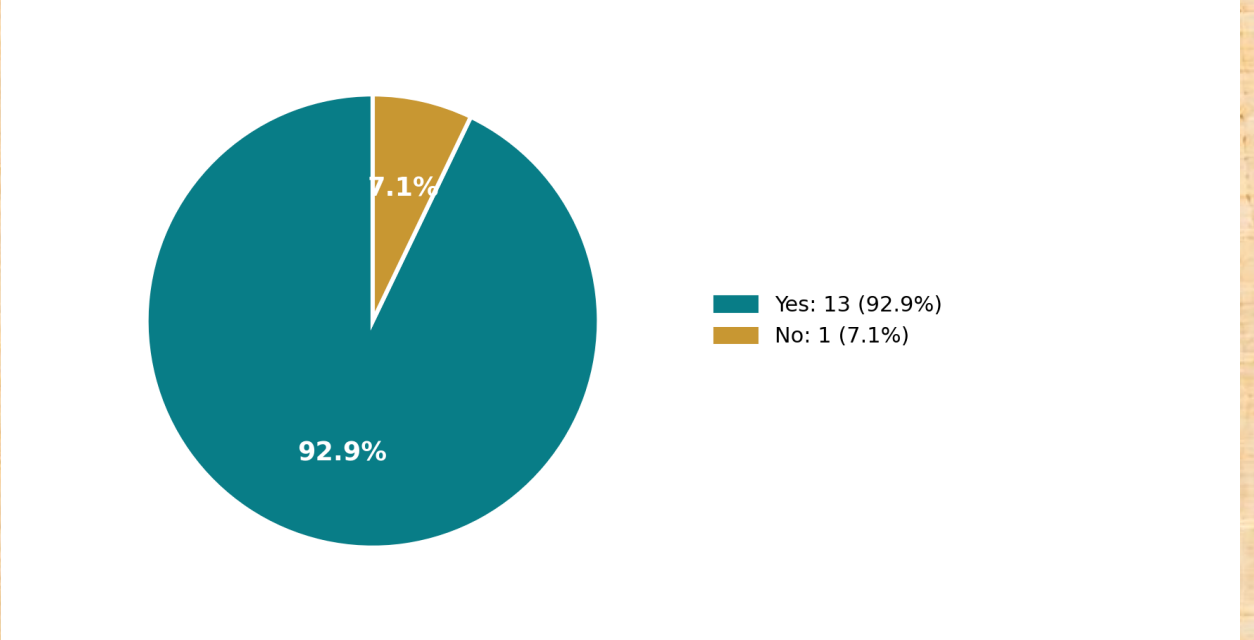
Figure A40. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 41

٤١. إذا كان لديك مساعد تستطيع أن تتدرب معه على التحدث والنطق في أي وقت دون إحراج، هل تعتقد أنك ستتدرب على الإنجليزية أكثر؟

Would embarrassment-free speaking and pronunciation practice at any time encourage you to practise more?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

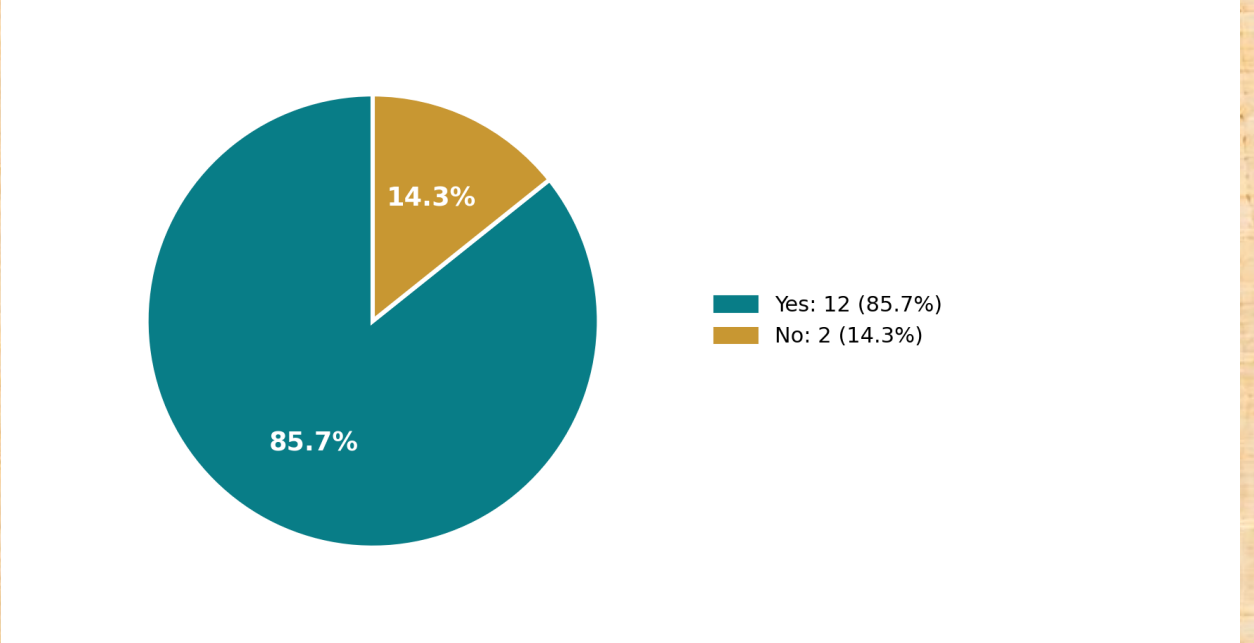
Figure A41. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 42

٤٢. إذا كانت فرصتك في ممارسة التحدث بالإنجليزية أقل من بعض الطلاب في مدارس أخرى، هل تعتقد أن وجود مساعد تستطيع التحدث والتدرب معه في أي وقت سيساعدك على تحسين مستواك؟

Could an anytime speaking assistant help if you have fewer opportunities to practise than students in other schools?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	12	85.7%
لا	2	14.3%

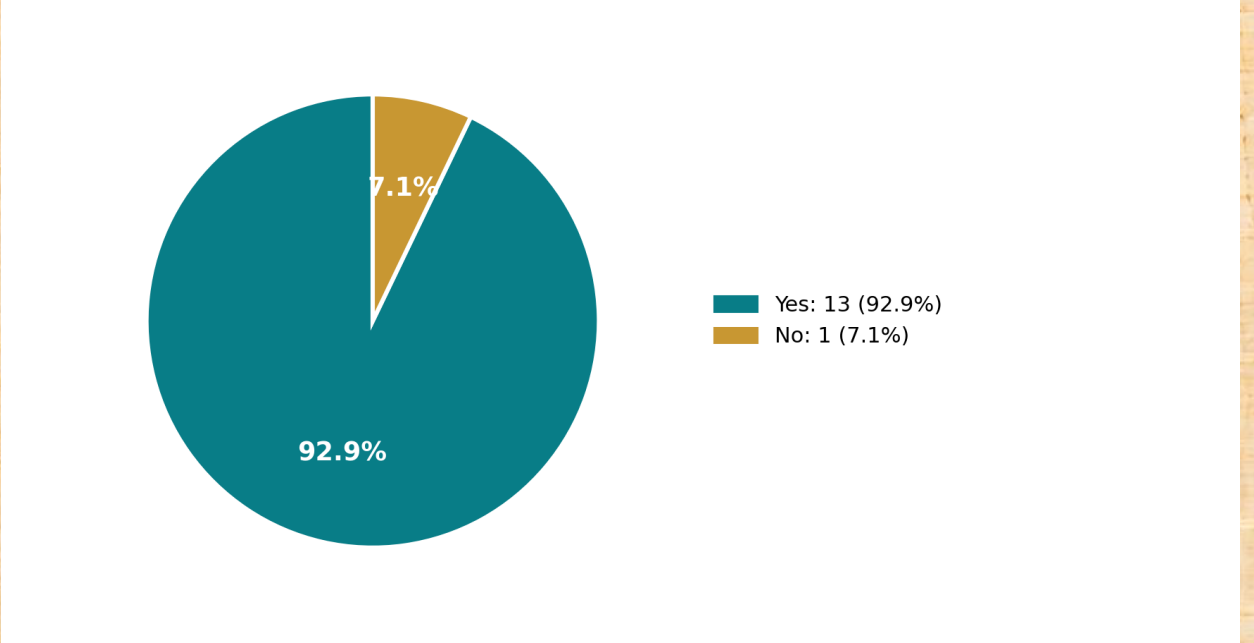
Figure A42. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 43

٤٣. إذا كان المساعد مرتبطًا بمنهجك، ويشرح أخطاءك، ويتابع تقدمك، هل يمكن أن تعتمد عليه كمساعد للتعلم بجانب المدرسة والمدرس؟

Could you rely on a curriculum-linked assistant that explains errors and tracks progress alongside school and your teacher?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

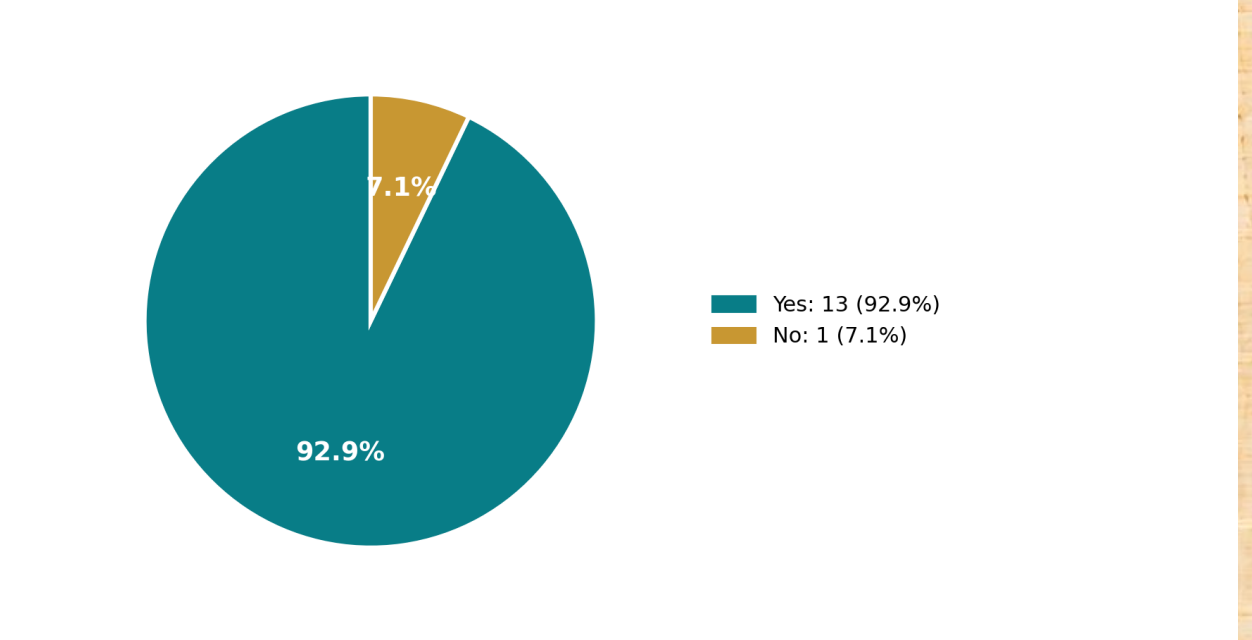
Figure A43. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 44

٤٤. هل ترغب في الاستمرار في استخدام مساعد كهذا بعد المرحلة الثانوية لتطوير الإنجليزية في الجامعة أو العمل؟

Would you want to continue using such an assistant after secondary school for university or work?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

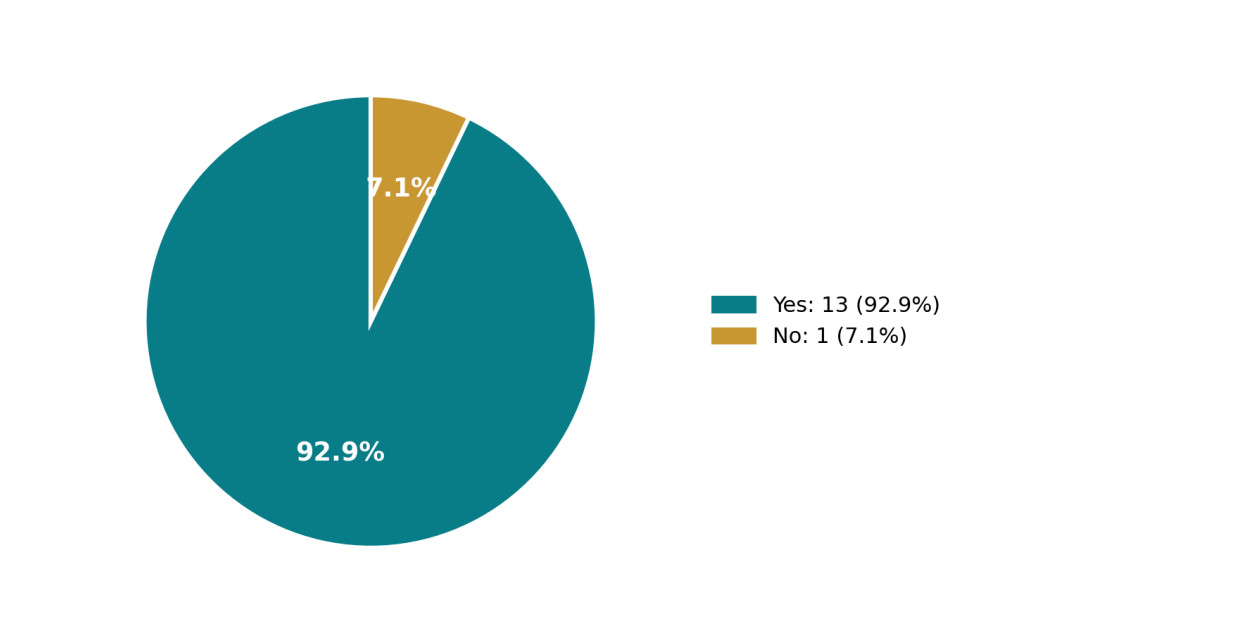
Figure A44. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 45

٤٥. إذا كان هذا المساعد متاحًا لك الآن بهذه الخصائص، هل ستستخدمه؟

Would you use this assistant now if these features were available?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	13	92.9%
لا	1	7.1%

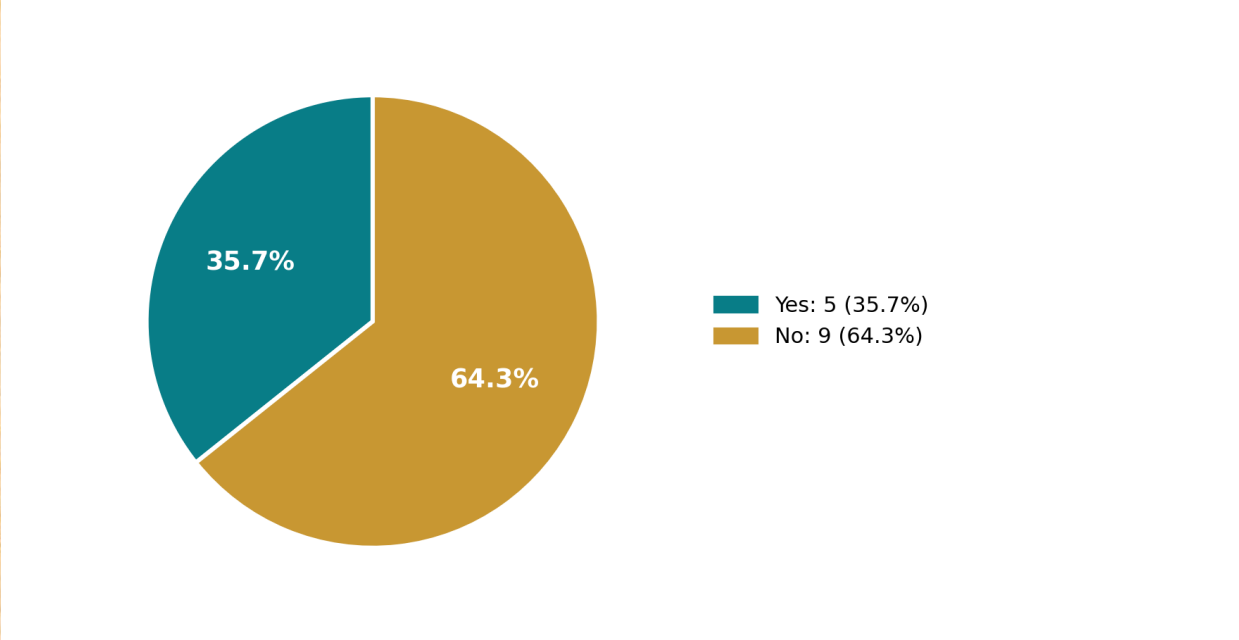
Figure A45. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Question 46

٤٦. هل توجد خاصية أخرى تتمنى وجودها في هذا المساعد؟

Is there another feature you would like?

Single response per respondent. Valid responses: 14.



Original response option	Count	Percentage
نعم	5	35.7%
لا	9	64.3%

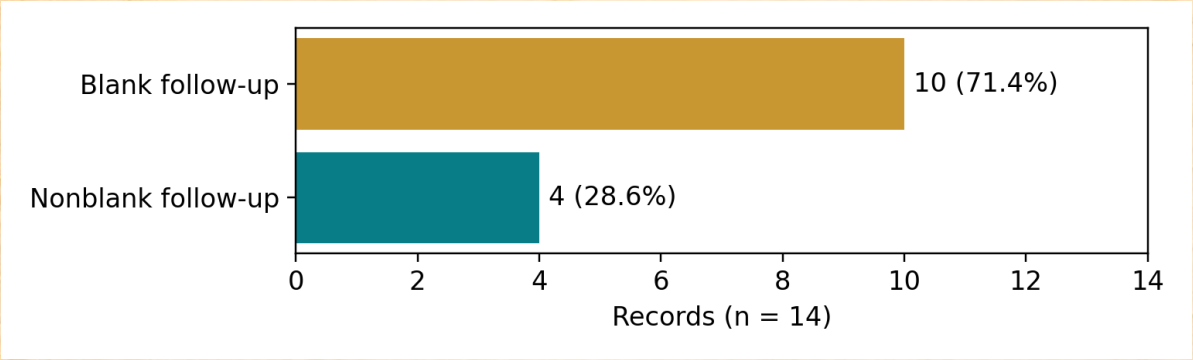
Figure A46. Source: raw survey export; denominator 14. Counts, including zero responses, are shown above.

Appendix A Open follow up

إذا كانت إجابتك نعم، اكتبها باختصار:

If yes, briefly describe it.

Q46 recorded 5 yes responses. Four nonblank follow-up texts were supplied; 10 records are blank. The mismatch is a completion limitation, not a fifth invented suggestion. Blanks are not coded as no additional needs.



Original text	Interpretation for planning
فهم الشخصيه ومحاوله تحسين أجواء النقاش	Understand personality and improve the conversational atmosphere.
Like app courses	A course-like app.
Normal conversations but using topics that are based on the curriculum	Ordinary conversations using curriculum-based topics.
it should be for free to all stidents	Free availability for all students.

Each theme is based on one text. Interpretations are descriptive; spelling in original responses is preserved and no prevalence beyond these four texts is claimed.